Making the Invisible Visible: The Impact of Revealing Indoor Air Pollution on Behavior and Welfare*

Robert Metcalfe
Columbia University & NBER

Sefi Roth
London School of Economics

October 16, 2026

Abstract

Exposure to ambient air pollution harms health and productivity and has motivated many mitigation policies. However, as people spend 90% of their time indoors, it is essential to understand exposure to Indoor Air Pollution (IAP) and, if high, ways to reduce it. We design and implement a field experiment in London that monitors households' IAP and randomly reveals their IAP in real time. At baseline, IAP exceeds ambient pollution when residents are home and is above WHO standards 38% of the time. We also find a steep household income-IAP gradient—larger than the income-ambient pollution gradient. During the experiment, real-time feedback reduces IAP by 17% (1.9 $\mu g/m^3$) overall and 34% (5 $\mu g/m^3$) during occupancy. The mechanism is behavioral: households increase natural ventilation (e.g., opening windows and doors) in response to feedback. In terms of welfare, we find: (i) a willingness to pay of £4.8 ($6) per 1 $\mu g/m^3$ reduction in indoor PM2.5; (ii) higher willingness to pay for mitigation than for full information; (iii) a price elasticity of demand for IAP monitors around -0.75; and (iv) that a subsidy for an IAP monitor or air purifier yields an infinite marginal value of public funds—a Pareto improvement.

*Acknowledgments: We thank Tom Parkes and Natalie Curd at Camden Council for helping us implement the project. We also thank Lillian Bartholomew, Kofi Broadhurst, Cora Buentjen, Navya Kumar, Victor Mylonas, Lieuwe Troelstra and Lotte Turnbull for their excellent research assistance. We thank Doug Almond, Patrick Baylis, Susanna Berkouwer, Nick Bloom, Marshall Burke, Olivier Deschenes, Teevrat Garg, Joshua Graff Zivin, Michael Greenstone, Nathan Hendren, Charlie Hutchinson, Nic Kuminoff, Stephen Jarvis, Akshaya Jha, Matthew Neidell, and Ben Sprung-Keyser for excellent comments and critical discussions, as well as seminar participants at the University of Chicago, London School of Economics, Stockholm School of Economics, University of Southern California, University of Birmingham, MCC Berlin, Institute of Fiscal Studies, University of Exeter, Southern Methodist University, University of California Santa Barbara, University of Wisconsin-Madison, Georgia Tech, Arizona State University, U.S. EPA, the AERE Summer conference, and the Advances in Field Experiments Conference. LSE Research Ethics Committee No 41072. AEA Registry: AEARCTR-0009912. We received no funding from any external organization for the research.

1 Introduction

The adverse impacts of ambient air pollution and its associated costs have received substantial policy and academic attention (Chay and Greenstone, 2003, 2005; Luechinger, 2009; Currie and Walker, 2011; Graff Zivin and Neidell, 2012; Currie et al., 2015; Chang et al., 2016; Schlenker and Walker, 2016; Ebenstein et al., 2016; Deschenes et al., 2017; Zhang et al., 2018; Deryugina et al., 2019; Clay et al., 2021; Bishop et al., 2023; Borgschulte et al., 2024; Miller et al., 2024). However, Indoor Air Pollution (IAP) has received considerably less attention, particularly in developed economies. Although pollution levels are generally lower than in many low- and middle-income countries recent empirical work has documented elevated indoor pollutant concentrations in these settings as well (Lebel et al., 2022; Burke et al., 2022; Kashtan et al., 2024). In fact, the U.S. EPA has long noted that indoor pollutant concentrations are often two to five times higher than outdoor levels (EPA, 1987). This is particularly concerning given that individuals in the developed world spend approximately 80 (90) percent of their time in their home (indoors) (Klepeis et al., 2001; Sundell, 2004; Spalt et al., 2016), a figure which has been increasing since 2020 with the trend to work-from-home (Barrero et al., 2023; Morris et al., 2023, 2024; Zarate et al., 2024). Therefore, as most air pollution exposure occurs indoors, the full welfare impact of air pollution may be substantially underestimated (Jeuland et al., 2015).

The relatively limited academic research on this subject has shown that IAP is correlated with a range of health outcomes, including respiratory diseases, heart disease, cancer, and milder effects such as headaches, dizziness, and fatigue (Monn, 2001; Pope III et al., 2011; Zheng et al., 2015; Li et al., 2017), as well as reduced cognitive performance (Stafford, 2015; Künn et al., 2023; Xu et al., 2024).[1] It has also been argued that IAP is likely to be as dangerous for human health as ambient air pollution (WHO et al., 2010), and studies show that IAP contains a variety of inflammatory and carcinogenic metals (Bandowe et al., 2021). Despite these potentially substantial impacts on mortality, morbidity, productivity, and well-being, the true scale of IAP and whether it can be mitigated remains uncertain due to insufficient monitoring, reporting, and experimentation. Recent advances in residential IAP measuring technology have led to high validity and

[1] There is also a large literature suggesting that ambient air pollution is detrimental to cognitive ability, human capital formation, and productivity (Currie et al., 2009; Graff Zivin and Neidell, 2012; Davis et al., 2013; Ebenstein et al., 2016; Bharadwaj et al., 2017; Chang et al., 2019; Huang et al., 2020; Bedi et al., 2021; Carneiro et al., 2021; Persico and Venator, 2021; Duque and Gilraine, 2022; Krebs and Luechinger, 2024; Chen, 2025; La Nauze and Severnini, 2025).

reliability of pollution monitors (Wang et al., 2020),[2] so estimating the true scale of the issue and assessing potential solutions is now possible.[3]

Indoor air quality is a largely unobserved good, one where both awareness and information are limited, yet technological advancements hold potential for significant improvements. An important characteristic of indoor air pollution is that it is largely shaped by household production decisions, including cooking, heating, and ventilation. As a result, providing information about indoor pollution may affect behavior through different mechanisms than those studied in the outdoor air pollution literature, since households can directly influence many of the activities that generate pollution within the home. A large literature has shown that improved information can substantially alter individual behavior and market outcomes, whether through technological innovation (Jensen, 2007; Jessoe and Rapson, 2014), information about others' beliefs (Cantoni et al., 2019; Bursztyn et al., 2020), or advances in predictive modelling (Fairweather et al., 2024). A key feature of the IAP context is that information is becoming more observable due to declining monitoring costs. This cost reduction, coupled with rising demand for air quality, has likely spurred innovation (Acemoglu and Linn, 2004), enabling households to monitor IAP affordably. However, since this information remains private, governments cannot directly monitor or regulate it. While individuals now have access to better IAP data, broader societal gains may come from policies that promote monitor adoption and motivate IAP reductions.

In this study, we conduct a field experiment in London to achieve three primary objectives. First, to document essential descriptive facts about IAP, including its scale, its relationship with ambient pollution, and the key predictors of IAP based on household and dwelling characteristics. Second, to investigate how the provision of real-time indoor air pollution information can motivate behavioral changes that improve indoor air quality, and thereby enhance overall health and well-being. Third, to understand households' willingness to pay for IAP monitoring and mitigation.

Achieving these objectives is empirically challenging for several reasons. A major hurdle is data availability: since monitors are not mandatory in homes, IAP levels often remain invisible. Even data from individuals who purchase monitors—such as crowd-

[2]The cost of IAP monitors has also declined over time (U.S. Environmental Protection Agency, 2025). This reduction can be attributed to factors including learning-by-doing in production, the development of low-cost sensors, and a growing market for IAP monitors, which spurs innovation (Acemoglu and Linn, 2004), as awareness increases about poor air quality from sources such as wildfires.

[3]The World Health Organization (WHO) estimates that IAP accounts for around three percent of the burden of disease and causes over three million premature deaths annually across a sample of developing countries (*WHO*, 2021).

sourced data—introduces selection bias, as middle- to low-income households are typically under-represented (Graff Zivin et al., 2024). Existing data also do not allow us to assess the magnitude of the problem across heterogeneous households or to estimate the causal effects of interventions to reduce IAP. The diverse sources and physical complexities of IAP make it extremely difficult—if not impossible—to accurately model indoor air pollution without indoor data.[4] The interaction between indoor and outdoor pollutants, combined with unobserved and unpredictable human behaviors (e.g., opening windows), adds further complexity. Together, these factors create a highly dynamic environment, making precise prediction and effective IAP management exceptionally difficult without direct indoor exposure data.

We overcome these empirical challenges by partnering with the local government of the London Borough of Camden, which distributed recruitment letters to a randomly selected sample of Camden's residents. Individuals who were interested in the topic completed an online survey on demographics, dwelling specifics, and health and were then randomly assigned to either a control or treatment group. Control group households received air pollution monitors that recorded real-time IAP levels but had their displays blinded. Treatment group households received the same monitors to collect baseline information, but after two weeks, the real-time IAP information became visible on their monitors. Additionally, households in this group received a two-page information sheet on interpreting the data and possible ways to reduce indoor pollution.

The treatment is a full knowledge treatment of IAP. This consists of many channels, such as real-time information on IAP (which has both knowledge and salience components), Hawthorne effects from monitoring, and information on how bad IAP is for human health. While we do not have experimental variation to separate these channels apart, we will provide evidence that the real-time feedback is what is driving any treatment effects we find. Moreover, both the control and treatment group understand that their IAP is being measured, so there is no difference in the intensity or incidence of monitoring between the groups,[5] and they placed the monitors in the same location in their homes.

Our treatment can be seen as "making the invisible visible," as the air pollution that we measure is undetectable to households without specialized monitoring equipment. By

[4]Indoor pollution arises from numerous sources, such as cooking, heating (e.g., fireplaces), smoking, and the burning of candles and incense.

[5]Previous research has shown that monitoring can itself have important consequences on behavior, especially in environmental markets (Gosnell et al., 2020; Zou, 2021).

providing real-time feedback in the treatment group, our intervention can be framed as full knowledge of the risks from indoor pollution, which can enter into the utility maximizing health production model (Pattanayak and Pfaff, 2009; Graff Zivin and Neidell, 2013). Individuals will choose the optimal risk-averting behaviors, which will be a function of time, resources, and knowledge of the averting behaviors. We posit that many individuals do not even have knowledge about how to measure IAP, less the averting behaviors that reduce IAP. So our treatment is a direct test of the the knowledge function in these health models, since if people are fully aware and knowledgeable, the treatment will have no impact on behavior and indoor pollution levels.

Our field experiment allows us to monitor indoor fine Particulate Matter (PM2.5), the US Air Quality Index (US AQI), temperature and relative humidity, for a duration of four weeks among 258 households. We also elicit a whole range of beliefs about the incidence of IAP, people's subjective health status, household and building composition, a range of behaviors and residents' willingness to pay for information on and mitigation of indoor air pollution. Our findings concern short-run responses in a self-selected sample from one London borough with relatively low pollution levels, which should be kept in mind when considering their broader applicability.

1.1 Our Findings

We begin our empirical analysis by documenting four novel descriptive facts from the IAP data. First, for our random sample in London, we find that average indoor PM2.5 concentrations (10.5 $\mu g/m^3$) are very similar to ambient PM2.5 concentrations (10.8 $\mu g/m^3$), with both exceeding The WHO and US EPA annual standards of 5 and 9 $\mu g/m^3$ respectively.[6] Second, we find that during occupancy hours (16:00-23:59), when residents are actually exposed to the air quality in their dwellings, indoor pollution levels are significantly higher than outdoor pollution (14.6 vs. 12.2 $\mu g/m^3$), likely due to household activities such as cooking, cleaning, heating, smoking, using candles, etc. Since the health and productivity effects are driven by exposure rather than average concentrations, we demonstrate that during time spent at home when exposure is very high, IAP is signifi-

[6]The Environmental Targets Regulations (2023) for PM2.5 in England require that the annual average of 10 $\mu g/m^3$ for PM2.5 will not exceeded at any monitoring station by the end of 2040. We use annual targets for comparison, as our study captures exposure over multiple weeks at different seasons of the year, making annual rather than daily exposure levels a more appropriate basis for comparison.

cantly higher than ambient air pollution.[7]

Third, our analysis reveals that ambient air pollution is not a main predictor of IAP. While we find a positive correlation between hourly indoor and hourly outdoor PM2.5 levels, outdoor levels do not explain indoor variations very well ($R^2 < 0.1$) and do not explain overall averages, emphasizing the important role of indoor PM2.5 sources. Fourth, there are several important predictors of indoor PM2.5, such as smoking, household income, and property ownership. This suggest a socioeconomic dimension to IAP, where lower-income households face higher exposure to indoor PM2.5, which has also been documented in previous studies (Ferguson et al., 2020; Burke et al., 2022). This relationship can be explained by numerous factors, such as poor housing quality, and thus can exacerbate existing inequalities in health, education, and productivity. To put this into perspective, households with below-median income levels have indoor PM2.5 concentrations that are 20 $\mu g/m^3$ higher than those of above-median income households. Expressed differently, for every £1,000 decrease in income, PM2.5 concentrations rise by 0.1 $\mu g/m^3$ (a 1% increase in income is associated with a 0.034% reduction in PM2.5).[8]

Turning to the results from the field experiment itself, we have five main findings. First, we find that randomized real-time feedback leads to a significant reduction in overall PM2.5 concentrations by 1.9 $\mu g/m^3$, translating to an 17% decrease from baseline levels. We also find that the treatment effect is much larger during occupancy hours, when people are typically at home and exposed to PM2.5, with a significant reduction of 5.0 $\mu g/m^3$, or 33.9% reduction from baseline. These reductions are very meaningful, especially considering the WHO annual mean standard for PM2.5 is 5 $\mu g/m^3$ and that our observed reductions exceed the effect of many large-scale interventions aimed at reducing air pollution.[9] When we re-weight our estimates to match our experimental sample to the population sample in terms of income, we find an even larger reduction in PM2.5

[7] These results are in contrast to the work by Krebs et al. (2021), who document that outdoor concentrations of PM are higher than indoor concentrations using crowd-sourced data from California.

[8] In addition, we find that cooking technologies and ventilation opportunities are also important predictors of IAP. These findings highlight the complexity of IAP and reveal that, while behavioral factors are significant predictors of IAP, ambient air pollution is not a primary determinant. This is in contrast to some of the older in-situ measurements of IAP from small selected samples (Hanley et al., 1994; Riley et al., 2002; Chen and Zhao, 2011).

[9] For example, the 2005 US Clean Air Act amendment for PM2.5 and the congestion pricing in Stockholm resulted in reductions of 3% and 15%, respectively, in particulate matter pollution (Sager and Singer, 2022; Simeonova et al., 2021). Specifically, Sager and Singer (2022) found a 0.4 $\mu g/m^3$ reduction in PM2.5, while Simeonova et al. (2021) observed a 4.56 $\mu g/m^3$ decline in PM10.

concentrations during occupancy time.[10]

Second, we show that the mechanism for these large reductions in PM2.5 came primarily through improved home ventilation. We show this in three different ways. First, we surveyed the treatment group about their actions to reduce PM2.5 levels. An overwhelming majority (72%) stated that they increased ventilation as their primary strategy to reduce pollution mainly through the opening of windows. Second, we utilize our data on indoor and outdoor air temperature before and after the treatment and show that the treatment group's indoor air temperature is more correlated with outdoor air temperature during the treatment period compared to the control group. This suggests that people are indeed opening windows to ventilate and dilute PM2.5 concentrations in their homes. Lastly, we examine the characteristics of peak indoor pollution events to determine whether the intervention led participants to modify their pollution-generating activities or implement strategies to curb high pollution levels. We do not find that the number of peak PM2.5 events is different between the treatment and control. However, their intensity, which we defined as the maximum PM2.5 level of an event, was reduced.

These results suggest that most people do not stop conducting their daily activities, such as cooking and heating behaviors that emit high levels of PM2.5 in the home (these activities are integral to their utility). Instead, households are changing how they respond to the high levels or taking preventive action to avoid a high level of pollution(e.g., opening windows during or after cooking). This result is a novel test of the averting behaviors entering the utility function directly–i.e., people still want to cook as it provides utility. Given the universality of this ventilation mechanism across all households, our results provide some external validity and possibility of scaling to other samples in the developed world (List, 2020, 2024).

The above result on ventilation at peak times also allows us to say something about the channels of the treatment effect. Treatment does not reduce baseline level of IAP or change the number of peak events–which would be consistent with an overall health knowledge channel from our treatment. But treatment changes the response to a peak PM2.5 event and that this response is only possible with real-time feedback. Therefore, we believe the main channel is the salient real-time feedback.

Third, we find that residents in the treatment group updated their beliefs about their

[10]This increase in the treatment effect with re-weighting suggests that our experimental sample estimates might be lower than the estimate if everyone in the country adopted the IAP monitor. However, there will be selection into adoption (with and without subsidies) so our experimental sample might be more externally-valid than extrapolating the average consumer.

home's IAP, realizing it is significantly worse than they originally believed. Interestingly, the effectiveness of our treatment in not only altering participants' perceptions of air quality but also affecting the confidence level of these perceptions, offering valuable insights into the psychological impact of air quality awareness interventions. For outdoor air quality, we find no significant effects, which is reassuring since we only provided information about indoor pollution.

Fourth, we conduct a comprehensive welfare analysis by examining the intervention's impact on human health, estimating households' WTP for clean air, and evaluating the welfare implications of subsidies for adopting an IAP monitor or an air purifier. Using our average treatment effect estimates, we show that the mortality savings from implementing the intervention across all UK households could reach £40 billion (£582 per capita). In terms of WTP, we find that in the control group, the average WTP is £37 for an IAP monitor and £45 for an air purifier (which reduces IAP to zero) over five years. In the treatment group, exposure to real-time air quality information increases these values to £39 for the monitor and £50 for the purifier over the same period. These findings indicate that households value mitigation measures (air purifiers) more than simply acquiring information on their indoor air quality. Translating these into a WTP for 1 $\mu g/m^3$ reduction, we find values of £4.06 for the control group and £5.54 for the treatment group (with perfect information), suggesting that real-time information enhances the perceived value of air quality improvements.

Finally, we assess the welfare impacts of subsidies for adopting an IAP monitor or an air purifier. We incorporate the price elasticities of demand derived from our WTP estimates, combined with our health and productivity impact estimates, reductions in healthcare spending by the National Health Service (NHS), and increased tax revenue from higher earnings to estimate a marginal value of public funds (MVPF) of a £1 subsidy for an IAP monitor or purifier (Hendren and Sprung-Keyser, 2020; Hahn et al., 2024). Under a range of different scenarios and assumptions, a £1 change in a subsidy for an IAP monitor or air purifier leads to an outcome where the productivity increases that flow back to the government more than offsets the upfront subsidy cost. That is, a £1 subsidy pays for itself just through the benefits to the government from reductions in IAP. This indicates that these subsidies would constitute a Pareto improvement by fully paying for themselves and this result occurs because the fiscal externalities from reducing PM2.5 are larger than the cost of the subsidy. There are two things to note here with our estimates. First, even if our treatment effect decreased by 90% (which is outside of our

95% confidence intervals), we would still estimate an MVPF of infinity for the £1 government subsidy for the IAP monitor. Second, even if people meaningfully underestimate the health benefits from reducing PM2.5 concentrations, the MVPF remains infinite.

1.2 Relationship to Existing Literature

Our study connects and contributes to several strands of research. In the context of IAP, the study closest to ours is by Greenstone et al. (2021), who conducted a field experiment in Delhi, India. They randomly assigned indoor air quality monitors to households and offered the opportunity to rent subsidized air purifiers. Their findings indicated that access to indoor air pollution monitors did not affect the take-up of subsidized air purifier rentals. However, their study faced significant survey non-response and attrition issues, and the authors themselves noted that the results should be interpreted with caution and considered suggestive. It is also worth highlighting that their study was conducted in Delhi, which is very different from London in terms of many characteristics, including housing stock, population, income, and pollution levels.[11]

A substantial literature has also examined household air pollution associated with solid-fuel use in low- and middle-income countries and documented its substantial health burden (Smith et al., 2014). Within this literature, several field experiments have examined the effects of cleaner cooking technologies on emissions and household well-being, with mixed results (Smith et al., 2011; Duflo et al., 2008; Mobarak et al., 2012; Bensch and Peters, 2015; Hanna et al., 2016; Barron and Torero, 2017; Pattanayak et al., 2019; Berkouwer and Dean, 2022; Beltramo et al., 2023). However, these studies generally do not directly measure indoor PM2.5 exposure. A notable exception is Berkouwer and Dean (2023), who use mobile PM2.5 monitors to measure personal exposure for 48 hours and examine how cookstove technologies affect that exposure.

Furthermore, our study relates to the extensive literature showing that access to ambient air quality information induces avoidance behavior, reducing pollution exposure and

[11] There are two other related studies. First, Sater et al. (2021) installed IAP monitors in French homes with wood-burning fireplaces, providing static feedback on PM2.5. They found that personalized information reduced indoor pollution by over 20%. Our study differs in key ways: (1) we analyze a general population sample, while they focused on fireplace users; (2) our recruitment was via randomized government letters, while they used targeted outreach; (3) we provide real-time feedback, while they used weekly delayed reports; (4) we examine the mechanisms behind pollution reduction through revealed behavior, whereas they rely on subjective reports; and (5) we assess the economic welfare effects of IAP changes. Second, Baquie et al. (2024) conducted an RCT in Tbilisi testing how different air pollution information affects awareness, avoidance behaviors, and self-reported health. Although their findings highlight the role of awareness in behavioral change, they did not study the impact on actual indoor pollution concentrations.

improving health outcomes (Neidell, 2009; Zivin and Neidell, 2009; Moretti and Neidell, 2011; Janke, 2014; Greenstone et al., 2022; Jha and Nauze, 2022; Barwick et al., 2024; Ahmad et al., 2022). Like this literature, our intervention makes an otherwise invisible environmental risk visible to households. However, whereas previous work has focused primarily on how information about outdoor air pollution affects avoidance behavior, we examine whether providing personalized information about indoor air pollution similarly changes household behavior. In doing so, we study a setting in which households directly influence many of the activities that determine indoor air quality. Related work also shows that ambient air quality information can generate political pressure for improved environmental quality (Axbard and Deng, 2024; Li et al., 2024; Yang et al., 2024). [12] We also relate to the work of Burke et al. (2022) and Lunderberg et al. (2023), who show that wildfire smoke can infiltrate homes and raise PM2.5 levels to 3-4 times the health-based guidelines. Notably, our findings in London demonstrate that indoor PM2.5 levels can reach 10 times the health-based guidelines for extended periods in many households. However, we find that ambient air pollution is not the primary source of this indoor pollution in London, rather it is generated by the behaviors of the residents themselves.

We also contribute to the literature on the valuation of clean air. To the best of our knowledge, we are the first to estimate the WTP for reductions in indoor PM2.5. The only other study estimating WTP for indoor air pollution more broadly is Ito and Zhang (2020) which examined the willingness to pay for indoor air quality improvements in China by leveraging data on air purifiers' effectiveness in reducing indoor PM10 and the price elasticity of demand. Ito and Zhang (2020) estimated an annual WTP of $1.34 to remove 1 $\mu g/m^3$ of PM10.[13]

We also relate to the literature on environmental justice. There are existing papers showing the income-pollution relationship for ambient air pollution (Banzhaf et al., 2019; Hsiang et al., 2019; Colmer et al., 2020; Jbaily et al., 2022; Currie et al., 2023; Colmer

[12] There is also research from Mexico (Hanna et al., 2021) and Uganda (Bassi et al., 2022) showing that there is generally a lack of information or knowledge about PM2.5 concentrations, meaning that people are not adequately compensated for the damage. However, information provided by government sources on PM2.5 concentrations to citizens is not always trusted, as Imtiaz et al. (2025) show for Pakistan. This is a potential benefit of private indoor IAP monitors (which is beyond the scope of our study)–the inability of government or external sources to manipulate the data.

[13] Relatedly, Chowdhury et al. (2025) measure households' willingness to pay for air purifiers in Dhaka and find the value to be under 10% of retail cost, rising by 26% after two months of combined monitor and purifier ownership. There have also been a range of hedonic estimates of the valuation of ambient air pollution (Smith and Huang, 1995; Bayer et al., 2009; Freeman et al., 2019). However, IAP is one of the environmental amenities that cannot be estimated from hedonic markets because most IAP exposure is generated from indoor sources and not location-specific sources (unless pollution comes from the ground, such as radon (Pinchbeck et al., 2023)).

et al., 2024). We show that lower-income households have significantly higher IAP levels than above median-income levels (whilst controlling for ambient air pollution), suggesting that environmental justice is not just an ambient phenomena, it is an indoor phenomena as well. Furthermore, the size of the gradient of the income-IAP relationship is much larger than the size of the income-ambient pollution relationship. As such, our paper leaves open the possibility that the observed differences in health impacts across income levels from ambient PM2.5 could, at least in part, be attributable to disparities in IAP exposure.

Finally, our paper relates to work showing that knowledge is a costly input in utility-maximizing health production models, with real costs to monitoring IAP and learning averting behaviors. This supports prior research suggesting that knowledge and awareness are critical to improving population health (Smith et al., 1990; Jalan and Somanathan, 2008; Pattanayak and Pfaff, 2009; Ashraf et al., 2013; Brown et al., 2017; Keskin et al., 2017; Bennett et al., 2018; Goeb et al., 2020; Ito and Zhang, 2020; Weitz et al., 2020). From our field experiment, these costs appear real and potentially large. Whether policymakers nudge adoption of technologies that provide real-time feedback (Loewenstein et al., 2007) or offer subsidies or standards to reduce monitoring costs, there is a clear need to make the invisible visible. These policies will likely pay for themselves.

2 Background, Experimental Design, and Data

In this section, we provide some background on indoor PM pollution (section 2.1), the design of the field experiment (section 2.2), and details on sampling, data, randomization, and estimation (section 2.3).

2.1 Background on Indoor Particulate Matter Pollution

Particulate matter, especially fine particles (PM2.5), is a key air pollutant due to its prevalence and substantial impact on public health. PM2.5 consists of a complex mixture of tiny particles and liquid droplets suspended in the air with a diameter of 2.5 micrometers or less (about three percent of the diameter of a human hair). The size of the particles is important for three main reasons: First, these tiny particles are able to bypass the body's natural defenses and penetrate deep into the lungs and even the bloodstream, posing serious health risks. There is overwhelming evidence that PM2.5 is incredibly harmful for

human health and well-being (Pope III and Dockery, 2006; Currie et al., 2014; Aguilar-Gomez et al., 2022). Second, PM2.5 is not visible to the naked eye, which means that individuals are often unaware of its presence and concentration levels in their immediate environment, leaving them unable to take measures to mitigate and adapt to exposure. Third, the small size of these particles allows them to better travel through small cracks and openings, enabling some outdoor-generated PM2.5 to infiltrate the indoor environment.

There are many sources of indoor PM2.5, including ambient (outdoor) pollution from both natural and anthropogenic sources such as transportation, industrial processes, wildfires, and dust storms. However, indoor PM2.5 is not simply a byproduct of ambient pollution—many indoor sources stem from everyday activities like cooking, heating, smoking, and even burning candles. Particulates can also be continually resuspended by daily physical activity in indoor settings (Tran et al., 2020). The majority of indoor PM emissions is believed to come from smoking, heating, and cooking (Tran et al., 2020).[14] Cooking appliances—including electric cookers, induction hobs, and toasters—contribute, as do the heating of pans and utensils, and especially the heating of food products like meats and oils (Wallace et al., 2004; Rohr and McDonald, 2016; Cheung et al., 2019). Cooking method matters: oil-based cooking (e.g., deep-frying, pan-frying, sautéing), particularly with low smoke-point oils, is a significant PM source. These activities drive indoor PM2.5 concentrations, which often exceed outdoor levels due to enclosed environments and inadequate ventilation—especially in colder months (O'Leary et al., 2019).

Interestingly, while some may believe indoor PM2.5 is "less dangerous" than its ambient counterpart due to potentially different composition, there is no scientific support for this claim. In fact, the WHO steering group on indoor air quality guidelines concluded there is no convincing evidence that particulate matter from indoor sources is less hazardous than outdoor sources (WHO et al., 2010).[15] While we understand ambient PM pollution well, we lack consistent indoor PM measurements. Still, we know ambient PM2.5 penetrates indoors, and we have limited evidence that everyday indoor

[14] In the UK, domestic combustion is also a major source of ambient particulate matter. According to the UK Department for Environment Food and Rural Affairs, emissions from this source (mainly from burning wood in closed stoves and open fires) account for 29 percent of ambient PM2.5 emissions (DEFRA, 2024).

[15] Epidemiological and public health studies show a positive association between elevated indoor PM2.5 and various health problems, including asthma symptoms and medication use in children in Baltimore (McCormack et al., 2009, 2011), wheezing in children in New York (Jung et al., 2012), and exacerbated respiratory symptoms and COPD in the UK (Osman et al., 2007). See also Rohr and McDonald (2016) for a review linking cooking-related PM emissions to lung cancer and low birth weight.

activities generate especially harmful PM2.5 components. For example, PM from cooking is thought to include Organic Carbon (OC) (Klimont et al., 2017; Zhao et al., 2019; Alves et al., 2021), along with inflammatory and carcinogenic heavy metals, elements, and polycyclic aromatic hydrocarbons (PAHs) (See and Balasubramanian, 2008; Zhang et al., 2017; Cheung et al., 2019; Bandowe et al., 2021). Furthermore, the health impact of indoor PM2.5 may be greater than that of ambient PM because individuals spend most of their time indoors, leading to prolonged exposure to elevated levels.

2.2 Experimental Design

We conducted a field experiment in the London Borough of Camden, England, which is one of the 32 local authorities that make up the administrative area of Greater London. Camden presents an ideal setting for our study, due to its geography, diverse population, and wide range of dwelling types and characteristics. These factors provide a unique opportunity to examine how indoor air quality and our intervention vary across different households and dwelling types in a relatively large urban area. From 2021, we partnered with The Local Government of Camden to recruit participants for our study, using their mailing system for random distribution of recruitment letters to households across the whole borough. Interested individuals were prompted to complete an online survey gathering baseline data on demographic characteristics, dwelling specifics, and health and well-being assessments as seen in Appendix Figure A1. We followed up with a reminder letter for households that did not respond to the first invitation. This two-step communication strategy significantly improved our recruitment efforts, culminating in a response rate of about 20% (i.e., almost 20% of a representative population were interested in taking part in the research). Additionally, we also offered a payment of £20 for taking part, which may also have led to this good response rate (see Appendix A2 for the initial letter).

Following the collection of baseline data from the survey, participants were stratified and randomly assigned to either the control or treatment group, ensuring demographic and health balance in addition to balance across all beliefs about ambient and indoor air pollution.[16]

[16]We blocked on the following variables: large household, children, income, education, tenure, happiness, anxiety, health, pain, calm, energy, health, sleep, health condition, open plan kitchen, hob type, fireplace, indoor and outdoor air pollution belief and confidence, comparative indoor air quality belief. We then balanced on the same variables.

Control Group: Households in the control group were equipped with a Kaiterra air pollution monitor (and accompanying equipment) that recorded real-time indoor air pollution levels for four weeks and transmitted this information to us (the researchers). However, the screen display on the monitor was covered with a security sticker which prevented participants from viewing this data in real-time (see Appendix Figure A3).[17] These households were informed of their indoor air pollution levels at the very end of the experiment and after they completed the end-line survey.

Treatment Group: Households in the treatment group received the same onboarding procedure with the exact same technology and sticker.[18] However, after two weeks of baseline data collection, participants were given access to their real-time pollution readings through an identical pollution monitor which did not have the screen-obstructing security sticker and with an information sheet detailing some basic information about the interpretation of their monitor readings, the health effects of pollution exposure, and potential measures that can be adopted to reduce air pollution within the home (see Appendix Figure A4). This information sheet was based on data that is publicly available on the Camden and the Kaiterra websites but we make this information more salient.

In terms of the indoor air pollution monitor, households received a Kaiterra Laser Egg Air Pollution Monitor that measures and presents information on Fine Particulate Matter concentrations (PM2.5), temperature, and relative humidity but also measures PM10. Importantly, the monitor converts the pollution metrics into the United States (US) Air Quality Index (AQI), with the display altering its color based on AQI levels and explicitly indicating the AQI category, such as 0-50 AQI for 'Good', through to 'Hazardous' for higher values of above 300 AQI. See Figure A5 below which demonstrates the screen on good and bad levels of AQI.

The IAP monitor collects real-time information on the above air quality measures every second and feeds it into an online database which we use in our analysis. We used the Kaiterra Laser Egg series for several reasons. First, we verified that it is a very reliable monitor of PM via testing against other (more expensive) monitors and also through consultations with experts in the field.[19] Second, the monitor was relatively affordable,

[17] The use of the security sticker also enabled us to check if participants tried to remove it and reassuringly, we found no evidence of tempering.

[18] The use of the same sticker in both the control and treatment groups nullified any differences in Hawthorne effects or attention placed on monitoring.

[19] Following our decision to use this monitor we also found out that Greenstone et al. (2021) also used the same monitor in their experiment in India. This is very reassuring as we both choose the same monitor independently.

with a price tag of around $150 (2021 prices). Finally, the monitor has a very consumer friendly and clear display that transmits the air quality data in real-time to the consumer. This design was ideal for our research question of making the invisible visible, and we could remotely track all of the data from the monitors during the study period.

The field experiment was implemented in five sequential waves, necessitated by budgetary constraints that limited the number of air quality monitors we could procure and deploy at any given time. This staggered roll-out not only addressed logistical limitations but also allowed us to evaluate treatment effects across different seasons.[20] To ensure consistent and reliable data transmission, all participating households were equipped with dedicated Wi-Fi routers and SIM cards, as pre-testing revealed that household internet connectivity was insufficiently reliable. USB plugs and adapters were also supplied to facilitate simultaneous powering of the routers and monitors from a single socket—an insight informed by feedback from our pilot study.[21] To streamline logistics and maintain data security, a professional courier service delivered the initial kits, while the research team personally handled the retrieval and installation of treatment-group monitors. This hands-on approach ensured compliance with experimental protocols and minimized participant burden, contributing to the successful implementation of the intervention.

At the conclusion of the field experiment, all participants completed an endline survey designed to capture their experiences, behavioral adjustments, and any changes in beliefs, perceived health, and overall well-being. The survey also elicited participants' willingness to pay for air pollution mitigation technologies, providing insight into the potential welfare gains associated with these interventions. After survey completion and the return of monitoring equipment, participants in both the treatment and control groups received individualized debriefing reports detailing the levels of indoor air pollution recorded in their homes over the course of the study. To acknowledge their time and contribution, each participant was also provided with a £20 voucher as a token of appreciation.

[20] Appendix Table A1 provides details on the timing and household counts for each wave.

[21] We conducted a pilot with 40 households recruited via Camden Council's website and social media platforms. The pilot tested the monitors, refined survey instruments, and resolved procedural issues to inform the full-scale deployment.

2.3 Further Experimental Details

2.3.1 Sample and Data

Our study was initiated by dispatching recruitment letters to 3,000 random households in the London Borough of Camden. Overall, we received a response from 566 households, which translates to a response rate of 19%. This figure aligns with the expected response for studies of this nature, where participants are required to commit to a data collection process in their own homes. Importantly, this response rate also offers insights into policy implications, particularly regarding household willingness to engage with air quality monitoring initiatives.[22] In our approach, we offered air monitoring equipment for free for four weeks, which sheds light on the potential uptake of such devices if the government would subsidize a similar initiative. This aspect of our findings could inform policymakers about the feasibility and public interest in subsidized air quality monitoring programs within residential settings.

From the pool of respondents, we selected 262 households that were available during the study period (i.e., not on holiday) and contributed to optimizing the balancing process (see below for more details). The study experienced minimal attrition, with only three households failing to complete the study in accordance with our research protocol. Additionally, we excluded one participant who was identified as an extreme outlier in terms of PM2.5 levels. As a result, our final sample consists of 127 households in the control group and 131 households in the treatment group.

Table A2 provides a comparison of various characteristics between our study sample and the broader population of Camden, thereby facilitating an evaluation of our study's external validity (Camden Council, 2023). The table reveals notable differences that illustrate the composition of our sample relative to the general demographic and socioeconomic fabric of Camden. Specifically, our sample is characterized by a higher proportion of individuals with advanced educational attainment and higher earners, compared to the general Camden population. Furthermore, there is a reduced presence of social housing occupants and single households within our sample, alongside a higher prevalence of homeownership. In terms of heating, our sample exhibits more gas heating and less electric heating in comparison to the broader Camden population.

[22] There is selection into the research based on their preferences and constraints, which means that we have a framed field experiment (Harrison and List, 2004). We will understand the importance of this selection when we re-weight our estimates to match the population sample.

While these discrepancies do not undermine the validity of our findings within the sample context (the internal validity), they do suggest caution when generalizing our main results to the entire Camden population and indicate that our study's insights may be most applicable to similar demographic segments. Despite this limitation in representativeness, it is important to acknowledge that our study represents a significant advancement over previous research efforts in the field of indoor air quality monitoring. Prior studies have often relied on samples comprising households that independently decided to purchase air monitors or recruited participants exclusively via social media channels. These methods are likely to introduce a much larger selection bias into the study, limiting the diversity and generalizability of the research findings. In contrast, our approach, while not perfectly mirroring the Camden population, offers a more systematic and controlled method of participant selection, thereby reducing the likelihood of such biases. Moreover, the characteristics of our sample provide valuable insights into the types of households that are willing to engage with air quality monitoring initiatives. This understanding can inform future outreach and engagement strategies for air quality monitoring projects, ensuring they effectively reach and resonate with a broader segment of the population. Finally, in the robustness section, we address this external validity threat directly by reweighting our estimate based on observable demographic characteristics, ensuring it is more representative of the entire Camden population.

We complemented our indoor air quality data which we collected ourselves with information on ambient pollution and weather from government-regulated measuring stations. These stations are managed by the UK Environment Agency on behalf of the Department for Environment Food and Rural Affairs (Defra) and the Devolved Administrations. Given the extensive network of monitors in London, we utilised four different measuring sites for this information which are situated within relatively close proximity to the households in our study (median distance of 2km). Overall, our comprehensive dataset encompasses 150,079 hours of air pollution and weather data across 258 households.

2.3.2 Randomization

In each wave of our study, the procedure for randomizing households was as follows. Initially, four seeds were chosen through a random selection process. Subsequently, households were categorized (blocked on) based on a variety of characteristics to ensure a balanced distribution across treatment and control groups. These characteristics included

household size (specifically identifying households with more than three members), the presence of children, levels of education and income, the gender composition and number of occupants, housing tenure, responses to surveys on beliefs and welfare, kitchen layout (notably the presence of an open-plan design), the type of heating system and main cooking appliance (the kind of fireplace used and cooking hob), whether an extractor hood was installed, the total number of rooms and windows, existing air purifiers, and whether smokers resided within the household.

To achieve an optimal randomization balance concerning these factors, we employed a block randomization technique, executing 5,000 iterations for each of the four seeds. In each iteration, we replaced the previous assignment if it showed an improved balance across our specified blocking variables. Upon completing the randomization for a wave, we checked for balance with the previous waves that had already been completed and selected the treatment assignment among the four options that best maintained balance across waves.

Table 1 presents the number of observations and means of a range of characteristics across the treatment and control groups. The final column reports the p-values from statistical tests assessing whether the mean differences between the two groups are statistically significant. The results show that there are no significant differences between the treatment and control groups, confirming that the randomization process was successful. This balance across key characteristics underscores the robustness of our randomization process and reinforces the internal validity of our intervention's effects analysis.

2.3.3 Estimation

Our main empirical strategy employs a differences-in-differences (DiD) methodology to estimate the causal effect of air quality information on household behavior and subsequent indoor air pollution levels. Formally, we estimate the following model:

$$\begin{aligned} PM_{2.5,it} = \beta_0 + \beta_1 \text{Treatment}_i + \beta_2 \text{Post}_t + \beta_3 (\text{Post} \times \text{Treatment})_{it} \\ + \beta_4 \text{AmbientPM}_{it} + \delta_t + \varepsilon_{it} \end{aligned} \tag{1}$$

where $PM_{2.5,it}$ represents indoor PM2.5 for household i at time (hour) t, Treatment_i is a dummy indicating that the household is in the treatment group, Post_t is a dummy indicating the post treatment period, $(\text{Post} \times \text{Treatment})_{it}$ is the DiD interaction term,

AmbientPM$_{it}$ is ambient PM2.5 concentrations for household i at time t, δ_t is time (day of week and month) fixed effects, and ε_{it} is an idiosyncratic error term. Standard errors are heteroskedastic-consistent and (two way) clustered by household and date to account for the fact that pollution is serially correlated across both space and time. [23]

The inclusion of ambient PM2.5 allows us to control for external air quality conditions, ensuring that our estimates are not confounded by outdoor pollution levels that may independently affect indoor PM2.5 levels. The day of week fixed effect captures any systematic variation in air quality that could be attributed to different days of the week, thereby controlling for weekly patterns in household activities that could affect pollution levels. Additionally, the inclusion of month fixed effects helps address seasonality, which can influence pollution through factors such as ventilation practices for example.

We employ the differences-in-differences (DiD) methodology for several important reasons that align with the specific constraints and objectives of our study. First, the DiD approach inherently controls for the Hawthorne Effect, the phenomenon where individuals alter their behavior simply because they are aware they are being observed, which is a concern in any experimental study. By comparing changes over time between the treatment and control groups, any behavioral modifications among participants due to their awareness of being studied will be differenced out in the estimation process. Second, the reliance on differential changes over time as the basis for estimation helps to circumvent potential measurement errors that could distort the analysis. Since DiD estimates are derived from changes in the dependent variable, rather than its levels, any constant measurement error is differenced out, thereby purifying our estimates from such noise. This attribute of the DiD methodology is particularly important given the complexities associated with accurately measuring indoor air pollution levels. Third, our sample is sufficiently large but perhaps not large enough to completely eradicate all concerns about randomization and the potential for the imbalance between treated and control groups. Our balancing test reported in Table 1 suggests that our treatment and control groups are balanced on observables but the DiD methodology allows us also to control for unobservable factors that are constant over time, thereby mitigating potential biases.

[23] For robustness, we also cluster standard errors at the household level only, and the standard errors remain nearly identical.

2.3.4 Pre-registration

Prior to the start of the study, we archived a pre-registration describing the experimental design, primary outcomes, and planned sample size.[24] Although this registration was completed before data collection, it was a study registration rather than a fully specified pre-analysis plan. While it described the primary treatment analysis and target sample size, it did not pre-specify all estimation choices, secondary analyses, or robustness exercises reported in this paper. Consequently, some analyses involve greater researcher discretion than would have been the case under a fully specified pre-analysis plan. To improve transparency, we report multiple-testing adjustments, present simple ("vanilla") specifications alongside our preferred specifications, and report results for the originally planned sample in the Online Appendix.

The original registration specified a target sample of 150 households (75 treatment and 75 control), which was powered for the primary treatment-effect analysis. Following additional funding obtained during the implementation phase of the project, we expanded recruitment to improve statistical precision, particularly for heterogeneity analyses and willingness-to-pay outcomes. The final sample therefore consists of 258 households (131 treatment and 127 control). The decision to expand recruitment was made during study implementation and was not based on interim analyses or statistical significance. In addition to the increase in sample size, the only material deviation from the original registration concerns the study timeline. The registration incorrectly stated that the post-treatment observation period would last four weeks. In practice, all participant-facing recruitment materials consistently described the study as lasting one month in total, including the pre-treatment period, and the experiment was implemented accordingly.

To assess the sensitivity of our findings to the original study design, the Online Appendix also reports results using the first three recruitment waves (175 households), an additional 150-household sample matching the original recruitment target, and the corresponding vanilla specifications. The results are highly similar across these alternative samples and specifications.

[24] The study pre-registration can be accessed at: https://www.socialscienceregistry.org/trials/9912.

3 Descriptive Statistics and Baseline Analysis

In this section, we conduct a descriptive analysis focusing first on the comparison and interactions between ambient and indoor air pollution (Section 3.1). We then estimate the determinants of IAP, focusing on household and dwelling characteristics as predictors (Section 3.2).

3.1 The Relationship Between Indoor and Ambient Air Pollution

We start by examining the relationship between indoor and outdoor air pollution. Table A3 provides a summary of 150,079 hours of air pollution and weather data collected from 258 households. The left side of the table presents information for the full sample, covering the entire four-week duration of the field experiment, while the right side focuses exclusively on the pre-treatment period (the first two weeks). This distinction is critical, as the treatment is designed to reduce indoor pollution, making the pre-treatment data vital for establishing baseline conditions. Column 4 aggregates data across all hours of the day and indicates that indoor and ambient PM2.5 concentrations are relatively similar during the pre-treatment period, averaging 10.85 µg/m³ and 11.61 µg/m³, respectively. These two numbers are important because they highlight that, despite London being a modern developed city with ambitious and effective policies to reduce pollution (mainly traffic related policies such as the Low Emission Zone, expanded cycling infrastructure, and strict emission standards for taxis) pollution levels remain above recommended international health guidelines. To put it in perspective, the annual US EPA and WHO guidelines are currently set at 9 and 5 g/m³, respectively. In England, the Environmental Targets Regulations 2023 mandate reducing PM2.5 concentrations to 10 g/m³ or lower by 2040, but this higher target must be achieved across all Automatic Urban and Rural Network (AURN) monitoring sites in England, not just on average.

In columns 5 and 6 of Table A3 we further analyze the pre-treatment pollution data by stratifying it into two distinct time frames: occupancy time (16:00-23:59) and non-occupancy time (all other hours). This temporal distinction is essential because we are particularly interested in periods when people are at home, active, and near the pollution monitor. The results reveal an important pattern: indoor pollution levels are significantly

higher during occupancy hours compared to non-occupancy hours (14.6 vs. 8.9 $\mu g/m^3$).[25] This 65 percent increase in pollution during occupancy hours is critical for several reasons. First, elevated pollution levels during occupancy time are particularly concerning, as this is when individuals are most likely to be at home (back from school and/or work) and actively using their home. Because participants were instructed to place their monitors in the room where they spend most of their awake time, the data also closely approximate their real-world exposure to indoor air pollution, providing meaningful insights into potential health and well-being impacts. [26]

Second, the notable increase in pollution during active household hours suggests that human behavior significantly contributes to the observed air quality degradation (note that ambient pollution levels do not exhibit the same sizable increase in concentrations and are not very well correlated with IAP as we document more formally below). As mentioned before, activities, such as cooking and heating, can release pollutants, intensifying indoor air pollution during these hours. Moreover, ventilation may also play a crucial role in the evolution of indoor air quality, with its effectiveness potentially being dependent on human behavior as well, such as the opening and closing of windows. This behavior-driven pattern of pollution highlights the influence of daily activities on indoor air quality, suggesting a clear link between human behavior and environmental health impacts. These findings emphasize the need for targeted strategies to mitigate exposure to indoor air pollutants, especially during peak times when people are at home and engaged in activities that contribute to their indoor air quality.

Our temporal analysis also reveals that during occupancy hours (16:00-23:59), when residents are actually exposed to the air quality in their dwellings, indoor pollution levels are in fact *higher* than the pollution outside (14.6 vs. 12.2 g/m3), with indoor concentrations exceeding outdoor levels for a significant proportion of the time. This is evident in Table A3 as described above and also in Figure A9 where we show a histogram of the hourly difference between indoor and ambient pollution during occupancy hours. The figure reveals that for 22% of the time, indoor PM2.5 is higher than ambient PM2.5 during this time.[27] This result suggests that pollution spikes during these episodes are not

[25] We also plot the average PM2.5 levels by hour in Figure A6 to illustrate how PM2.5 fluctuates throughout the day. The figure clearly shows a sharp peak in indoor PM2.5 during the evening hours. Ambient PM2.5 also rises during this time, possibly due to vehicle emissions, though to a much lesser extent.

[26] We also present histograms of indoor PM2.5 concentrations for the entire day and specifically during occupancy in Figures A7 and A8, respectively.

[27] Figure 1, which presents the histogram of the hourly difference between indoor and ambient pollution for the entire day (not just during occupancy hours), reveals that indoor pollution exceeds outdoor pollution 16% of the time.

only sharp enough to meaningfully elevate the overall average but also frequent enough to be a major concern. While there are not many studies that can accurately compare the two measures, our finding is consistent with the work of Greenstone et al. (2021) for India. However, our result is in contrast to the work by Krebs et al. (2021), who document that outdoor concentrations of PM are higher than indoor concentrations using crowd-sourced data from the PurpleAir Real-Time Air Quality Monitoring Network in California. There could be many reasons as to why the Krebs et al. (2021) study differs from ours and the Greenstone et al. (2021) study.[28] Overall, our indoor finding demonstrates the severity of indoor air pollution levels in a developed city and their potential health impacts. This highlights the urgent need for interventions to reduce exposure to particulate matter in indoor environments, which may be an even more pressing issue than addressing ambient pollution.

Following our analysis of temporal pollution patterns and the discussion around the potential role of human behavior in shaping indoor air quality, we further delve into the relationship between indoor and outdoor pollution levels. This examination is visualised in Figure 2, where we plot hourly outdoor PM2.5 concentrations on the x-axis against hourly indoor PM2.5 concentrations on the y-axis. To illustrate the correlation between these two variables, we have fitted a linear regression line, depicted in green, against a 45-degree black dashed line, which would indicate a direct one-to-one relationship. This analysis reveals a positive relationship between indoor and outdoor PM2.5 levels, suggesting that outdoor pollution does contribute to indoor pollution levels to some extent. However, the relatively low coefficient of determination associated with the fitted line (R^2=0.098) indicates that outdoor PM2.5 levels explain very little of the variations in indoor PM2.5 concentrations. This finding implies that while there is a correlation as suggested by prior studies (Krebs et al., 2021; Burke et al., 2022), a significant portion of indoor air pollution can be attributed to sources within the indoor environment itself.[29]

[28] The divergence between the our findings and Krebs et al. (2021) could be because: (1) theirs is based in a different location with different climatic conditions, housing stock, and human activity indoors; (2) their sample is based on a very selected (but important) group of people who decided to purchase an air pollution monitor whilst our sample is based on a more represented pool of households; and (3) some combination of both.

[29] Given that ambient air monitors provide pollution concentration measurements for specific locations, which may not accurately reflect individual exposure due to the distance between the monitor and the individual's actual environment (Fowlie et al., 2019), we investigate the outdoor-indoor pollution relationship in Appendix Figure A10 but separately for households located below and above the median distance to the nearest ambient pollution monitor. This allows us to test whether proximity to the monitor explains the low explanatory power of this relationship, despite the relatively close location of the monitors to the households in our study. The results indicate that even for households in closer proximity to the ambient monitors, the R^2 value remain very low, suggesting that proximity alone does not account for the weak relationship observed.

Figure 2 further supports the statement that indoor air pollution is not that related to ambient air pollution conclusion in two notable ways. First, the scale of the axes is markedly different, with indoor pollution levels reaching up to 999 µg/m³, whereas outdoor levels remain well below 100 µg/m³. This disparity underscores the potential for indoor activities and sources to significantly elevate pollution levels beyond what is observed outdoors. Second, on the left-hand side of the figure, we observe many cases where very high levels of indoor pollution occur while ambient pollution remains relatively low. This also suggests that the source of the pollution in these cases is likely to be indoors. Together, these insights highlight the complexity of indoor air pollution dynamics and the critical role that indoor sources and activities play in determining the healthiness of indoor environments.

We continue by econometrically estimating the extent to which ambient air pollution infiltrates homes, contributing to IAP, and the duration of its impact on indoor pollution levels. To do so, we replicate Krebs et al. (2021), which investigates the penetration rates of ambient PM2.5 into the indoor environment. Using their methodology, we regress indoor PM2.5 levels on lagged outdoor PM2.5 values (up to 12 hours) while controlling for outdoor temperature, dew point temperature, wet-bulb temperature, and including fixed effects for hour, day, and month.[30] This is displayed in figure A11, where we plot the coefficients of the lagged outdoor PM2.5 values. We get very similar results to Krebs et al. (2021), with bigger point estimates. Consistent with their findings, the primary influence of the outdoor environment lasts for approximately 5 hours and dissipates entirely within 9 hours. We also look at this over the first two weeks of our data - the pre-treatment period, to avoid the effect of the experiment, and we get an identical looking graph as seen in Figure A12. Furthermore, we investigate the effect of the treatment of this experiment in the post-treatment period on penetration rates. This can be seen in Figure A13a and A13b, where the penetration rates are higher in the control group and the effects last longer. This indicates that some households in the treatment group have altered their behavior and ensured that outdoor pollution affects them less.

We also look at this effect on different time periods to show how outside air temperature affects penetration rates. Similar to Krebs et al. (2021), we show that penetration rates are much higher in warmer weather. We can see this when comparing Winter and Summer in Figure A14b and Figure A14a. In these graphs, it is clear that penetration rates are higher in summer, and that indoor PM2.5 stays higher for longer. This could be

[30]The regression equation we use is $PM2.5_{it} = \sum_j \beta_j \overline{AmbientPM2.5}_{it-j} + \gamma X_{it} + \delta_t + \theta_i + \varepsilon_{it}$. Standard errors are clustered at the household and day levels.

due people keeping their windows open for longer hours.

3.2 Predictors of Indoor Air Pollution

We next turn our attention to identifying the principal predictors of indoor PM2.5 levels, as detailed in Tables A4 and A5. These tables, which focus solely on the pre-treatment occupancy period, examine the influence of various household and dwelling characteristics on indoor air pollution concentrations. We begin by examining household characteristics in Table A4. Our descriptive analysis reveals that smoking is the most significant predictor of elevated indoor PM2.5 levels. This finding highlights the substantial impact that indoor smoking can have on air quality and is aligned with existing research in this area which tends to be based on a much smaller sample size.[31]

Following smoking, the household's income level (measured as above and below median for the area) is the next biggest predictor of higher IAP. We further explore the income dimension in Figure A15, which shows PM2.5 concentrations across different income groups during our pre-treatment period. The figure reveals a pronounced downward-sloping relationship between income and PM2.5 concentrations, highlighting how pollution exposure decreases significantly as income levels rise. In particular, we find that for every £1000 increase in income, indoor PM2.5 falls by $0.1\mu g/m^3$ (a 1% increase in income is associated with a 0.034% reduction in PM2.5). To put this in perspective, below median income households have a PM2.5 level 20 $\mu g/m^3$ units higher than above median income households. Although similar patterns have been observed in ambient pollution (Jbaily et al., 2022; Colmer et al., 2020), the relationship between income and pollution is much larger for indoor than ambient. It is important to document that this disparity exists for indoor pollution in the developed world but also potentially larger than ambient.[32]

Our analysis in Table A4 also shows that renters experience higher levels of indoor

[31] For example, a study by Semple et al. (2015) in Scotland analyzed PM2.5 concentrations in homes with smokers compared to those without and revealed that the average PM2.5 levels in the 93 smoking homes were about ten times higher than in the 17 non-smoking homes. Importantly, the findings reported by the study in Scotland and our findings presented here not only highlight the substantial role of cigarette smoking in elevating indoor PM2.5 concentrations but also the severe exposure risks for non-smokers residing in these environments. According to Semple et al. (2015), non-smokers living in smoking households typically experienced average PM2.5 exposure levels more than three times higher than the World Health Organization's (WHO) guidance for annual PM2.5 exposure.

[32] When we look at ambient PM2.5 we find no evidence for statistically and economically significant relationship between income and ambient pollution over the course of the data collection period. The ambient PM2.5 gradient shows that for every £1000 increase in income, outdoor PM2.5 falls by 0.0073.

PM2.5 compared to homeowners. This disparity could reflect differences in the quality of living conditions, including the age and maintenance of rented versus owned properties, as well as potential restrictions on modifications that could improve indoor air quality, or it could be personal behaviors associated with renters. Overall, these results suggest a socioeconomic dimension to indoor air quality, where lower-income households (renters and smokers who are more prevalent among disadvantaged groups (Auld, 2005; Hiscock et al., 2012; DeCicca et al., 2022)) may face higher exposure to indoor PM2.5.

Next, we examine how dwelling characteristics are linked with indoor PM2.5 levels in Table A5. Notably, certain household appliances emerge as significant predictors of indoor air pollution levels. In particular, we find that having an electric or gas stove is positively correlated with higher indoor air pollution levels compared to induction hobs. These appliances, commonly used for cooking and food preparation, contribute to the indoor emission of particulates, thereby elevating PM2.5 concentrations. Interestingly, induction hobs appear to have a relatively lower impact on indoor PM2.5 levels. This finding aligns with their design, as induction hobs heat cookware directly through electromagnetic fields, reducing heat loss and cooking time. This efficiency minimizes overheating, which in turn produces lower emissions of PM2.5 compared to electric hobs. Furthermore, induction hobs lack exposed heating elements, preventing particulate emissions from surface oxidation. Their efficient heating also reduces ambient heat buildup, a factor supported by our observation of a positive correlation between higher indoor temperatures and elevated PM2.5 levels. However, this may be due to selection bias as induction hobs are also more expensive and are used by higher income households.

In contrast to the impact of cooking appliances, the presence of more windows in a dwelling is associated with lower baseline pollution levels. This can be attributed to improved air circulation and the dispersion of indoor-generated pollutants, effectively mitigating indoor air pollution. Interestingly, we do not find a statistically significant association between the presence of an air purifier and indoor PM2.5 levels. While air purifiers are known to reduce indoor pollution when properly used and maintained, this result may reflect factors the importance human behavior, as ownership does not guarantee regular use or adequate maintenance. [33]

Overall, our results highlight the complexity of factors affecting indoor air quality and underscore the significance of household and dwelling characteristics in influencing indoor air quality. As such, interventions to improve indoor air quality might require a

[33] 44 households reported that they had at least one air purifier at baseline.

tailored approach through architectural design, lifestyle modifications, informed appliance choices, and personalized information about indoor pollution.

4 Main Results

4.1 Impact of Real-time Feedback on Indoor Air Quality

In this section, we investigate the impact of our intervention designed to provide residents with real-time, personalized feedback on indoor PM2.5 pollution levels within their homes. Our primary aim is to assess whether such feedback can effectively reduce indoor PM2.5 concentrations. Figure 3 shows our main point estimates and confidence intervals for three regression equations.[34] The left estimate shows the average treatment effect of our intervention on total PM2.5 concentrations. As evident from the figure, we find that providing real-time indoor pollution information reduces overall pollution concentration by 1.9 $\mu g/m^3$ (microgram per cubic meter), representing an effect size of 17.3% of mean PM2.5 of the control group post treatment or 17.6% of baseline pollution (i.e. pre-treatment). Notice that this estimate captures a reduction relative to the control-group counterfactual, rather than a before-and-after decline of the same magnitude among treated households.

In the next two estimates presented in Figure 3, we differentiate (as in the previous section) between occupancy time (16:00-23:59) and non-occupancy time (all other hours) respectively. This differentiation is crucial, given that our occupancy hours typically represent time when individuals are at home, active and near their pollution monitor which gives us a very good indication to their actual pollution exposure during this time. Our results demonstrate a much more pronounced effect during occupancy hours, compared to our overall estimate reported for the all day. This estimate is also statistically significant at the 1 percent level and very large. In particular, we find that our intervention reduces indoor PM2.5 concentrations by 5 $\mu g/m^3$, equivalent to a 34% or 0.11 of a standard deviation reduction from baseline during the same hours or 32.5% reduction compared to mean PM2.5 in the control group post treatment. This is a sizable reduction, especially when compared to other prominent interventions. For example, the 2005 US Clean Air Act amendment for PM2.5 achieved a reduction of only 3% (or 0.4 $\mu g/m^3$) over five

[34] Appendix Table A6 presents these results in a table format, including versions with and without bootstrapped standard errors for robustness.

years, congestion pricing in Stockholm reduced PM10 pollution by 10-15%, and the introduction of Low Emission Zones (LEZ) in Germany reduced PM10 levels by 9%. (Sager and Singer, 2022; Simeonova et al., 2021; Wolff, 2014). Conversely, the effect observed during non-occupancy hours is much smaller and not statistically significantly different from zero, suggesting the intervention's effectiveness is only heightened when residents are likely to be home and awake.

We further delve into the temporal dynamics of the intervention's efficacy through Figure 4, which shows the average treatment effects segmented by hour of the day. This graphical representation corroborates the findings outlined in Table A6, particularly highlighting the concentration of negative effects during occupancy hours. These temporal insights enrich our understanding of the intervention's impact, emphasizing the significance of daily household routines on indoor air quality. We also explore the effect over time in Figures 5 and A16. These graphs indicate that (1) our parallel trends assumption holds reasonably well and (2) the effect size remains consistently high throughout the post-treatment period, with a more pronounced and concentrated impact observed during occupancy hours. In Figure A17 we further analyze this temporal dimension and find that the intervention's impact during the first and second weeks is nearly identical, highlighting once more the consistency of the treatment effect over time. Finally, in Figure 6, we examine the impact of our intervention by season and find that our results are entirely driven by the intervention during the winter months. During this period, dwellings typically have lower levels of ventilation (due to closed windows) and higher usage of heating sources, both of which contribute to elevated indoor pollution levels.

Next, we explore the potential non-linear effects of our intervention through a probit model analysis, focusing on the time households spend within various Air Quality Index (AQI) categories. The US AQI, a scale that quantifies air quality, spans from 0 to 500, categorizing air quality into six levels: Good (0-50), Moderate (51-100), Unhealthy for Sensitive Groups (101-150), Unhealthy (151-200), Very Unhealthy (201-300), and Hazardous (301-500), with lower scores indicating healthier air quality. A unique feature of our intervention is the air quality monitor's ability to visually display AQI categories, altering its color and displaying category names (e.g., "Good" in green for PM2.5 levels at 8 $\mu g/m^3$). We hypothesize that this visual and categorical representation of air quality may nudge households to improve their indoor air to the safest AQI category achievable.

Our analysis, detailed in Figure 7 and Appendix Table A7, specifically investigates these effects during occupancy hours, a period of heightened indoor activity and poten-

tial pollutant exposure as we documented before. The findings suggest that the intervention effectively encourages households to maintain air quality within the "Good" and "Moderate" PM2.5 ranges (below 100 AQI which is equivalent to 35.4 $\mu g/m^3$), increasing the likelihood of air quality falling within these safer ranges. Conversely, there is a significant decrease in the time spent in all higher AQI categories, such as "Unhealthy" and "Hazardous." In particular, we find that the intervention reduced the likelihood of time spent in the Unhealthy, Very Unhealthy, or Hazardous ranges by 0.06, 0.05, and 0.04 standard deviations, respectively, compared to pre treatment baseline. Overall, these results highlight the intervention's efficacy in nudging household behavior towards maintaining healthier indoor air quality levels and more generally the efficacy of providing real-time pollution feedback in significantly lowering indoor PM2.5 concentrations, especially during critical periods when individuals are present and active in their homes.

In order to understand the variation in household responses to our treatment, we also perform a heterogeneity analysis to investigate the Average Treatment Effect (ATE) across different household characteristics. The results, presented in Figure 8, reveal that households with below-median income experience the largest reductions in indoor pollution exposure. Specifically, the ATE for these households is 19.6 $\mu g/m^3$ during occupancy time, compared to 5 $\mu g/m^3$ for the full sample. This substantial impact is likely because these households started with significantly higher levels of indoor pollution in the pre-treatment period, as highlighted in Section 3.2 above, requiring more drastic actions to address their pollution problem. As such, we also examine the relative effects (percentage reduction) from baseline and find that below-median income households achieved a staggering 89.3% reduction, compared to 8.7% among above-median income household.

We further explore this dimension in Table A8, where we stratify results based on whether households had above- or below-median PM2.5 levels at baseline (the first two weeks). The findings for households with above-median PM2.5 levels are consistent to those for below-median-income households, strongly supporting the argument that households with higher pre-treatment pollution levels require more significant reductions. Interestingly, households with below-median PM2.5 levels at baseline showed an increase in pollution exposure post-treatment. We hypothesize that this increase may be due to households learning through the intervention that their pollution levels were lower than they had initially perceived and consequently adjusting their behaviors. In other words, these households might have been operating below their efficient pollution level and used the information provided by the treatment to re-optimize. Later in the

paper (in Section 4.3), we present empirical evidence from our survey data on beliefs, which supports this hypothesis. Beyond the income-related findings, we also observe significantly higher reductions in indoor pollution (exceeding 5 $\mu g/m^3$) among specific demographic groups, including households without a college education, renters, single-occupant households, households without children, and those with pre-existing health conditions.

4.2 Mechanisms

Following our main results showing how the treatment reduced indoor pollution, we move to empirically explore the mechanisms driving our main results by examining participant-stated answers and revealed actions in detail. We begin with the former as we elicited direct insights from the participants themselves. In particular, individuals in the treatment group were surveyed about the specific steps, if any, they undertook to mitigate indoor pollution levels. This inquiry is pivotal, as it provides first-hand accounts of the behavioral changes or preventive measures adopted by residents when faced with immediate feedback on their indoor air quality. Figure A18, which summarizes the answers we received from participants, shows that a significant majority, 72.1%, reported increasing ventilation as their primary strategy to reduce pollution. This mainly involved opening windows to allow fresh air to circulate and dilute indoor pollution, which highlights the intuitive and accessible nature of enhancing ventilation as a potential first-line response to air quality concerns. Conversely, a small segment of the sample, 9.3%, reported decreasing ventilation as their main strategy to reduce pollution. This might seem a contradictory response at first glance but as mentioned above, indoor pollution could be influenced by external factors such as outdoor air quality (e.g. a window opening into a major road) or weather conditions, suggesting a complex decision-making process regarding air quality management and the importance of real-time monitoring.

We also find that 11.6% of respondents indicated they had reduced activities known to contribute to indoor pollution, such as cooking, dusting, and the use of candles and aerosols. This reduction reflects a conscious effort to decrease the generation of pollutants at the source. Interestingly, only 2.3% of participants turned to air purifiers as a solution, leveraging technology to filter out pollutants from indoor air. Finally, 4.7% of the respondents adopted multiple strategies, combining various measures to tackle indoor pollution effectively. These findings highlight the range of actions individuals are willing to take to

improve their living environments, demonstrating the critical role of real-time air quality feedback in empowering residents to make informed decisions about their indoor air quality.

Whilst the analysis above provides great insight into the behavior responses, it is important to note that the response rate for this specific survey question was 32%. As such, we proceed with two additional methods to examine the behavioral response, both of which utilize data collected from all participants in our study and focus on revealed rather than stated behaviors. First, we analyze our collected data on indoor and outdoor temperatures as an indirect measure of ventilation habits. By evaluating the absolute difference between these two measures of temperatures before and after the intervention, we aim to deduce changes in household ventilation strategies. A narrowing of this temperature gap suggests an increase in activities like opening windows, a straightforward method for reducing indoor pollutant concentrations by allowing in cleaner outdoor air. Table 2 shows the results of such analysis, utilizing our main empirical approach (DiD) but with absolute temperature difference (between indoor and outdoor) as the outcome variable instead of Indoor PM2.5. The results, which are in Table 2, clearly show a narrowing of this temperature gap, echoing the results found in Figure A18.

Finally, we examine the characteristics of peak indoor pollution events, focusing on both their intensity and frequency. We define these events by applying a rolling 5-hour window to the monitor data and identifying the local maximum within each window. Each local maximum is recorded as a potential peak. If the PM2.5 level at the peak exceeds 35 $\mu g/m^3$—surpassing the threshold for a "good" air quality rating—we define the surrounding period during which the monitor reports non-good air quality as a full pollution event. This approach allows us to assess whether the intervention prompted households to alter pollution-generating activities or implement mitigating behaviors such as increased ventilation. A reduction in the intensity or frequency of events would suggest that households responded to or anticipated high pollution levels and took effective action.

We begin with intensity. Column 1 of Table 3 shows that the maximum PM2.5 level during peak events falls by 36.9 $\mu g/m^3$ on average following the intervention. This substantial reduction suggests that households either took preventive measures (e.g., opening windows in advance) or responded quickly once alerted by high readings. Column 2 tests for changes in the frequency of pollution events, to assess whether the intervention altered indoor activities such as cooking or heating. We find no statistically significant

effects, suggesting no change in the underlying behaviors that generate pollution.

Taken together, these results reinforce our earlier conclusion that ventilation is the primary behavioral margin through which real-time feedback influences indoor air quality. Understanding this mechanism is critical for evaluating the broader applicability and effectiveness of such interventions in promoting public health via improved indoor environments.

4.3 Impact on Beliefs

We also study the impact of our treatment on participants' perceptions and confidence regarding indoor and outdoor air quality, as measured through the difference in air quality beliefs between the baseline and end-line surveys. In particular, our "Air Quality Belief" variable captures the participants' perceived level of air quality using a spectrum from "Good" with no health risk, to "Hazardous", both indoors and outdoors, while our "Confidence" variable assesses the certainty with which participants hold these beliefs.

Table 4 shows the estimated coefficients of the treatment effect, quantifying how the intervention influenced participants' beliefs about air quality in their home environments and their confidence. The results show that residents updated their beliefs, realizing that their indoor air is in fact worse than they originally thought. This estimate is statistically significant at the 5% level and is fairly large - this translates to 33% of the standard deviation of the variable (mean is 0.17, sd is 0.79). Importantly, the effectiveness of our treatment in not only altering participants' perceptions of air quality but also affecting the confidence level of these perceptions, offering valuable insights into the psychological impact of air quality awareness interventions. For ambient (outdoor) air quality, the results differ. The estimates do not show statistically significant effects, implying that the intervention did not alter beliefs or confidence regarding outdoor air quality. This distinction in findings between indoor and outdoor air quality perceptions makes sense as we only provided information about indoor pollution and not ambient pollution. In fact, given that most of the indoor pollution we observed in our study is generated indoors in conjunction with the fact that we only provided information on indoor pollution, one can describe this analysis as a placebo test and the results of this exercise are very reassuring.

Finally, we revisit our earlier findings from Section 4.1, which show that households with initially low levels of pollution actually slightly increased their pollution exposure following the treatment. We hypothesized that this occurred because they learned their

pollution levels were better than they had previously believed and, as a result, adjusted their air quality at home (via behavioral change) to align with their efficient level of pollution. Since we have data on their initial beliefs, we can now empirically test this explanation. In Figure A19, we present the correlation between pre-treatment beliefs about indoor air quality and the actual pre-treatment indoor air quality for households below- and above-median PM2.5 levels separately. The results reveal that households with below-median PM2.5 levels tend to believe their pollution levels are worse than they actually are, while those with above-median PM2.5 levels perceive their pollution to be better than it truly is. This finding strongly supports (though does not causally prove) our hypothesis that households with low baseline PM2.5 levels adjust their behavior to increase pollution exposure in an effort to reach their optimal level of pollution.

4.4 Robustness

We conduct a series of robustness checks to validate the reliability of our primary findings. These checks are designed to ensure that our results are not driven by specific model specifications or sample selections. We employ four key strategies: adding household fixed effects to our preferred DiD estimator, omitting ambient pollution controls, using two-way clustering technique without bootstrapped standard errors and re-evaluating our analysis using a reduced sample size as outlined in our pre-regitration plan. Given that the majority of the effects observed in our study occur during occupancy evening hours when residents are typically at home, we will focus on this time period in the subsequent analysis and we will also examine our estimates when we modify this definition.

We begin by re-estimating our model using household fixed effects to account for unobservable household-specific characteristics. In particular, this approach controls for all time-invariant heterogeneity across households. As a robustness test, we aim to determine whether this alternative specification yields results consistent with our original findings. Figure A20 and Table A10 show that the household fixed effects model produces an estimate of 3.45 $\mu g/m^3$ during occupancy time, which is similar to our main estimate reported in Table A6.

We also re-run our main analysis using a reduced sample of households, as speci-

fied in our pre-registration plan (N=150).[35] Aside from necessary adjustments to the specified date ranges due to logistical constraints, this is the only deviation from the original plan.[36] The primary reasons for increasing the sample size were to enable heterogeneity analysis and to ensure sufficient statistical power for our willingness-to-pay analysis. Additionally, the original choice of 150 households was based on preliminary results from our pilot study, as we did not have prior literature to guide our power calculations. Notably, the pilot study employed a different recruitment process-advertising on social media rather than randomly distributing letters-which resulted in significantly higher estimates, possibly due to selection bias. By adhering to this predefined sample and comparing it with the full dataset, we rigorously evaluate the stability of our findings across different sample configurations and ensure transparency. The results, presented in Column 2 of Table 5, closely mirror those obtained from the full sample, which is very reassuring.

The final three columns of Table 5 present an analysis of our estimates across alternative time windows of the day. This analysis produces very similar results, with the effect being primarily concentrated during the PM hours. Finally, Panel B of Table A6 reports the main results without bootstrapped standard errors. While the estimates are less precisely estimated in this specification, the ATE during occupancy hours remains highly statistically significant at the 5 percent level.

The outcomes from all the above robustness checks - the household fixed effects model, the omission of ambient pollution controls, and the analysis with a reduced sample size and our different clustering - consistently align with our main results, reaffirming the strong effect of our intervention. The persistence of the treatment effect across these varied specifications and sample criteria underscores the robustness of our primary findings. Notably, the treatment effect for the occupancy hours remains statistically significant and economically meaningful in all scenarios tested, bolstering the credibility of our original conclusions and the internal validity of our study. These analyses are intended as robustness exercises rather than preferred specifications, as all observations in the full sample were collected under the same experimental protocol. Using the full sam-

[35] The results presented here are actually based on the first three waves of our experiment (N=175). However, we also conducted the same analysis after randomly dropping 25 households from the third wave to exactly match the pre-specified sample size of 150 households. The results remained virtually identical. We present the findings using the slightly larger sample size for reproducibility purposes.

[36] Our pre-registration mistakenly stated that we would run the experiment for four weeks post-treatment. However, this was never our actual intention for this study. As evident from our recruitment letters to residents, we consistently communicated that the study duration was one month overall (including the pre-treatment period).

ple also ensures consistency across the analyses reported throughout the paper, rather than relying on different samples for different outcomes.

Finally, as we mentioned earlier in the paper, our study sample does not fully align with the demographic composition of the broader Camden population. Table A2 highlights these differences (see section 2.3 for a full discussion). While these discrepancies do not compromise the internal validity of our findings, they do raise concerns about the external validity of our results. To address this threat head-on, we re-weight our estimates using detailed knowledge of our sample's demographics and the demographic characteristics of Camden as a whole, ensuring that our results are more representative of the broader population. If we re-weight on income, assuming the other characteristics of households are relatively similar across the UK, we can see that the whole day ATE increases to -4.43 and the occupancy time rises to -9.91 (compared to -1.9 and -5.0 in the unweighted sample).[37] This requires the assumption that the population of Camden is similar to the rest of England and air pollution levels would be similar since we do not have data on people who did not participate in our study. When re-weighting by the other variables in Table A2 we find that the whole day ATE estimates range from -1.67 to -1.95 and the occupancy time ATEs range from -3.02 to -4.86.

5 Welfare Analysis

In this section, we examine the impact of the intervention on human health, residents' willingness to pay for improvements in indoor air and information about it, and the overall marginal value of public funds (MVPFs) for a government subsidy of indoor air pollution monitors.

5.1 Impact on Health

We first focus on the potential mortality benefits. To analyze this dimension, we follow Carozzi and Roth (2023) and apply Concentration-Response (C-R) functions from the existing literature to our estimated pollution reduction. Krewski et al. (2009) and Lepeule et al. (2012) report increases in mortality risk of 6% and 14%, respectively, per $10\mu g/m^3$ increase in PM2.5. We use these estimates to calculate the proportional reduction in mor-

[37] This is calculated by re-weighting the sample to ensure 50% of households have income below the median level, rather than 24% as in our sample.

tality risk and calculate the potential mortality benefit, using the UK Value of a Prevented Fatality of £2.4 million in 2023 prices.

Using our main estimates from Figure 3, and assuming that this effect remains constant (as suggested by our analysis in Figure A17), we evaluate the impact of providing all households in the UK with the same treatment implemented in our study. Our analysis considers two key reductions in PM2.5 concentrations: a 5 $\mu g/m^3$ reduction observed during occupancy hours and a 1.9 $\mu g/m^3$ reduction calculated as the all-day average. Using these reductions, our results suggest that the annual mortality benefit of such an intervention, based on the high and low C-R functions would be £102.4 billion and £46.4 billion (or £1,501 and £680 per capita), respectively, for the 5 $\mu g/m^3$ reduction. For the smaller 1.9 $\mu g/m^3$ reduction, the annual mortality benefits are estimated at £39.73 billion and £17.79 billion (or £582.05 and £260.63 per capita), respectively.

Given the substantial uncertainty surrounding the C-R functions, we adopt a midpoint C-R function of 1.10. Furthermore, there is also uncertainty surrounding the correct estimated pollution reduction that we should use for our calculation, as we do not observe individuals' exposure throughout the entire day. For instance, individuals might be at school or work during certain hours, where indoor air quality may differ significantly from that observed in their homes. To address this, we adjust our ATE by weighting the observed reduction during occupancy hours (16:00–23:00), which represents one-third of the day. Using this adjusted ATE and the midpoint C-R function, the estimated annual mortality benefit is £25.52 billion.

The above calculations rely on several assumptions. In particular, we assume that the estimated pollution reductions persist beyond the two-week treatment period and generalize from our sample to the UK population. We also assume that the C-R relationships estimated in US ambient-pollution studies apply to residential indoor pollution in the UK. As discussed in the background section, the WHO advisory group found no compelling evidence that indoor particulate matter poses a lower health risk than outdoor particulate matter (WHO et al., 2010). Finally, we hold the coefficient constant in log mortality risk across the concentration range considered, including below the range observed in the underlying studies where relevant. The resulting valuations therefore represent potential mortality benefits conditional on these assumptions.

5.2 Impact on Economic Welfare

In this section, we estimate the willingness to pay (WTP) for an indoor air pollution monitor and an air purifier, and examine the welfare implications of subsidizing these technologies. This setting is particularly well-suited for valuation, as treatment group households possess real time information about their indoor air quality, eliminating issues of misinformation or biased beliefs.

To mitigate common valuation biases (e.g., hypothetical bias), we used an incentive-compatible Becker-DeGroot-Marschak (BDM) mechanism (Becker et al., 1964). Participants were asked how much of a £100 voucher they would forgo to acquire each product, with bids compared against a randomly drawn price if selected for implementation. Figure A21 shows demand curves for both products. Participants placed higher value on the air purifier—likely due to its direct mitigation effect—than on the monitor, which only provides information.[38]. Figures A22a and A22b show demand by treatment status. Average WTP for the monitor was £36.65 in the control group and £38.84 in the treatment group, while corresponding values for the air purifier were £44.65 and £50.39. Dividing purifier WTP by mean PM2.5 concentrations gives a sample average of £4.80 per $\mu g/m^3$, conditional on complete removal of measured pollution and linearity in WTP.

To our knowledge, this is the first study to experimentally estimate WTP for reductions in indoor PM2.5. The most closely related study is Ito and Zhang (2020), which uses a revealed preference approach based on air purifier sales in China to estimate WTP for PM10. They find households are willing to pay approximately $6.30 (£5.10) per $\mu g/m^3$ of PM10. Adjusting for purchasing power parity (PPP), our UK WTP estimate of £4.80 corresponds to £1.91 in China, suggesting Chinese households value air pollution reduction more—potentially due to greater pollution severity and awareness. This discrepancy is unlikely to stem from pollutant differences, as PM2.5 is a subset of PM10 and filtered by most purifiers.[39]

We next calculate the marginal value of public funds (MVPF) for a £1 subsidy for indoor pollution monitor using the WTP estimates above and the framework in Hahn et al. (2024). The MVPF is the ratio of the societal WTP for the subsidy to the net cost incurred by the government in providing that subsidy In our calculation we use the estimated demand elasticity of 0.75 for the monitor and our daily ATE of –1.9 $\mu g/m^3$. We

[38] Note: the Philips purifier also has a real-time air quality light.

[39] PM10 includes particles up to 10 micrometers in diameter, encompassing PM2.5.

also use the cost estimates from mortality dose-response functions discuss above to calculate per capita health WTP. We combine the mortality valuations with earnings and crime effects estimated by Borgschulte et al. (2024) and Bondy et al. (2020). Again, as mentioned above we assume that relationships estimated for ambient pollution extend to the exposure changes induced by our intervention. On the fiscal side, we account for additional tax revenue and savings in NHS and police-related expenditures.

Table 6 and Figure 9 summarize MVPF estimates. Our baseline scenario—ATE of –1.9 over 20 years, low C-R function—yields total private benefits of £47.08 per £1 subsidy: £1 transfer, £0.75 health benefit, £44.83 income benefit, and £0.50 crime benefit. Net government gains are £13.02 per £1 subsidy: £13.69 in additional tax revenue, £0.18 NHS savings, and £0.15 in reduced crime expenditures. The resulting MVPF is effectively infinite. Figure 9 shows negative net government costs due to large fiscal benefits. Even omitting income effects (Borgschulte et al. (2024)) yields an MVPF of 3.4-well above typical values reported by Hahn et al. (2024). For more details on the calculations and sensitivity analyses, see the online appendix.

It is rare to observe infinite MVPFs. Hendren and Sprung-Keyser (2020) document such outcomes for select child health and education programs, but no environmental policy evaluated by Hahn et al. (2024) exhibits an infinite MVPF. To our knowledge, this is the first demonstration that an environmental intervention—specifically a subsidy for IAP mitigation—can pay for itself.

6 Conclusion

There is a well-established literature on ambient air pollution and its health and welfare impacts, which has shaped environmental policy globally. In contrast, we know relatively little about indoor air pollution (IAP)—its levels, determinants, and how it can be causally reduced. Using a field experiment in London, we document IAP levels and their association with household and dwelling characteristics, and estimate the effects of providing real-time pollution information. The intervention reduces indoor PM2.5 concentrations by 34% (5 $\mu g/m^3$) during evening hours when residents are typically at home.

Our evidence points to increased natural ventilation as an important mechanism through which households respond to information. The treatment also changes beliefs about indoor air quality, and our willingness to pay estimate imply a valuation of £4.80

per $1\mu g/m^3$ reduction in PM2.5. We further show that subsidies for IAP monitors or purifiers have an infinite Marginal Value of Public Funds (MVPF), conditional on assumptions about the persistence of pollution reductions and their effects on health and other outcomes.

Our study leaves several questions for further research. The experiment involves a self-selected sample of households in one London borough, with participants generally more affluent and educated than the wider population. The relatively low pollution levels in this setting and the two-week treatment period should also be kept in mind when interpreting the results. Studies involving larger and more diverse samples, different pollution settings, and longer follow-up periods would help establish how widely these findings apply, whether the observed responses persist, and whether they translate into sustained reductions in exposure and improvements in health.

Overall, our findings offer timely insights into the magnitude, sources, and tractability of IAP, and the welfare benefits of reducing it. For policymakers, this highlights an urgent need to address IAP through effective interventions. It also underscores a potential policy trade-off: energy-efficiency measures that reduce ventilation (e.g., insulation) may unintentionally raise IAP exposure. We recommend empirically evaluating this trade-off and investing in technological solutions that support both climate and public health goals.

References

Acemoglu, D. and Linn, J. (2004), 'Market size in innovation: theory and evidence from the pharmaceutical industry', *The Quarterly Journal of Economics* **119**(3), 1049–1090.

Aguilar-Gomez, S., Dwyer, H., Graff Zivin, J. and Neidell, M. (2022), 'This is air: The "nonhealth" effects of air pollution', *Annual Review of Resource Economics* **14**(1), 403–425.

Ahmad, H., Gibson, M., Nadeem, F., Nasim, S. and Razaee, A. (2022), 'Forecasts: Consumption, production, and behavioral responses'.

Alves, C. A., Vicente, E. D., Evtyugina, M., Vicente, A. M., Sainnokhoi, T.-A. and Kováts, N. (2021), 'Cooking activities in a domestic kitchen: Chemical and toxicological profiling of emissions', *Science of the Total Environment* **772**, 145412.

Ashraf, N., Jack, B. K. and Kamenica, E. (2013), 'Information and subsidies: Complements or substitutes?', *Journal of Economic Behavior & Organization* **88**, 133–139.

Auld, M. C. (2005), 'Smoking, drinking, and income', *Journal of Human Resources* **40**(2), 505–518.

Axbard, S. and Deng, Z. (2024), 'Informed enforcement: Lessons from pollution monitoring in china', *American Economic Journal: Applied Economics* **16**(1), 213–252.

Bandowe, B. A. M., Lui, K., Jones, T., BeruBe, K., Adams, R., Niu, X., Wei, C., Cao, J.-J., Lee, S., Chuang, H.-C. et al. (2021), 'The chemical composition and toxicological effects of fine particulate matter (pm2.5) emitted from different cooking styles', *Environmental Pollution* **288**, 117754.

Banzhaf, S., Ma, L. and Timmins, C. (2019), 'Environmental justice: The economics of race, place, and pollution', *Journal of Economic Perspectives* **33**(1), 185–208.

Baquie, S., Behrer, A. P., Du, X., Fuchs Tarlovsky, A. and Nozaki, N. K. (2024), Know thy foe: Information provision and air pollution in tbilisi, Technical report, The World Bank.

Barrero, J. M., Bloom, N. and Davis, S. J. (2023), 'The evolution of work from home', *Journal of Economic Perspectives* **37**(4), 23–49.

Barron, M. and Torero, M. (2017), 'Household electrification and indoor air pollution', *Journal of Environmental Economics and Management* **86**, 81–92.

Barwick, P. J., Li, S., Lin, L. and Zou, E. Y. (2024), 'From fog to smog: The value of pollution information', *American Economic Review* **114**(5), 1338–1381.

Bassi, V., Kahn, M. E., Gracia, N. L., Porzio, T. and Sorin, J. (2022), Searching for customers, finding pollution, Technical report, National Bureau of Economic Research.

Bayer, P., Keohane, N. and Timmins, C. (2009), 'Migration and hedonic valuation: The case of air quality', *Journal of Environmental Economics and Management* **58**(1), 1–14.

Becker, G. M., DeGroot, M. H. and Marschak, J. (1964), 'Measuring utility by a single-response sequential method', *Behavioral science* **9**(3), 226–232.

Bedi, A. S., Nakaguma, M. Y., Restrepo, B. J. and Rieger, M. (2021), 'Particle pollution and cogni-

tion: Evidence from sensitive cognitive tests in brazil', *Journal of the Association of Environmental and Resource Economists* **8**(3), 443–474.

Beltramo, T., Blalock, G., Stephen, H., Levine, D. I. and Simons, A. M. (2023), 'The effects of fuel-efficient cookstoves on fuel use, particulate matter, and cooking practices: Results from a randomized trial in rural uganda', *The Energy Journal* **44**(6), 43–70.

Bennett, D., Naqvi, A. and Schmidt, W.-P. (2018), 'Learning, hygiene and traditional medicine', *The Economic Journal* **128**(612), F545–F574.

Bensch, G. and Peters, J. (2015), 'The intensive margin of technology adoption–experimental evidence on improved cooking stoves in rural senegal', *Journal of Health Economics* **42**, 44–63.

Berkouwer, S. B. and Dean, J. T. (2022), 'Credit, attention, and externalities in the adoption of energy efficient technologies by low-income households', *American Economic Review* **112**(10), 3291–3330.

Berkouwer, S. B. and Dean, J. T. (2023), Private actions in the presence of externalities: The health impacts of reducing air pollution peaks but not ambient exposure, Technical report, National Bureau of Economic Research.

Bharadwaj, P., Gibson, M., Zivin, J. G. and Neilson, C. (2017), 'Gray matters: Fetal pollution exposure and human capital formation', *Journal of the Association of Environmental and Resource Economists* **4**(2), 505–542.

Bishop, K. C., Ketcham, J. D. and Kuminoff, N. V. (2023), 'Hazed and confused: the effect of air pollution on dementia', *Review of Economic Studies* **90**(5), 2188–2214.

Bondy, M., Roth, S. and Sager, L. (2020), 'Crime is in the air: The contemporaneous relationship between air pollution and crime', *Journal of the Association of Environmental and Resource Economists* **7**(3), 555–585.

Borgschulte, M., Molitor, D. and Zou, E. Y. (2024), 'Air pollution and the labor market: Evidence from wildfire smoke', *Review of Economics and Statistics* **106**(6), 1558–1575.

Brown, J., Hamoudi, A., Jeuland, M. and Turrini, G. (2017), 'Seeing, believing, and behaving: Heterogeneous effects of an information intervention on household water treatment', *Journal of Environmental Economics and Management* **86**, 141–159.

Burke, M., Heft-Neal, S., Li, J., Driscoll, A., Baylis, P., Stigler, M., Weill, J. A., Burney, J. A., Wen, J., Childs, M. L. et al. (2022), 'Exposures and behavioural responses to wildfire smoke', *Nature Human Behaviour* **6**(10), 1351–1361.

Bursztyn, L., González, A. L. and Yanagizawa-Drott, D. (2020), 'Misperceived social norms: Women working outside the home in saudi arabia', *American Economic Review* **110**(10), 2997–3029.

Butera, L., Metcalfe, R., Morrison, W. and Taubinsky, D. (2022), 'Measuring the welfare effects of shame and pride', *American Economic Review* **112**(1), 122–168.

Camden Council (2023), 'State of the borough report'.

URL: *https://www.wemakecamden.org.uk/wp-content/uploads/2023/03/State-of-Borough-report-web.pdf*

Cantoni, D., Yang, D. Y., Yuchtman, N. and Zhang, Y. J. (2019), 'Protests as strategic games: experimental evidence from hong kong's antiauthoritarian movement', *The Quarterly Journal of Economics* **134**(2), 1021–1077.

Carneiro, J., Cole, M. A. and Strobl, E. (2021), 'The effects of air pollution on students' cognitive performance: evidence from brazilian university entrance tests', *Journal of the Association of Environmental and Resource Economists* **8**(6), 1051–1077.

Carozzi, F. and Roth, S. (2023), 'Dirty density: Air quality and the density of american cities', *Journal of Environmental Economics and Management* **118**, 102767.

Chang, T., Graff Zivin, J., Gross, T. and Neidell, M. (2016), 'Particulate pollution and the productivity of pear packers', *American Economic Journal: Economic Policy* **8**(3), 141–169.

Chang, T. Y., Graff Zivin, J., Gross, T. and Neidell, M. (2019), 'The effect of pollution on worker productivity: evidence from call center workers in china', *American Economic Journal: Applied Economics* **11**(1), 151–172.

Chay, K. Y. and Greenstone, M. (2003), 'The impact of air pollution on infant mortality: evidence from geographic variation in pollution shocks induced by a recession', *The Quarterly Journal of Economics* **118**(3), 1121–1167.

Chay, K. Y. and Greenstone, M. (2005), 'Does air quality matter? evidence from the housing market', *Journal of Political Economy* **113**(2), 376–424.

Chen, C. and Zhao, B. (2011), 'Review of relationship between indoor and outdoor particles: I/o ratio, infiltration factor and penetration factor', *Atmospheric Environment* **45**(2), 275–288.

Chen, P. (2025), 'Industrialization and pollution: The long-term impact of early-life exposure on human capital formation', *Journal of Public Economics* **241**, 105270.

Cheung, P. K., Jim, C. Y. and Siu, C. T. (2019), 'Air quality impacts of open-plan cooking in tiny substandard homes in hong kong', *Air Quality, Atmosphere & Health* **12**, 865–878.

Chowdhury, A., Garg, T., Jagnani, M. and Mattsson, M. (2025), 'Misbeliefs, experience, and technology adoption: Evidence from air purifiers in bangladesh', *Available at SSRN 5284948* .

Clay, K., Muller, N. Z. and Wang, X. (2021), 'Recent increases in air pollution: evidence and implications for mortality', *Review of Environmental Economics and Policy* **15**(1), 154–162.

Colmer, J., Hardman, I., Shimshack, J. and Voorheis, J. (2020), 'Disparities in pm2. 5 air pollution in the united states', *Science* **369**(6503), 575–578.

Colmer, J., Qin, S., Voorheis, J. and Walker, R. (2024), The changing nature of pollution, income, and environmental inequality in the united states, *in* 'AEA Papers and Proceedings', Vol. 114, American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 41–46.

Currie, J., Davis, L., Greenstone, M. and Walker, R. (2015), 'Environmental health risks and hous-

ing values: evidence from 1,600 toxic plant openings and closings', *American Economic Review* **105**(2), 678–709.

Currie, J., Hanushek, E. A., Kahn, E. M., Neidell, M. and Rivkin, S. G. (2009), 'Does pollution increase school absences?', *The Review of Economics and Statistics* **91**(4), 682–694.

Currie, J., Voorheis, J. and Walker, R. (2023), 'What caused racial disparities in particulate exposure to fall? new evidence from the clean air act and satellite-based measures of air quality', *American Economic Review* **113**(1), 71–97.

Currie, J. and Walker, R. (2011), 'Traffic congestion and infant health: Evidence from e-zpass', *American Economic Journal: Applied Economics* **3**(1), 65–90.

Currie, J., Zivin, J. G., Mullins, J. and Neidell, M. (2014), 'What do we know about short-and long-term effects of early-life exposure to pollution?', *Annu. Rev. Resour. Econ.* **6**(1), 217–247.

Davis, D. A., Akopian, G., Walsh, J. P., Sioutas, C., Morgan, T. E. and Finch, C. E. (2013), 'Urban air pollutants reduce synaptic function of ca 1 neurons via an nmda/n pathway in vitro', *Journal of Neurochemistry* **127**(4), 509–519.

DeCicca, P., Kenkel, D. and Lovenheim, M. F. (2022), 'The economics of tobacco regulation: a comprehensive review', *Journal of Economic Literature* **60**(3), 883–970.

DEFRA (2024), 'Emissions of air pollutants in the uk – particulate matter (pm10 and pm2.5)'.
URL: *https://www.gov.uk/government/statistics/emissions-of-air-pollutants/emissions-of-air-pollutants-in-the-uk-particulate-matter-pm10-and-pm25*

Deryugina, T., Heutel, G., Miller, N. H., Molitor, D. and Reif, J. (2019), 'The mortality and medical costs of air pollution: Evidence from changes in wind direction', *American Economic Review* **109**(12), 4178–4219.

Deschenes, O., Greenstone, M. and Shapiro, J. S. (2017), 'Defensive investments and the demand for air quality: Evidence from the nox budget program', *American Economic Review* **107**(10), 2958–2989.

Duflo, E., Greenstone, M. and Hanna, R. (2008), 'Indoor air pollution, health and economic well-being', *SAPI EN. S. Surveys and Perspectives Integrating Environment and Society* (1.1).

Duque, V. and Gilraine, M. (2022), 'Coal use, air pollution, and student performance', *Journal of Public Economics* **213**, 104712.

Ebenstein, A., Lavy, V. and Roth, S. (2016), 'The long-run economic consequences of high-stakes examinations: Evidence from transitory variation in pollution', *American Economic Journal: Applied Economics* **8**(4), 36–65.

EPA, U. (1987), 'The total exposure assessment methodology (team) study: Summary and analysis'.

Fairweather, D., Kahn, M. E., Metcalfe, R. D. and Olascoaga, S. S. (2024), Expecting climate change: A nationwide field experiment in the housing market, Technical report, National Bureau of Economic Research.

Ferguson, L., Taylor, J., Davies, M., Shrubsole, C., Symonds, P. and Dimitroulopoulou, S. (2020), 'Exposure to indoor air pollution across socio-economic groups in high-income countries: A scoping review of the literature and a modelling methodology', *Environment International* **143**, 105748.

Fowlie, M., Rubin, E. and Walker, R. (2019), Bringing satellite-based air quality estimates down to earth, *in* 'AEA Papers and Proceedings', Vol. 109, American Economic Association 2014 Broadway, Suite 305, Nashville, TN 37203, pp. 283–288.

Freeman, R., Liang, W., Song, R. and Timmins, C. (2019), 'Willingness to pay for clean air in china', *Journal of Environmental Economics and Management* **94**, 188–216.

Goeb, J., Dillon, A., Lupi, F. and Tschirley, D. (2020), 'Pesticides: What you don't know can hurt you', *Journal of the Association of Environmental and Resource Economists* **7**(5), 801–836.

Gosnell, G. K., List, J. A. and Metcalfe, R. D. (2020), 'The impact of management practices on employee productivity: A field experiment with airline captains', *Journal of Political Economy* **128**(4), 1195–1233.

Graff Zivin, J. and Neidell, M. (2012), 'The impact of pollution on worker productivity', *American Economic Review* **102**(7), 3652–73.

Graff Zivin, J. and Neidell, M. (2013), 'Environment, health, and human capital', *Journal of Economic Literature* **51**(3), 689–730.

Graff Zivin, J. S., Krebs, B. and Neidell, M. J. (2024), Private adoption of public good technologies: The case of purpleair, Technical report, National Bureau of Economic Research.

Greenstone, M., He, G., Jia, R. and Liu, T. (2022), 'Can technology solve the principal-agent problem? evidence from china's war on air pollution', *American Economic Review: Insights* **4**(1), 54–70.

Greenstone, M., Lee, K. and Sahai, H. (2021), Indoor air quality, information, and socioeconomic status: Evidence from delhi, *in* 'AEA Papers and Proceedings', Vol. 111, pp. 420–24.

Hahn, R. W., Hendren, N., Metcalfe, R. D. and Sprung-Keyser, B. (2024), A welfare analysis of policies impacting climate change, Technical report, National Bureau of Economic Research.

Hanley, J. T., Ensor, D. S., Smith, D. D. and Sparks, L. (1994), 'Fractional aerosol filtration efficiency of in-duct ventilation air cleaners', *Indoor Air* **4**(3), 169–178.

Hanna, R., Duflo, E. and Greenstone, M. (2016), 'Up in smoke: the influence of household behavior on the long-run impact of improved cooking stoves', *American Economic Journal: Economic Policy* **8**(1), 80–114.

Hanna, R., Hoffmann, B., Oliva, P. and Schneider, J. (2021), The power of perception limitations of information in reducing air pollution exposure, Technical report, IDB Working Paper Series.

Harrison, G. W. and List, J. A. (2004), 'Field experiments', *Journal of Economic literature* **42**(4), 1009–1055.

Heeks, M., Reed, S., Tafsiri, M. and Prince, S. (2018), 'The economic and social costs of crime second edition', *Home Office Research Report 99* .

Hendren, N. and Sprung-Keyser, B. (2020), 'A unified welfare analysis of government policies', *The Quarterly Journal of Economics* **135**(3), 1209–1318.

Hiscock, R., Bauld, L., Amos, A., Fidler, J. A. and Munafò, M. (2012), 'Socioeconomic status and smoking: a review', *Annals of the New York Academy of Sciences* **1248**(1), 107–123.

Hsiang, S., Oliva, P. and Walker, R. (2019), 'The distribution of environmental damages', *Review of Environmental Economics and Policy* .

Huang, J., Xu, N. and Yu, H. (2020), 'Pollution and performance: Do investors make worse trades on hazy days?', *Management Science* **66**(10), 4455–4476.

Imtiaz, I., Nakamura, S., Nasim, S. and Rezaee, A. (2025), 'Beliefs, signal quality, and information sources: Experimental evidence on air quality in pakistan'.

Ito, K. and Zhang, S. (2020), 'Willingness to pay for clean air: Evidence from air purifier markets in china', *Journal of Political Economy* **128**(5), 1627–1672.

Jalan, J. and Somanathan, E. (2008), 'The importance of being informed: Experimental evidence on demand for environmental quality', *Journal of Development Economics* **87**(1), 14–28.

Janke, K. (2014), 'Air pollution, avoidance behaviour and children's respiratory health: Evidence from england', *Journal of health economics* **38**, 23–42.

Jbaily, A., Zhou, X., Liu, J., Lee, T.-H., Kamareddine, L., Verguet, S. and Dominici, F. (2022), 'Air pollution exposure disparities across us population and income groups', *Nature* **601**(7892), 228–233.

Jensen, R. (2007), 'The digital provide: Information (technology), market performance, and welfare in the south indian fisheries sector', *The Quarterly Journal of Economics* **122**(3), 879–924.

Jessoe, K. and Rapson, D. (2014), 'Knowledge is (less) power: Experimental evidence from residential energy use', *American Economic Review* **104**(4), 1417–1438.

Jeuland, M., Pattanayak, S. K. and Bluffstone, R. (2015), 'The economics of household air pollution', *Annu. Rev. Resour. Econ.* **7**(1), 81–108.

Jha, A. and Nauze, A. L. (2022), 'Us embassy air-quality tweets led to global health benefits', *Proceedings of the National Academy of Sciences* **119**(44), e2201092119.

Jung, K. H., Hsu, S.-I., Yan, B., Moors, K., Chillrud, S. N., Ross, J., Wang, S., Perzanowski, M. S., Kinney, P. L., Whyatt, R. M. et al. (2012), 'Childhood exposure to fine particulate matter and black carbon and the development of new wheeze between ages 5 and 7 in an urban prospective cohort', *Environment international* **45**, 44–50.

Kashtan, Y., Nicholson, M., Finnegan, C. J., Ouyang, Z., Garg, A., Lebel, E. D., Rowland, S. T., Michanowicz, D. R., Herrera, J., Nadeau, K. C. et al. (2024), 'Nitrogen dioxide exposure, health outcomes, and associated demographic disparities due to gas and propane combustion by us stoves', *Science advances* **10**(18), eadm8680.

Keskin, P., Shastry, G. K. and Willis, H. (2017), 'Water quality awareness and breastfeeding: Evidence of health behavior change in bangladesh', *Review of Economics and Statistics* **99**(2), 265–280.

Klepeis, N. E., Nelson, W. C., Ott, W. R., Robinson, J. P., Tsang, A. M., Switzer, P., Behar, J. V., Hern, S. C. and Engelmann, W. H. (2001), 'The national human activity pattern survey (nhaps): a resource for assessing exposure to environmental pollutants', *Journal of Exposure Science & Environmental Epidemiology* **11**(3), 231–252.

Klimont, Z., Kupiainen, K., Heyes, C., Purohit, P., Cofala, J., Rafaj, P., Borken-Kleefeld, J. and Schöpp, W. (2017), 'Global anthropogenic emissions of particulate matter including black carbon', *Atmospheric Chemistry and Physics* **17**(14), 8681–8723.

Krebs, B., Burney, J., Zivin, J. G. and Neidell, M. (2021), 'Using crowd-sourced data to assess the temporal and spatial relationship between indoor and outdoor particulate matter', *Environmental Science & Technology* **55**(9), 6107–6115.

Krebs, B. and Luechinger, S. (2024), 'Air pollution, cognitive performance, and the role of task proficiency', *Journal of the Association of Environmental and Resource Economists* **11**(4), 921–958.

Krewski, D., Jerrett, M., Burnett, R. T., Ma, R., Hughes, E., Shi, Y., Turner, M. C., Pope III, C. A., Thurston, G., Calle, E. E. et al. (2009), *Extended follow-up and spatial analysis of the American Cancer Society study linking particulate air pollution and mortality*, Vol. 140, Health Effects Institute Boston, MA.

Künn, S., Palacios, J. and Pestel, N. (2023), 'Indoor air quality and strategic decision making', *Management Science* **69**(9), 5354–5377.

La Nauze, A. and Severnini, E. (2025), 'Air pollution and adult cognition: evidence from brain training', *Journal of the Association of Environmental and Resource Economists* **12**(1), 221–255.

Lebel, E. D., Finnegan, C. J., Ouyang, Z. and Jackson, R. B. (2022), 'Methane and no x emissions from natural gas stoves, cooktops, and ovens in residential homes', *Environmental science & technology* **56**(4), 2529–2539.

Lepeule, J., Laden, F., Dockery, D. and Schwartz, J. (2012), 'Chronic exposure to fine particles and mortality: an extended follow-up of the harvard six cities study from 1974 to 2009', *Environmental health perspectives* **120**(7), 965–970.

Li, P., Lu, Y., Peng, L. and Wang, J. (2024), 'Information, incentives, and environmental governance: Evidence from china's ambient air quality standards', *Journal of Environmental Economics and Management* **128**, 103066.

Li, Z., Wen, Q. and Zhang, R. (2017), 'Sources, health effects and control strategies of indoor fine particulate matter (pm2. 5): A review', *Science of the Total Environment* **586**, 610–622.

List, J. A. (2020), Non est disputandum de generalizability? a glimpse into the external validity trial, Technical report, National Bureau of Economic Research.

List, J. A. (2024), 'Optimally generate policy-based evidence before scaling', *Nature* **626**(7999), 491–499.

Loewenstein, G., Brennan, T. and Volpp, K. G. (2007), 'Asymmetric paternalism to improve health behaviors', *JAMA* **298**(20), 2415–2417.

Luechinger, S. (2009), 'Valuing air quality using the life satisfaction approach', *The Economic Journal* **119**(536), 482–515.

Lunderberg, D. M., Liang, Y., Singer, B. C., Apte, J. S., Nazaroff, W. W. and Goldstein, A. H. (2023), 'Assessing residential pm2. 5 concentrations and infiltration factors with high spatiotemporal resolution using crowdsourced sensors', *Proceedings of the National Academy of Sciences* **120**(50), e2308832120.

McCormack, M. C., Breysse, P. N., Matsui, E. C., Hansel, N. N., Peng, R. D., Curtin-Brosnan, J., D'Ann, L. W., Wills-Karp, M., Diette, G. B., for Childhood Asthma in the Urban Environment, C. et al. (2011), 'Indoor particulate matter increases asthma morbidity in children with non-atopic and atopic asthma', *Annals of Allergy, Asthma & Immunology* **106**(4), 308–315.

McCormack, M. C., Breysse, P. N., Matsui, E. C., Hansel, N. N., Williams, D., Curtin-Brosnan, J., Eggleston, P., Diette, G. B. and for Childhood Asthma in the Urban Environment, C. (2009), 'In-home particle concentrations and childhood asthma morbidity', *Environmental health perspectives* **117**(2), 294–298.

Miller, N. H., Molitor, D. and Zou, E. (2024), The nonlinear effects of air pollution on health: Evidence from wildfire smoke, Technical report, National Bureau of Economic Research.

Mobarak, A. M., Dwivedi, P., Bailis, R., Hildemann, L. and Miller, G. (2012), 'Low demand for nontraditional cookstove technologies', *Proceedings of the National Academy of Sciences* **109**(27), 10815–10820.

Monn, C. (2001), 'Exposure assessment of air pollutants: a review on spatial heterogeneity and indoor/outdoor/personal exposure to suspended particulate matter, nitrogen dioxide and ozone', *Atmospheric Environment* **35**(1), 1–32.

Moretti, E. and Neidell, M. (2011), 'Pollution, health, and avoidance behavior: evidence from the ports of los angeles', *Journal of human Resources* **46**(1), 154–175.

Morris, E. A., Speroni, S. and Taylor, B. D. (2023), 'Going nowhere fast: Might changing activity patterns help explain falling travel?', *Journal of Transport Geography* **110**, 103620.

Morris, E. A., Speroni, S. and Taylor, B. D. (2024), 'Going nowhere faster: Did the covid-19 pandemic accelerate the trend toward staying home?', *Journal of the American Planning Association* pp. 1–19.

Neidell, M. (2009), 'Information, avoidance behavior, and health: the effect of ozone on asthma hospitalizations', *Journal of Human resources* **44**(2), 450–478.

O'Leary, C., Jones, B., Dimitroulopoulou, S. and Hall, I. P. (2019), 'Setting the standard: the

acceptability of kitchen ventilation for the english housing stock', *Building and Environment* **166**, 106417.

Osman, L. M., Douglas, J. G., Garden, C., Reglitz, K., Lyon, J., Gordon, S. and Ayres, J. G. (2007), 'Indoor air quality in homes of patients with chronic obstructive pulmonary disease', *American journal of respiratory and critical care medicine* **176**(5), 465–472.

Pattanayak, S. K., Jeuland, M., Lewis, J. J., Usmani, F., Brooks, N., Bhojvaid, V., Kar, A., Lipinski, L., Morrison, L., Patange, O. et al. (2019), 'Experimental evidence on promotion of electric and improved biomass cookstoves', *Proceedings of the National Academy of Sciences* **116**(27), 13282–13287.

Pattanayak, S. K. and Pfaff, A. (2009), 'Behavior, environment, and health in developing countries: evaluation and valuation', *Annu. Rev. Resour. Econ.* **1**(1), 183–217.

Persico, C. L. and Venator, J. (2021), 'The effects of local industrial pollution on students and schools', *Journal of Human Resources* **56**(2), 406–445.

Pinchbeck, E. W., Roth, S., Szumilo, N. and Vanino, E. (2023), 'The price of indoor air pollution: evidence from risk maps and the housing market', *Journal of the Association of Environmental and Resource Economists* **10**(6), 1439–1473.

Pope III, C. A., Burnett, R. T., Turner, M. C., Cohen, A., Krewski, D., Jerrett, M., Gapstur, S. M. and Thun, M. J. (2011), 'Lung cancer and cardiovascular disease mortality associated with ambient air pollution and cigarette smoke: shape of the exposure–response relationships', *Environmental Health Perspectives* **119**(11), 1616–1621.

Pope III, C. A. and Dockery, D. W. (2006), 'Health effects of fine particulate air pollution: lines that connect', *Journal of the air & waste management association* **56**(6), 709–742.

Riley, W. J., McKone, T. E., Lai, A. C. and Nazaroff, W. W. (2002), 'Indoor particulate matter of outdoor origin: importance of size-dependent removal mechanisms', *Environmental Science & Technology* **36**(2), 200–207.

Rohr, A. and McDonald, J. (2016), 'Health effects of carbon-containing particulate matter: focus on sources and recent research program results', *Critical Reviews in Toxicology* **46**(2), 97–137.

Sager, L. and Singer, G. (2022), *Clean Identification?: The Effects of the Clean Air Act on Air Pollution, Exposure Disparities and House Prices*, Grantham Research Institute on Climate Change and the Environment.

Sater, R. A., Perona, M., Huillery, E. and Chevallier, C. (2021), 'The effectiveness of personalised versus generic information in changing behaviour: Evidence from an indoor air quality experiment'.

Schlenker, W. and Walker, W. R. (2016), 'Airports, air pollution, and contemporaneous health', *Review of Economic Studies* **83**(2), 768–809.

See, S. W. and Balasubramanian, R. (2008), 'Chemical characteristics of fine particles emitted from different gas cooking methods', *Atmospheric Environment* **42**(39), 8852–8862.

Semple, S., Apsley, A., Ibrahim, T. A., Turner, S. W. and Cherrie, J. W. (2015), 'Fine particulate matter concentrations in smoking households: just how much secondhand smoke do you breathe in if you live with a smoker who smokes indoors?', *Tobacco Control* **24**(e3), e205–e211.

Simeonova, E., Currie, J., Nilsson, P. and Walker, R. (2021), 'Congestion pricing, air pollution, and children's health', *Journal of Human Resources* **56**(4), 971–996.

Smith, K. R., Bruce, N., Balakrishnan, K., Adair-Rohani, H., Balmes, J., Chafe, Z., Dherani, M., Hosgood, H. D., Mehta, S., Pope, D. et al. (2014), 'Millions dead: how do we know and what does it mean? methods used in the comparative risk assessment of household air pollution', *Annual Review of Public Health* **35**, 185–206.

Smith, K. R., McCracken, J. P., Weber, M. W., Hubbard, A., Jenny, A., Thompson, L. M., Balmes, J., Diaz, A., Arana, B. and Bruce, N. (2011), 'Effect of reduction in household air pollution on childhood pneumonia in guatemala (respire): a randomised controlled trial', *The Lancet* **378**(9804), 1717–1726.

Smith, V. K., Desvousges, W. H., Johnson, F. R. and Fisher, A. (1990), 'Can public information programs affect risk perceptions?', *Journal of Policy Analysis and Management* **9**(1), 41–59.

Smith, V. K. and Huang, J.-C. (1995), 'Can markets value air quality? a meta-analysis of hedonic property value models', *Journal of Political Economy* **103**(1), 209–227.

Spalt, E. W., Curl, C. L., Allen, R. W., Cohen, M., Adar, S. D., Stukovsky, K. H., Avol, E., Castro-Diehl, C., Nunn, C., Mancera-Cuevas, K. et al. (2016), 'Time–location patterns of a diverse population of older adults: the multi-ethnic study of atherosclerosis and air pollution (mesa air)', *Journal of Exposure Science & Environmental Epidemiology* **26**(4), 349–355.

Stafford, T. M. (2015), 'Indoor air quality and academic performance', *Journal of Environmental Economics and Management* **70**, 34–50.

Sundell, J. (2004), 'On the history of indoor air quality and health', *Indoor Air* **14**(s 7), 51–58.

Tran, V. V., Park, D. and Lee, Y.-C. (2020), 'Indoor air pollution, related human diseases, and recent trends in the control and improvement of indoor air quality', *International journal of environmental research and public health* **17**(8), 2927.

U.S. Environmental Protection Agency (2025), 'Low-cost air pollution monitors and indoor air quality'. Accessed: 24th January 2025.
URL: *https://www.epa.gov/indoor-air-quality-iaq/low-cost-air-pollution-monitors-and-indoor-air-quality*

Wallace, L. A., Emmerich, S. J. and Howard-Reed, C. (2004), 'Source strengths of ultrafine and fine particles due to cooking with a gas stove', *Environmental Science & Technology* **38**(8), 2304–2311.

Wang, Z., Delp, W. W. and Singer, B. C. (2020), 'Performance of low-cost indoor air quality monitors for pm2. 5 and pm10 from residential sources', *Building and Environment* **171**, 106654.

Weitz, J. S., Park, S. W., Eksin, C. and Dushoff, J. (2020), 'Awareness-driven behavior changes can

shift the shape of epidemics away from peaks and toward plateaus, shoulders, and oscillations', *Proceedings of the National Academy of Sciences* **117**(51), 32764–32771.

WHO (2021), World Health Organization. Accessed: 29th April 2024.
URL: *https://www.who.int/news-room/fact-sheets/detail/household-air-pollution-and-health*

WHO et al. (2010), *WHO guidelines for indoor air quality: selected pollutants*, World Health Organization. Regional Office for Europe.

Wolff, H. (2014), 'Keep your clunker in the suburb: low-emission zones and adoption of green vehicles', *The Economic Journal* **124**(578), F481–F512.

Xu, J., Zhao, H., Zhang, Y., Yang, W., Wang, X., Geng, C., Li, Y., Guo, Y., Han, B., Bai, Z. et al. (2024), 'Reducing indoor particulate air pollution improves student test scores: A randomized double-blind crossover study', *Environmental Science & Technology* **58**(19), 8207–8214.

Yang, L., Lin, Y., Wang, J. and Peng, F. (2024), 'Achieving air pollution control targets with technology-aided monitoring: Better enforcement or localized efforts?', *American Economic Journal: Economic Policy* **16**(4), 280–315.

Zarate, P., Dolls, M., Davis, S. J., Bloom, N., Barrero, J. M. and Aksoy, C. G. (2024), Why does working from home vary across countries and people?, Technical report, National Bureau of Economic Research.

Zhang, N., Han, B., He, F., Xu, J., Zhao, R., Zhang, Y. and Bai, Z. (2017), 'Chemical characteristic of pm2. 5 emission and inhalational carcinogenic risk of domestic chinese cooking', *Environmental pollution* **227**, 24–30.

Zhang, X., Chen, X. and Zhang, X. (2018), 'The impact of exposure to air pollution on cognitive performance', *Proceedings of the National Academy of Sciences* **115**(37), 9193–9197.

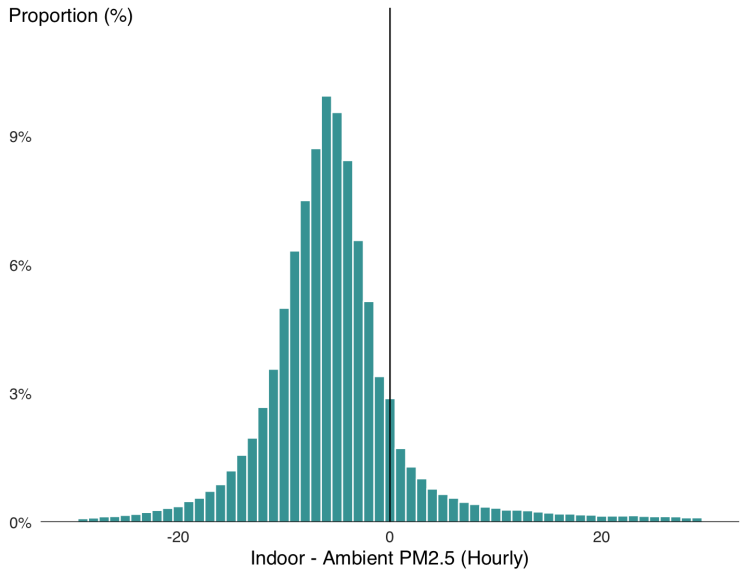
Zhao, Y., Chen, C. and Zhao, B. (2019), 'Emission characteristics of pm2. 5-bound chemicals from residential chinese cooking', *Building and Environment* **149**, 623–629.

Zheng, X.-y., Ding, H., Jiang, L.-n., Chen, S.-w., Zheng, J.-p., Qiu, M., Zhou, Y.-x., Chen, Q. and Guan, W.-j. (2015), 'Association between air pollutants and asthma emergency room visits and hospital admissions in time series studies: a systematic review and meta-analysis', *PloS one* **10**(9), e0138146.

Zivin, J. G. and Neidell, M. (2009), 'Days of haze: Environmental information disclosure and intertemporal avoidance behavior', *Journal of Environmental Economics and Management* **58**(2), 119–128.

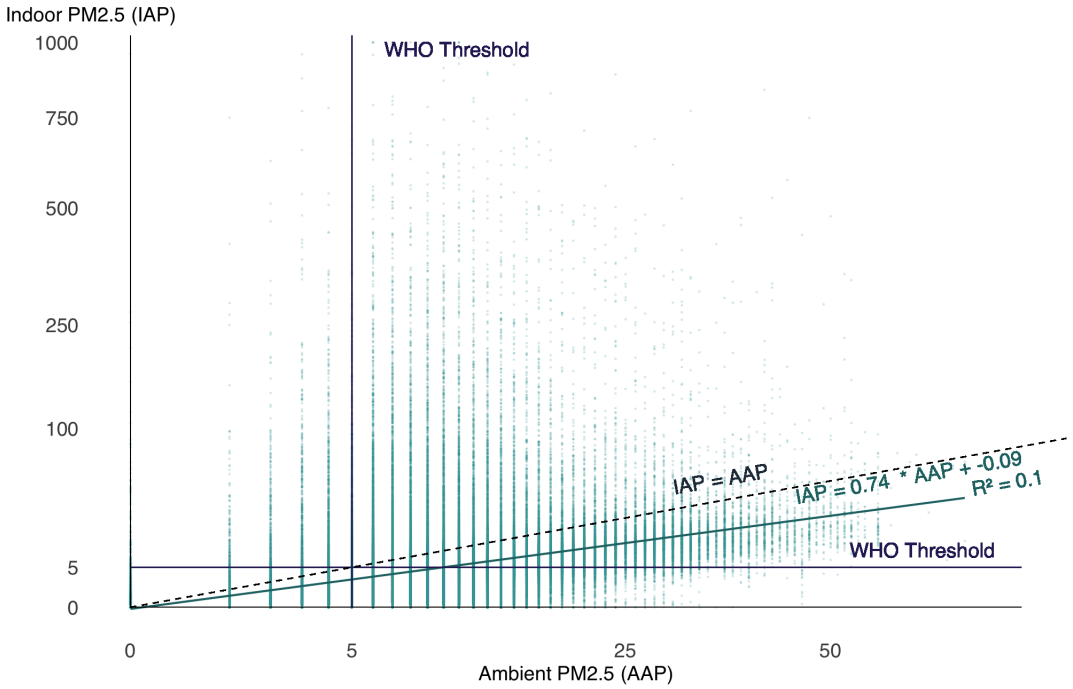
Zou, E. Y. (2021), 'Unwatched pollution: The effect of intermittent monitoring on air quality', *American Economic Review* **111**(7), 2101–2126.

Figure 1: Indoor PM2.5 minus Ambient PM2.5 per hour



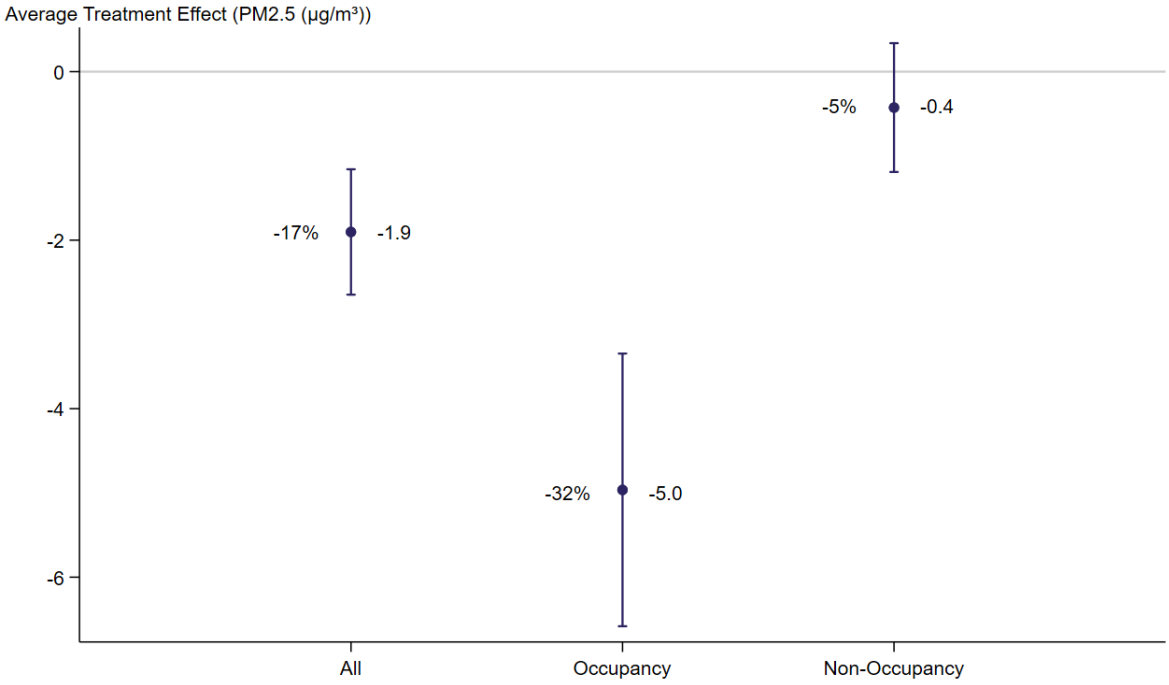
Note: This figure displays the distribution of the difference between indoor and ambient air pollution during the pre-treatment period.

Figure 2: The Relationship Between Indoor and Ambient Air Pollution



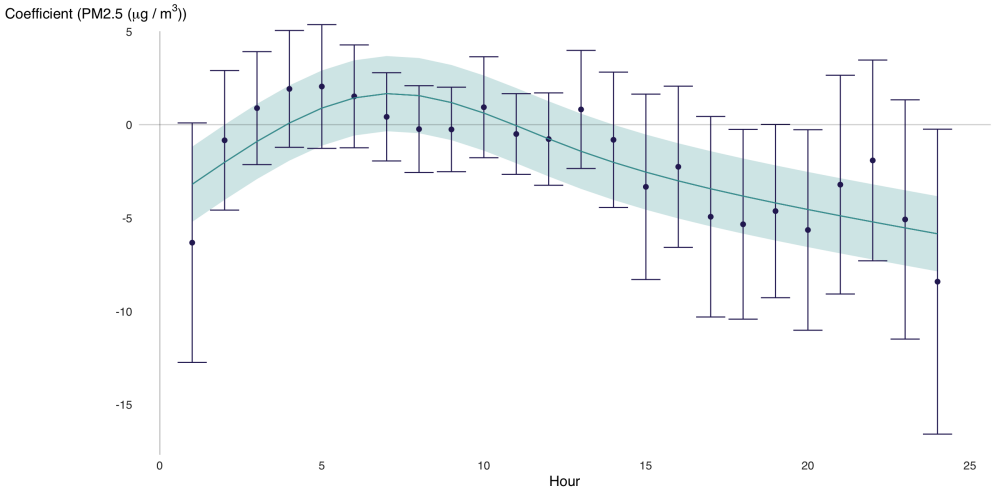
Note: This figure displays the relationship between indoor and ambient air pollution per hour during the pre-treatment period.

Figure 3: Average Treatment Effects of The Treatment (IAP Monitor)



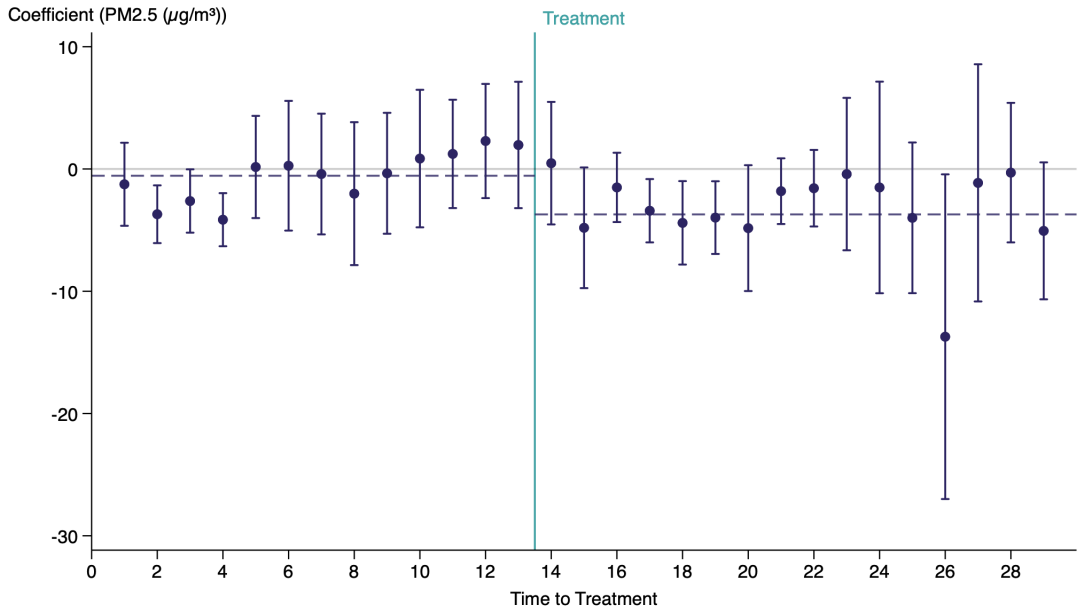
Note: This figure displays estimates of the coefficients of the average treatment effect by day respective to treatment time bootstrapped 1000 times. Ambient pollution levels as well as day and month fixed effects have been controlled for. 95% confidence intervals are presented clustered at a household and date level. Occupancy time here refers to 16:00-23:00.

Figure 4: Average Treatment Effects by Hour of The Day



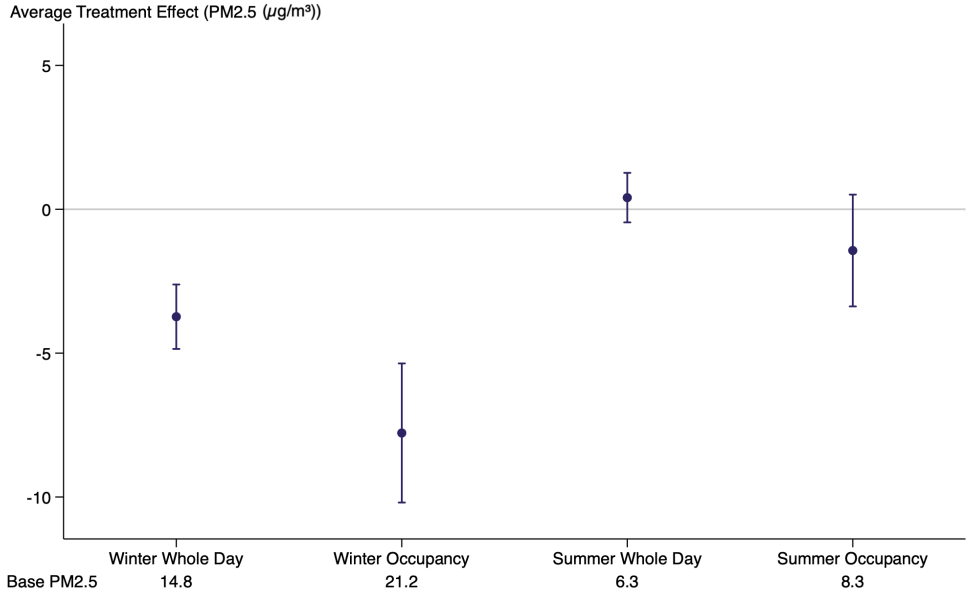
Note: This figure shows the coefficient on Treatment x Post, and uses our main specification, separated by hour of the day. 95% confidence intervals are presented, clustered at the household and date level. The light blue ribbon shows the bootstrapped standard errors.

Figure 5: Average Treatment Effect by Day



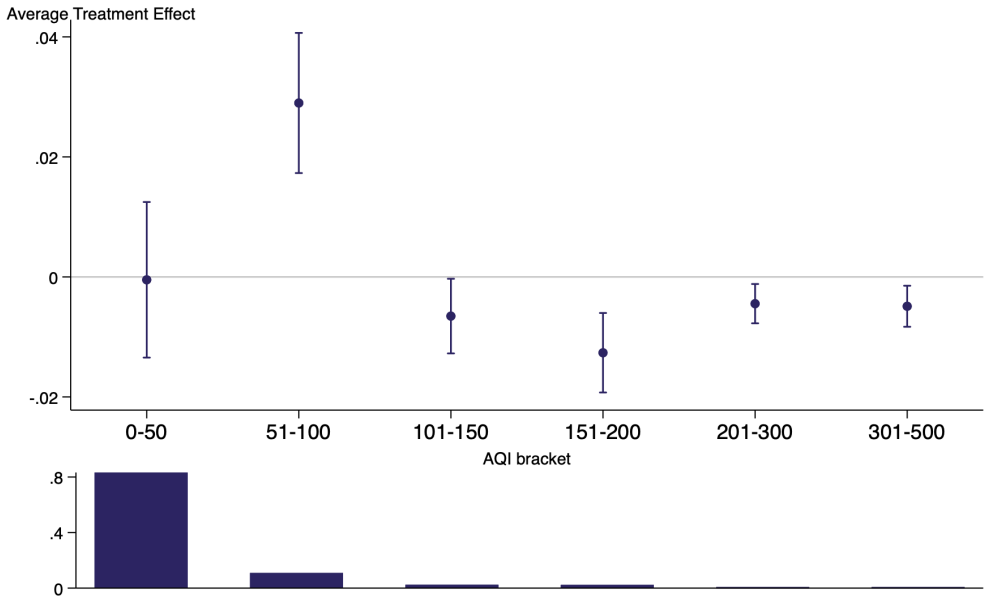
Note: This figure displays estimates of the coefficients of the average treatment effect by day respective to treatment time, controlling for ambient PM2.5 and day and month fixed effects. 95% confidence intervals are presented, clustered at the household and date level.

Figure 6: Average Treatment Effect by Season



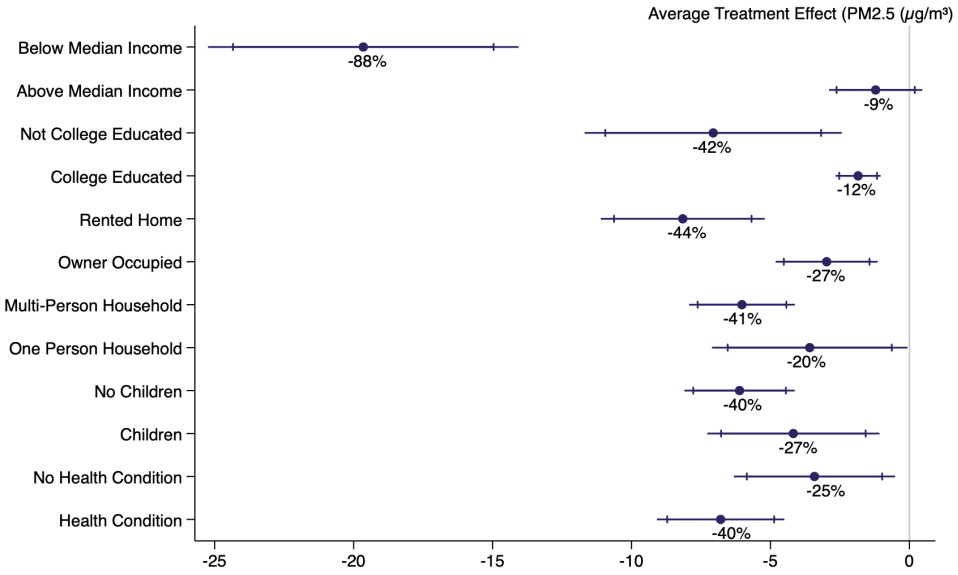
Note: This figure displays the average treatment effect by season and time of day bootstrapped 1000 times, controlling for ambient PM2.5 and day and month fixed effects. 95% confidence intervals are presented, clustered at the household and date level.

Figure 7: Average Treatment Effect by Air Quality Index (AQI) Bracket



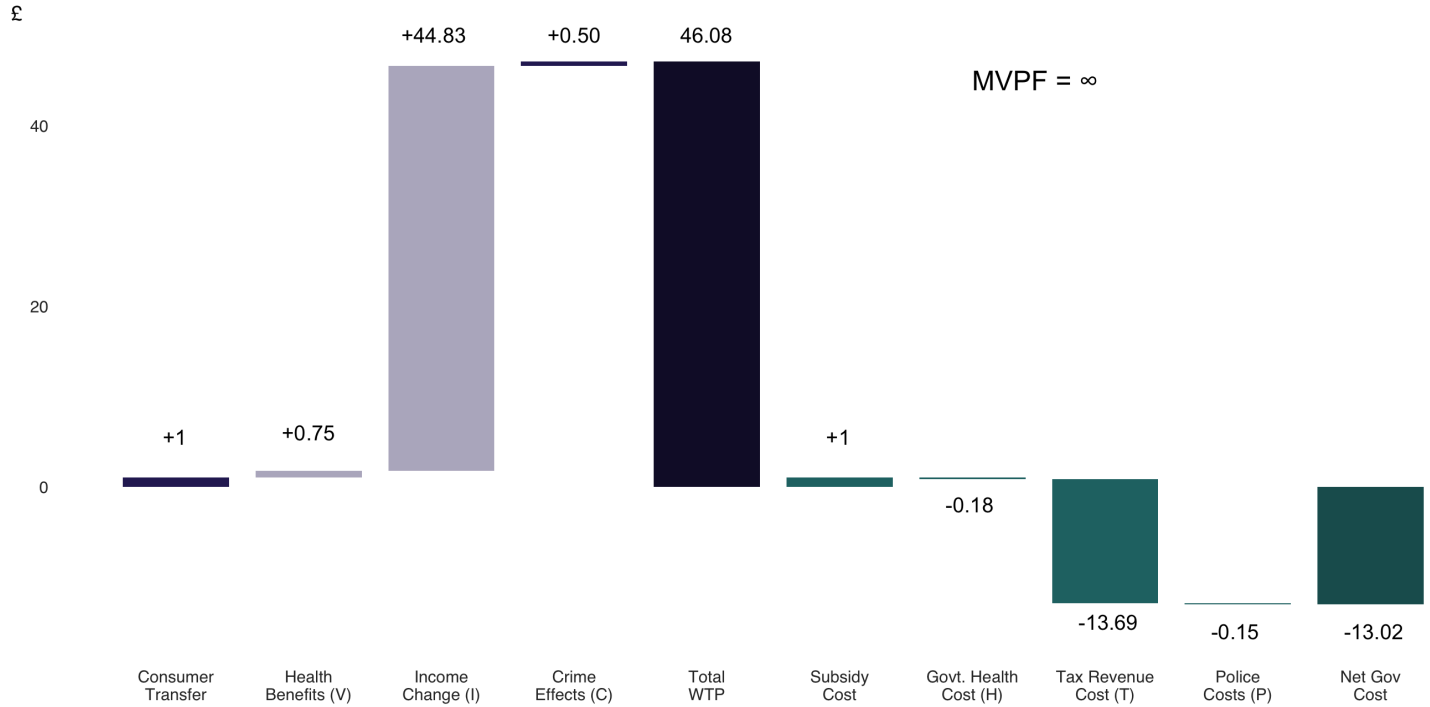
Note: This figure displays estimates of the coefficients of Treatment x Post on AQI brackets. This is a linear probability model, where each coefficient shows the probability that a household's AQI level will be within a certain bracket. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. The observations restricted to occupancy time (16:00-23:00) to reflect times when the average person is at home and near the monitor. Standard errors bootstrapped 1000 times and clustered two way at the household level and by date. 95% confidence intervals are presented.

Figure 8: Average Treatment Effect by Household Characteristics



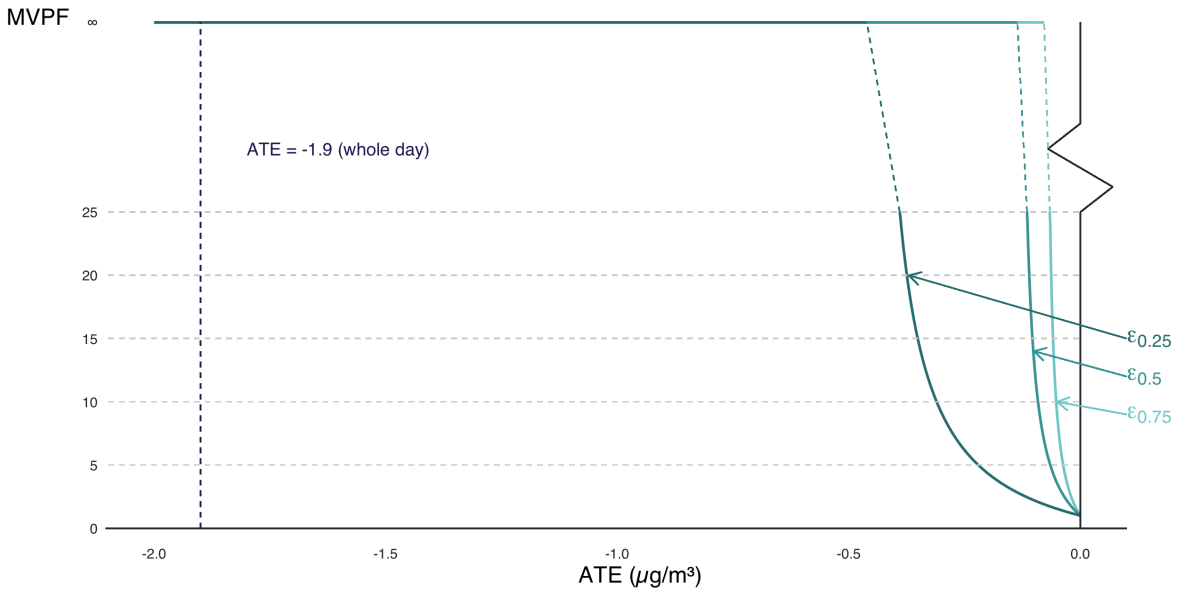
Note: This figure displays estimates of the coefficients of the average treatment effect for each characteristic in the occupancy period bootstrapped 1000 times, controlling for ambient PM2.5, all the other household characteristics (below median income dummy, number of smokers, asthma, children, number of residents, health conditions, if the house is owned by the resident and education). Day of week and month fixed effects are also included, with standard errors clustered at the household level and by date. 95% confidence intervals are presented.

Figure 9: Marginal Value of Public Funds for an Air Pollution Monitor Subsidy



Note: This figure displays the MVPF waterfall chart for the lower bound estimate of the MVPF of the whole day average treatment effect for the Kaiterra Monitor over a 20 year horizon. The health (V) and Income benefits (I) are depicted as a different shade of blue because there is uncertainty as to whether consumers fully value these in the purchase of the air monitor. Even if individuals have perfect information on these benefits (so zero H and I), the MVPF would remain being infinite.

Figure 10: MVPF by the ATE and price elasticity of IAP monitor demand



Note: This figure displays how the MVPF varies with the ATE for the IAP monitor across the interquartile ranges of elasiticies with the darkest shade referring to a PED of -0.22 and the lightest shade corresponding to a PED of -1.31. This is specifically relating to the third row of Table 6, as described in the text and shown in Figure 9. With our median elasticity of 0.75, we show that MVPF = ∞ for ATE = -1.9 over a 20 year horizon. These results are robust to a 10 year horizon too. This shows that an ATE as small as -0.14 can still have an infinite MVPF for the IAP monitor.

Table 1: Balance Test by Treatment

Variable	Control Mean	Control N	Treatment Mean	Treatment N	Difference	Std. Error
Females per Household	1.24	104	1.35	97	-0.11	(0.10)
Males per Household	1.31	103	1.25	87	0.06	(0.10)
Number of Rooms	4.67	118	4.81	108	-0.14	(0.33)
Number of Windows	7.23	115	6.94	103	0.29	(0.54)
Number of Smokers	0.13	118	0.10	109	0.03	(0.05)
Open Plan Kitchen	0.53	118	0.50	109	0.02	(0.07)
College Educated	0.87	119	0.86	111	0.02	(0.05)
Indoor PM2.5	11.11	125	10.41	120	0.70	(1.55)
Indoor AQI	31.89	125	27.81	120	4.07	(2.93)
Indoor Temperature	21.80	125	21.56	120	0.24	(0.45)
Outdoor PM2.5	12.05	125	11.35	120	0.71	(0.51)
Outdoor Temperature	13.91	125	13.53	120	0.38	(0.70)
Indoor PM2.5 Occupancy	14.40	125	14.90	120	-0.51	(2.49)
Indoor AQI Occupancy	38.44	125	34.97	120	3.47	(3.84)
Indoor Temperature Occupancy	22.19	125	21.97	120	0.22	(0.46)
Outdoor PM2.5 Occupancy	12.43	125	11.99	120	0.44	(0.40)
Outdoor Temperature Occupancy	16.10	125	15.72	120	0.38	(0.79)

Note: None of the differences are significant at the 10% level.

Table 2: Impact of treatment on indoor and outdoor temperature difference

	Total	Occupancy	Non-Occupancy
Treatment x Post	-0.714***	-0.570**	-0.767***
	(0.24)	(0.24)	(0.25)
Treatment	0.121	0.117	0.132
	(0.29)	(0.27)	(0.30)
Post	0.765***	0.580**	0.913***
	(0.26)	(0.28)	(0.28)
Ambient PM2.5	0.050***	0.026*	0.079***
	(0.01)	(0.01)	(0.01)
Constant	7.587***	6.299***	8.023***
	(0.31)	(0.32)	(0.32)
Base Indoor Temperature	21.21	21.61	21.02
Base Ambient Temperature	12.49	14.67	11.42
Observations	149,044	49,776	99,268

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on the temperature difference between indoor and outdoor temperatures in °C. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. The observations that are restricted to occupancy time (16:00-23:00) are to reflect times when the average person is at home. Base indoor and ambient temperature here reflect the average temperatures of the control group in the post-treatment period. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 3: IAP peak analysis

	Peak Height	Peak Frequency
Treatment x Post	-37.907*	-0.081
	(20.97)	(0.06)
Treatment	2.196	-0.038
	(17.97)	(0.12)
Post	12.994	0.081*
	(11.69)	(0.04)
Ambient PM2.5	-0.256	0.014***
	(0.39)	(0.00)
Constant	154.646***	-2.092***
	(10.59)	(0.11)
Observations	10,697	148,859

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Peaks are defined as local maximum levels of PM2.5 levels being above 35 over at least a 5 hour period. The first column only includes values where there where peaks, regressing the local maximum PM2.5 values of these peaks on the other variables. The second column displays a probit regression of a binary variable for the peaks.

Table 4: Belief Outcomes

	Indoor		Ambient	
	Air Quality Belief	Confidence	Air Quality Belief	Confidence
Treatment	0.259**	0.302**	0.155	0.115
	(0.10)	(0.13)	(0.12)	(0.14)
Constant	0.051	0.026	-0.000	0.076
	(0.07)	(0.09)	(0.08)	(0.10)
Observations	234	233	234	233

Note: This table displays estimates of the coefficients of treatment on the difference in air quality beliefs in the baseline and endline survey. Air quality belief represents a participants perception of indoor and outdoor air quality. The higher this belief is, the worse they believe their atmosphere is for their health. The options ranged from 1-5, with 5 being the worst. Confidence represents how certain a participant is about this level of air quality, also ranging from 1-5, with 5 being the most confident. We can see that participants in the treatment group updated their beliefs about their indoor air pollution and realized it is worse than they initially believed. Standard errors are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 5: Robustness regressions

	Occupancy			Non-Occupancy	12am-11am	12pm-11pm	Occupancy
	No Ambient	First Waves	All				Characteristic FEs
Treatment x Post	-4.495*	-6.528**	-4.964**	-0.426	-0.133	-3.798*	-5.210
	(2.34)	(2.84)	(2.33)	(1.06)	(1.03)	(1.96)	(2.67)
Treatment	1.007	0.439	1.261	-0.980	-1.292	0.852	-1.017
	(2.67)	(2.52)	(2.65)	(1.35)	(1.37)	(2.18)	(2.10)
Post	1.215	4.871**	2.525*	-0.129	-0.390	2.035	2.535
	(1.55)	(2.03)	(1.51)	(0.83)	(0.85)	(1.36)	(1.78)
Ambient PM2.5		0.560***	0.549***	0.519***	0.547***	0.554***	0.502***
		(0.07)	(0.08)	(0.03)	(0.03)	(0.07)	(0.07)
Constant	14.423***	5.370***	7.385***	3.969***	3.293***	6.411***	8.969***
	(1.51)	(1.38)	(1.55)	(0.97)	(1.11)	(1.28)	(1.38)
Observations	50,085	32,491	50,085	99,994	75,002	75,077	31,533

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. The first column doesn't control for outdoor PM2.5, the second column only includes the participants from the first 3 waves to match our pre-analysis plan. The third column is our normal specification as shown in 1 and the fourth column does this for non-occupancy time. The fifth and sixth two columns focus on time periods 12pm-11pm and 12am-11am respectively. The final column incorporates household characteristic fixed effects (eg. children, smoking, rooms, fireplaces, income bracket), wave fixed effects, hour-of-day fixed effects, month fixed effects and day-of-the-week fixed effects. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table 6: MVPF estimates for a $1 subsidy for either an IAP monitor or purifer

	Transfer	Health Benefits (V)	Income Change (I)	Crime (C)	Total WTP	Subsidy Cost	Govt. Health Cost Savings (H)	Tax Revenue (T)	Gov Crime Costs (P)	Net Government Cost	MVPF
IAP monitor: Whole Day ATE (-1.9)											
Lower Bound 10 Year	1	0.75	24.63	0.27	26.65	1	0.05	7.52	0.08	-6.65	∞
Upper Bound 10 Year	1	1.68	24.63	0.27	27.58	1	0.05	7.52	0.08	-6.65	∞
Lower Bound 20 Year	1	0.75	44.83	0.50	47.08	1	0.18	13.69	0.15	-13.02	∞
Upper Bound 20 Year	1	1.68	44.83	0.50	48.01	1	0.18	13.69	0.15	-13.02	∞
IAP monitor: Occupancy Time ATE (-5.0)											
Lower Bound 10 Year	1	1.96	64.81	0.72	68.49	1	0.14	64.81	0.21	-64.16	∞
Upper Bound 10 Year	1	4.33	64.81	0.72	70.86	1	0.14	64.81	0.21	-64.16	∞
Lower Bound 20 Year	1	1.96	117.97	1.31	122.24	1	0.46	36.04	0.38	-35.88	∞
Upper Bound 20 Year	1	4.33	117.97	1.31	124.61	1	0.46	36.04	0.38	-35.88	∞
IAP purifier: Whole Day ATE (-11.0)											
Lower Bound 10 Year	1	6.56	220.75	2.46	230.77	1	0.47	67.44	0.72	-67.63	∞
Upper Bound 10 Year	1	14.18	220.75	2.46	238.39	1	0.47	67.44	0.72	-67.63	∞
Lower Bound 20 Year	1	6.56	401.85	4.48	413.89	1	1.57	122.76	1.31	-124.64	∞
Upper Bound 20 Year	1	14.18	401.85	4.48	421.51	1	1.57	122.76	1.31	-124.64	∞
IAP purifier: Occupancy Time ATE (-15.4)											
Lower Bound 10 Year	1	9.07	309.05	3.44	322.56	1	0.65	94.41	1.01	-95.07	∞
Upper Bound 10 Year	1	19.31	309.05	3.44	332.80	1	0.65	94.41	1.01	-95.07	∞
Lower Bound 20 Year	1	9.07	562.58	6.27	578.92	1	2.20	171.86	1.83	-173.49	∞
Upper Bound 20 Year	1	19.31	562.58	6.27	589.16	1	2.20	171.86	1.83	-173.49	∞

Note: This table displays estimates of MVPFs from equation **??**. All the numbers are inflation adjusted to 2023£. We consider the 10 and 20 year horizons for the government health cost savings for the NHS and for the change in income tax revenue due to labor force participation. The upper and lower bound estimates are derived from the C-R mortality function, which tell us the WTP for a reduction in mortality. The varying degrees of ATE's is derived from the effectiveness we calculate of the Kaiterra Monitor through our experiment and is assumed for the Philips Air Purifier.

Appendices

Contents of Appendices

Appendices **61**

A Figures **61**

B Tables **83**

C Online Appendix **91**

C.1 Impact on Health 91

C.2 Impact on Economic Welfare 93

A Figures

Figure A1: Preliminary Survey Questions

Preliminary Survey

Thank you for participating in the Camden Household Air Monitoring Project (CHAMP), we're really excited to have you on board. Before we get started with the project, we just need a little more information from you, so please feel in as much as you can of this survey. It shouldn't take more than 10 minutes to complete.

Dr Sefi Roth
Department of Geography and Environment
London School of Economics and Political Science
geog.iaq@lse.ac.uk

1. What is the research about?

    • This study will observe the scale of indoor air pollution in homes throughout Camden for a four-week period. You will be given a small air pollution monitor for the main room in your home, and a 3G router to send the air pollution data from the device to our database.

    • You will also complete two short surveys. This first survey will determine your eligibility for the project, and the second will ask your experience with the monitors and the project. At the end of the project when you complete the second survey and give back the monitor, as a thank you, you will be rewarded with a £20 Amazon voucher.

    • We will also notify you of your indoor air pollution levels, how they compare to the average Camden resident in the study, and how to reduce the pollution levels.

2. What will my information be used for?

    We will use the collected information to write scientific papers on the topic of indoor air pollution and to guide future research in the area. The records from this study will be kept confidential. Only the researchers conducting the project will have access to the information you have provided. Your data will be anonymised – your name will not be used in any reports or publications resulting from this study.

3. What if I have a question or complaint?

    If you have any questions regarding this project, please contact the team on geog.iaq@lse.ac.uk. This study has undergone ethics review in accordance with the LSE Research Ethics Policy and Procedure. If you have any concerns or complaints regarding the conduct of this research, please contact the LSE Research Governance Manager via research.ethics@lse.ac.uk.

    You can withdraw from the study at any point.

    The air pollution monitors are the ownership of the London School of Economics, and we will be retrieving the monitors from all participants at the end of the four-week period.

    If you are happy to take part in this study, please tick the consent question below.

Thank so much!

Dr Sefi Roth
Department of Geography and Environment
London School of Economics and Political Science
geog.iaq@lse.ac.uk

Please tick the box: I have read and understood the study information and consent voluntarily to be a participant in this study. As part of this consent, I allow researchers at the London School of Economics to analyse the data for scientific purposes only.

I have read and understood the study information and consent voluntarily to be a participant in this study. As part of this consent, I allow researchers at the London School of Economics and University of Southern California (US), to analyse the data for scientific purposes only.

1. How many people usually live in your household?
   1
   2
   3
   4
   5
   6
   7+
   Prefer not to say

2. How old are the residents living in your household? (Resident 1)

3. How old are the residents living in your household? (Resident 2)

4. How old are the residents living in your household? (Resident 3)

5. How old are the residents living in your household? (Resident 4)

6. How old are the residents living in your household? (Resident 5)

7. How old are the residents living in your household? (Resident 6)

8. How old are the residents living in your household? (Residents 7 and above)

9. Of the adults in your household, how many are…
   Female
   Male
   Other

10. What is your highest level of education held?
    GCSE/O Levels or equivalent
    A-Levels or equivalent

Undergraduate degree
Postgraduate degree
Other
None

11. What is your gross (pre-tax) household income level?
    £0-£15,000
    £15,001-£35,000
    £35,001-£50,000
    £50,001-£65,000
    £65,001-£80,000
    £80,001-£95,000
    Over £95,000

12. What is your housing tenure?
    Owner-occupied
    Privately rented
    Socially rented
    Other

13. Overall, how happy did you feel yesterday?
    0-10 not at all happy – extremely happy)

14. Overall, how anxious did you feel yesterday?
    0-10 (not at all anxious – extremely anxious)

15. In general, what is the quality of your health?
    Very good
    Good
    Fair
    Bad
    Very bad

16. During the past 2 weeks, how much did pain interfere with your normal work including both work outside the home and housework? Did it interfere...
    Not at all
    A little bit
    Moderately
    Quite a bit
    Extremely

17. How much of the time during the past 2 weeks have you felt calm and peaceful?
    All of the time
    Most of the time
    Some of the time
    A little of the time
    None of the time

18. How much of the time during the past 2 weeks did you have a lot of energy?
    All of the time

Most of the time
Some of the time
A little of the time
None of the time

19. During the past 2 weeks, how much of the time has your physical health or emotional problems interfered with your social activities like visiting friends or relatives?
    All of the time
    Most of the time
    Some of the time
    A little of the time
    None of the time

20. During the past 2 weeks, how would you rate your sleep quality overall?
    Very good
    Fairly good
    Fairly bad
    Very bad

21. Do you or any of the people in your household have any underlying health conditions?
    Yes
    No
    Unsure
    Prefer not to say

22. What are therse conditions?

23. How many smokers are there in your household?
    None
    1
    2
    3
    4
    5+

24. In total, how many rooms are in your home?

25. Do you have an open plan kitchen (that is, where the kitchen and living room are not separated by a wall)?
    Yes
    No

26. How many windows can be opened in your home?

27. What is the primary appliance used for cooking in your home? (multiple answers allowed)
    Stovetop
    Oven
    Microwave

Toaster
Grill
Other

28. Do you have a hob in your kitchen?
    Yes
    No

29. What kind of hob do you have?
    Induction
    Gas
    Electric
    Other
    Unsure

30. Do you have an extractor hood in your kitchen?
    Yes
    No

31. Do you have a fireplace in your home?
    Yes
    No

32. What type of fireplace do you have?
    Open fireplace
    Wood burning stove
    Electric fireplace
    Gas fireplace
    Other

33. Do you have any working air purifiers in your home?
    Yes
    No

34. How many?

35. On average, do you think the air quality **in your home** is:
    Good (no health risk)
    Moderate (acceptable but some health concern)
    Unhealthy for sensitive people (such as children, asthmatics, and those with breathing difficulties)
    Unhealthy for all people (harmful for all people)
    Very unhealthy (emergency levels for all people)
    Hazardous

36. How confident are you about your answer to the question above?
    Very confident
    Somewhat confident
    Neither confident nor unconfident
    Somewhat unconfident

Very unconfident

37. On average, do you think the air quality **outside your home** is:
    - Good (no health risk)
    - Moderate (acceptable but some health concern)
    - Unhealthy for sensitive people (such as children, asthmatics, and those with breathing difficulties)
    - Unhealthy for all people (harmful for all people)
    - Very unhealthy (emergency levels for all people)
    - Hazardous

38. How confident are you about your answer to the question above?
    - Very confident
    - Somewhat confident
    - Neither confident nor unconfident
    - Somewhat unconfident
    - Very unconfident

39. In comparison to the average Camden home, do you think the air quality in your home is:
    - A lot better
    - Somewhat better
    - About the same
    - Worse
    - A lot worse
    - Unsure

40. In order to deliver your indoor air monitor, we need your contact details. Please provide below:
    - Full name
    - Full address
    - Mobile number
    - Preferred contact email

All done! Thank you very much for completing this survey. We'll be in touch shortly with more information on the next steps.

Figure A2: Initial Recruitment Letter

Air Quality
London Borough of Camden
5 Pancras Square
LONDON
N1C 4AG

Phone: 020 7974

camden.gov.uk

email: AirQuality@camden.gov.uk

Date:

Our reference: [ID]

Your reference: [ID}

OCCUPIER
[Add1]
[Add2]
[Add3]
[Add4]
[PostCode]

An invitation to participate in an indoor air pollution study

Dear Camden Resident,

You have been randomly selected to take part in an indoor air pollution research study taking place in Camden in a collaboration between the London School of Economics and Camden Council. The study will expand our scientific knowledge around indoor air pollution and help to improve your knowledge about the health and wellbeing effects of indoor air pollution.

We are looking for Camden residents like yourself to take part in this exciting project. You will be given a small device to monitor the air pollution in your home over a four week period. Installing the monitor is very straightforward, and we will provide technical support if needed. The information collected in this study will only be used for research purposes by approved researchers at the London School of Economics. Identifiable information will never be shared with any other party.

At the end of the study, you will receive a personalised report of the pollution levels within your home and some information around what this means for your health and wellbeing. Involvement in this study will put you at the forefront of scientific knowledge production and will be valuable in driving policy changes addressing the pollution problem and supporting public health. As a thank you for your time, you will also receive a voucher payment of £20.
If you are willing to participate in this study, please visit our website to register and to learn more. Scan the QR code below or visit the URL https://www.lse.ac.uk/geography-and-environment/news/indoor-air-pollution-in-camden. You can also email geog.iaq@lse.ac.uk or call 07472 740612 if you have any questions.

We look forward to hearing from you.

Kind regards,

The Camden Household Air Monitoring Project (CHAMP) team

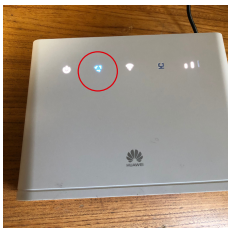
Figure A3: Monitor Instructions

HOW TO SET UP YOUR MONITOR

Set up your monitor in the room where you spend the most amount of time.



Open the WiFi router box. Insert the black cable at the back of the router and plug the cable into a socket. After a few minutes, you should see the second symbol at the front turn blue.



Take the pollution monitor out of the box. Insert the white cable into the back of the monitor and plug it into a nearby socket.



Press and hold the power button at the top of the monitor for three seconds to turn it on. It wont look like anything has changed, but it should be working.

IF YOU NEED ANY HELP OR HAVE ANY QUESTIONS PLEASE CALL **07472 740612** OR EMAIL **GEOG.IAQ@LSE.AC.UK**

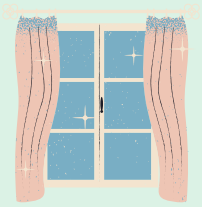
CHAMP Camden Household Air Monitoring Project · LSE · USC University of Southern California · Camden

Figure A4: Information Sheet

What can you do to improve air quality in your home?

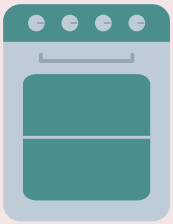
These simple steps can help you reduce indoor air pollution in your home to keep yourself and your family healthy.

Avoid opening windows during rush hour traffic if your home is close to a busy road.



Ventilate your kitchen when cooking using an extractor fan and opening windows.

This is especially important when using gas stoves and deep-frying.



Avoid burning candles or incense.



Ventilate rooms when cleaning or decorating.

Wait for paint or cleaning smells to subside before using the room again.



Cover pots and pans when cooking



Avoid Smoking inside and close to your home.



Consider using an air purifier if air pollution levels in your home are at dangerous levels.



Interested in learning more about how to improve air quality in your home? scan the QR code or follow this link for more information:
https://tinyurl.com/yfhas5x4

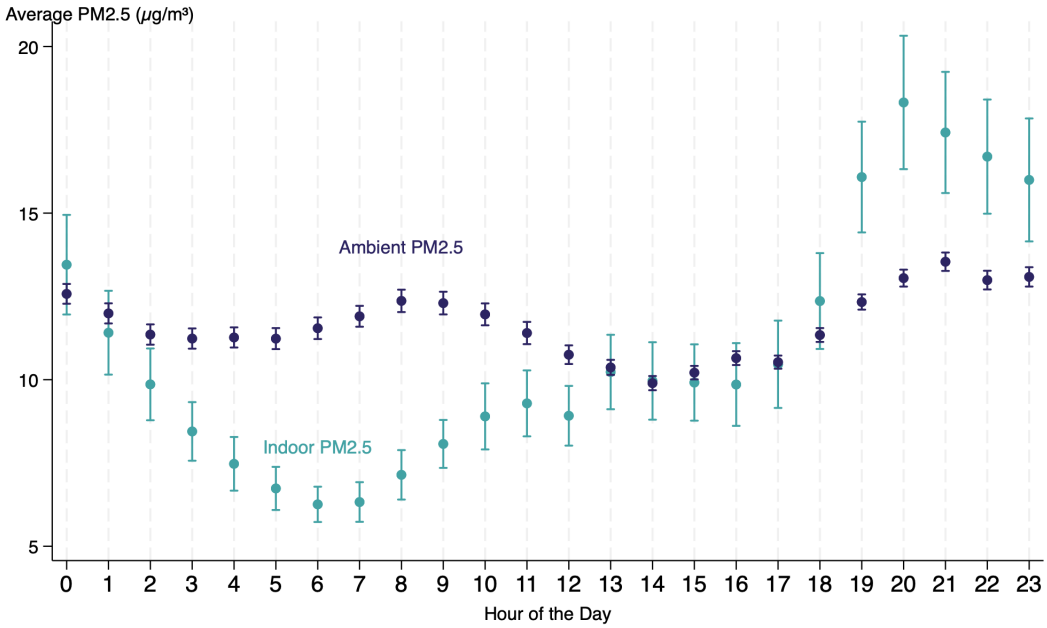
CHAMP Camden Household Air Monitoring Project · LSE · USC University of Southern California · Camden

Figure A5: Monitor Picture



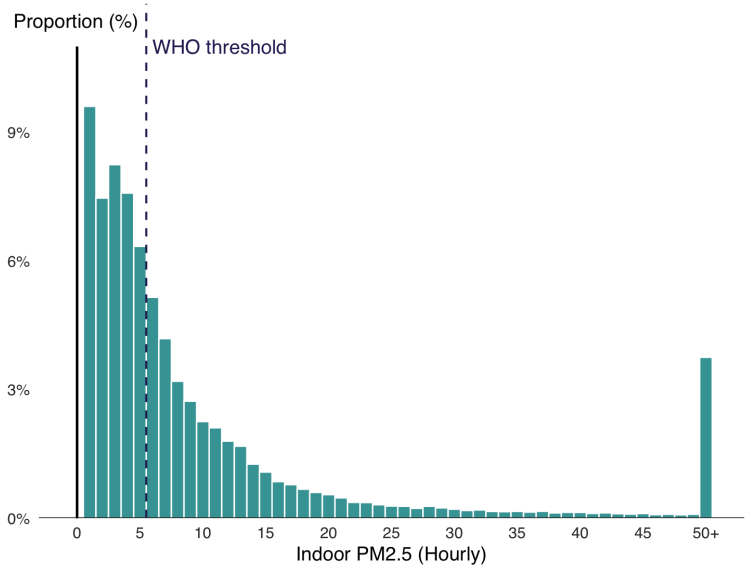
Note: This figure displays a picture of the Kaiterra Laser Egg Air Pollution Monitor.

Figure A6: Average PM2.5 by Hour of the Day



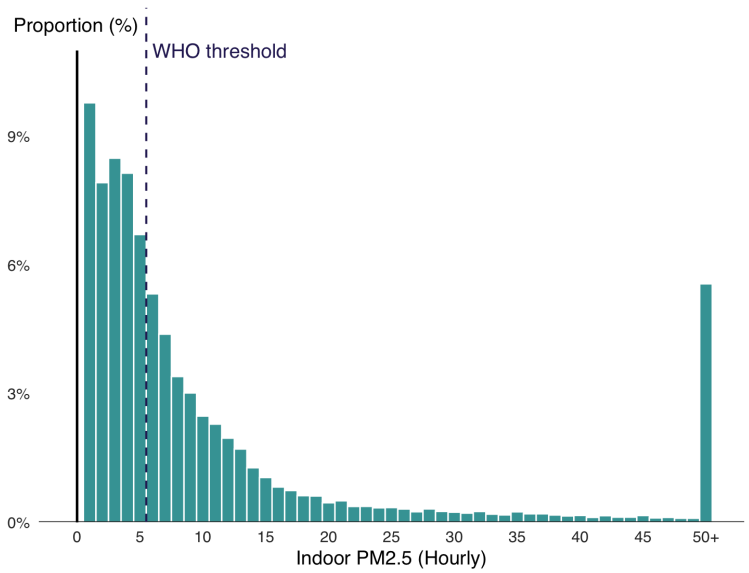
Note: This figure displays average indoor and ambient PM2.5 by hour of the day during the pre-treatment period.

Figure A7: Histogram of Indoor PM2.5 Whole Day



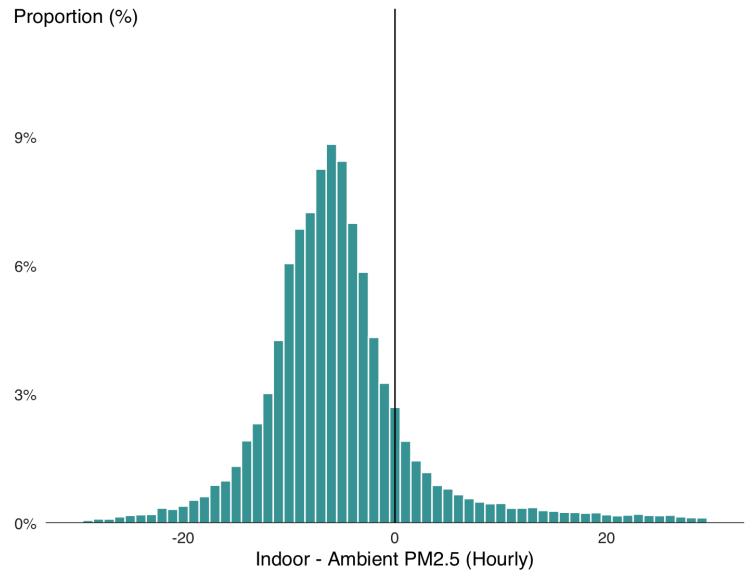
Note: This figure displays the distibution of the difference between IAP and AAP. Indoor PM2.5 is higher than ambient PM2.5 16% of the time.

Figure A8: Histogram of Indoor PM2.5 Occupancy Time



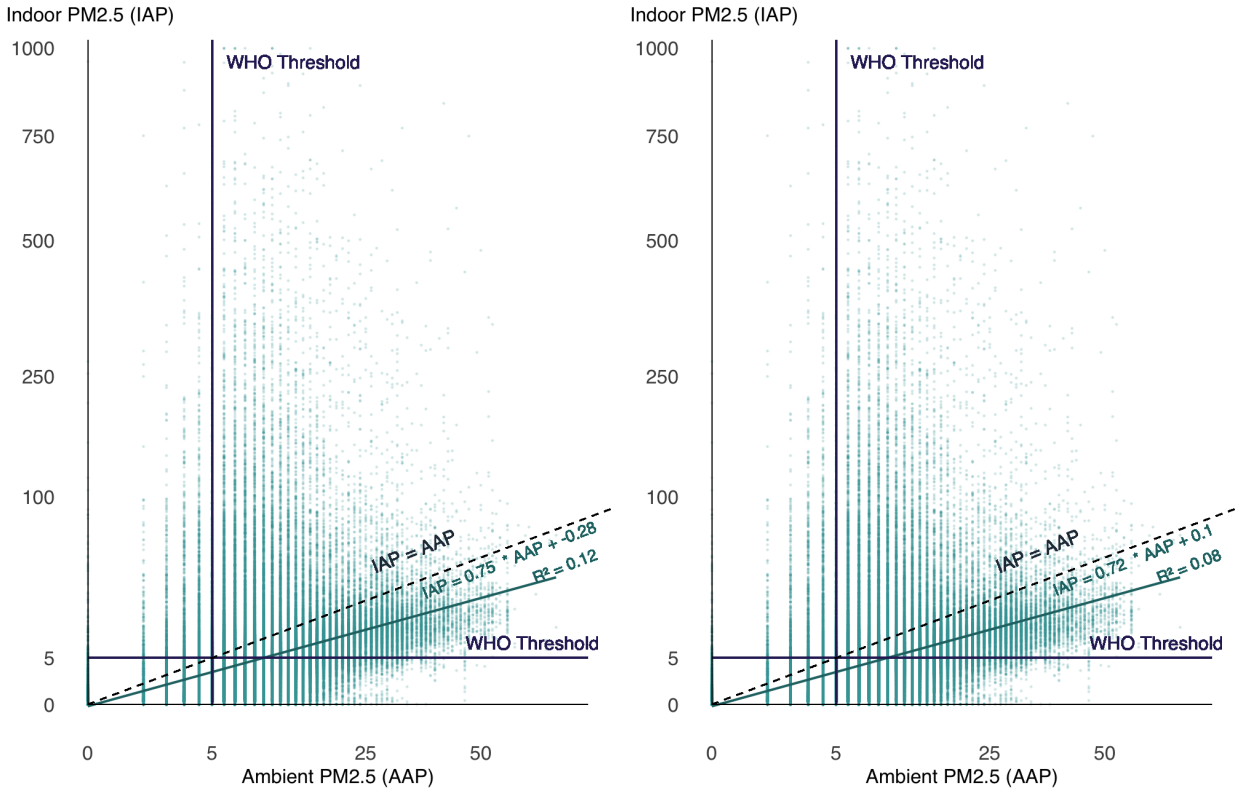
Note: These figure displays the distribution of the difference between IAP and AAP. Indoor PM2.5 is higher than ambient PM2.5 22% of the occupancy time.

Figure A9: Occupancy Time Indoor PM2.5 minus Ambient PM2.5



Note: These figure displays the distribution of indoor PM2.5 from 4pm-11pm.

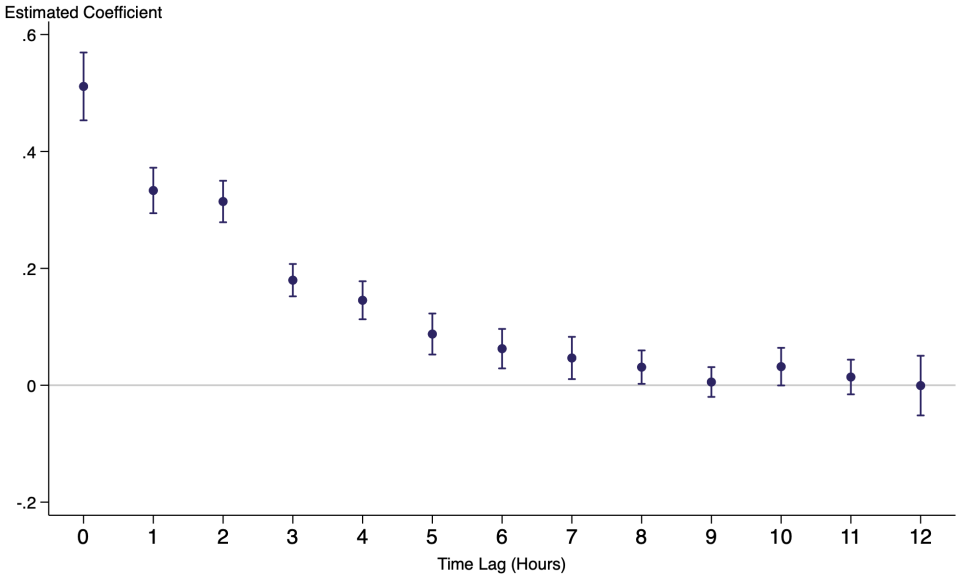
Figure A10: Scatter Plots of Ambient and Indoor PM2.5 by Distance to Air Pollution Monitor



(a) Below Median Distance (1.79km) (b) Above Median Distance (1.79km)

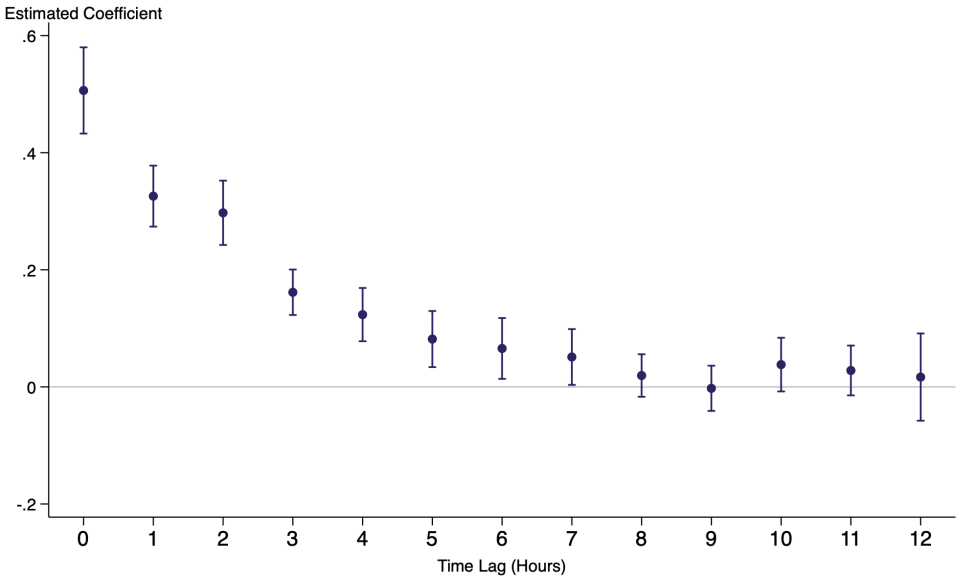
Note: These figures display scatter plots of ambient and indoor PM2.5. The first figure shows this relationship for households which live within 1.79km of their nearest air pollution monitor, and the second figure shows this for households who live further away.

Figure A11: How Ambient PM2.5 Affects Indoor PM2.5 Over Time



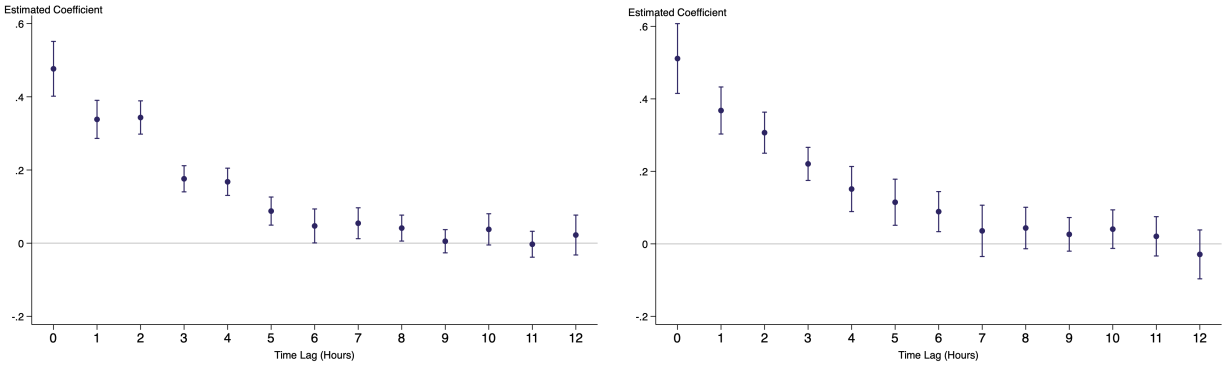
Note: This figure displays estimates of the coefficients of the effect of Ambient PM2.5 at different times on indoor PM2.5, controlling for outside temperature, dew point temperature, wet-bulb temperature, hour, day, month and household fixed effects. Standard errors clustered at the household level and by date. 95% confidence intervals are presented.

Figure A12: How Ambient PM2.5 Affects Indoor PM2.5 Over Time Pre-Treatment



Note: This figure displays estimates of the coefficients of the effect of Ambient PM2.5 at different times on indoor PM2.5 for the first two weeks of the experiment, controlling for outside temperature, dew point temperature, wet-bulb temperature, hour, day, month and household fixed effects. Standard errors clustered at the household level and by date. 95% confidence intervals are presented.

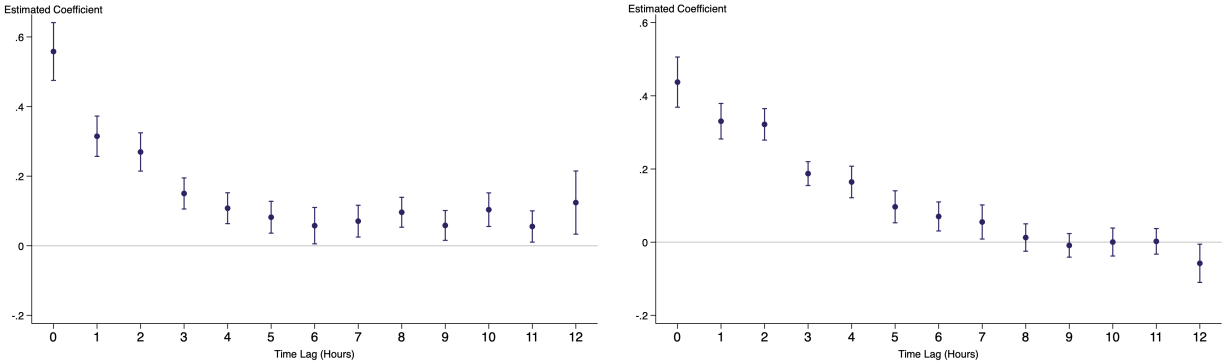
Figure A13: Post-Treatment Time Analysis



(a) Post-Treatment Time: Treatment (b) Post-Treatment Time: Control

Note: These figures display the penetration rate of ambient PM2.5 to indoor PM2.5 categorized by treatment and control.

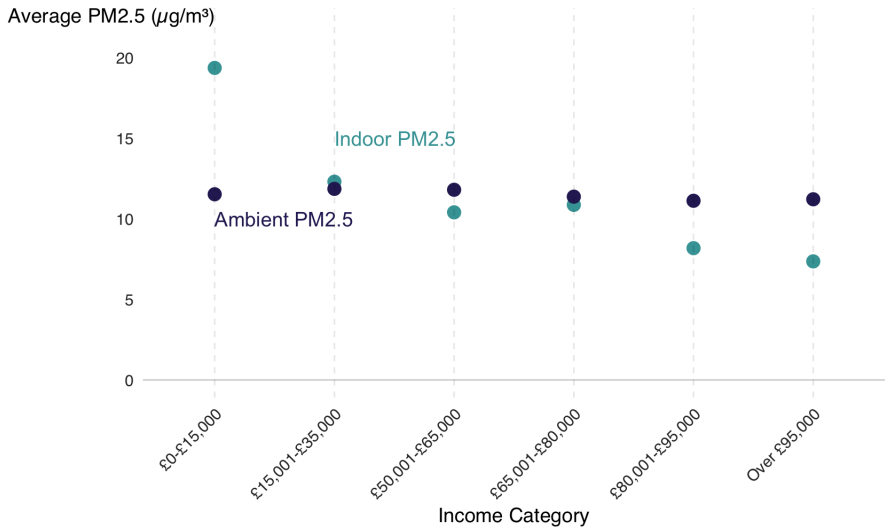
Figure A14: Seasonal Analysis of PM2.5 Penetration Rates



(a) Summer (b) Winter

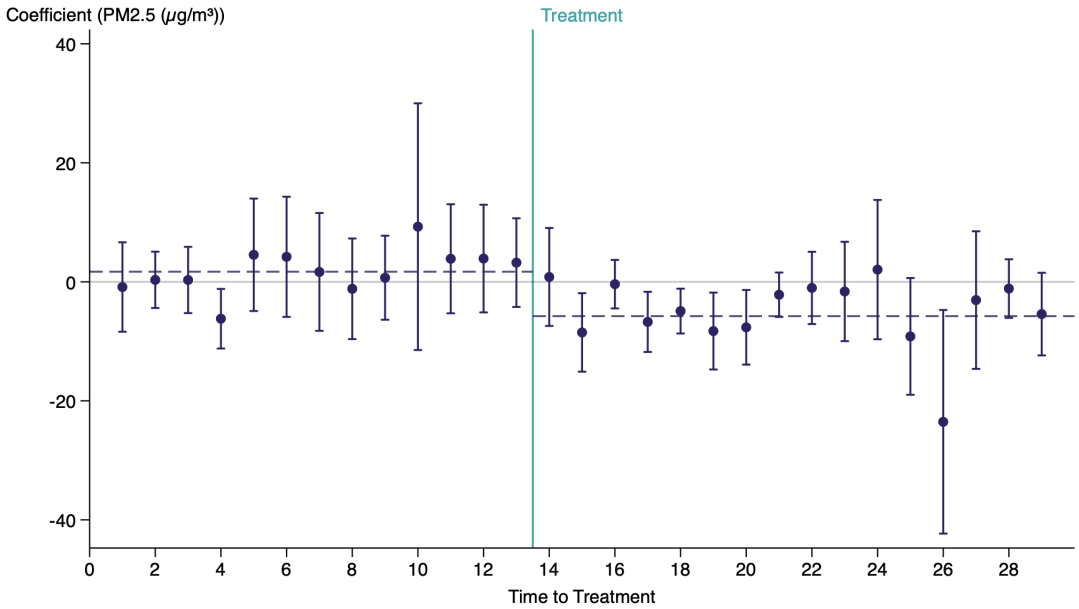
Note: These figures display the penetration rate of ambient PM2.5 to indoor PM2.5 categorized by different seasons.

Figure A15: Average PM2.5 by Income



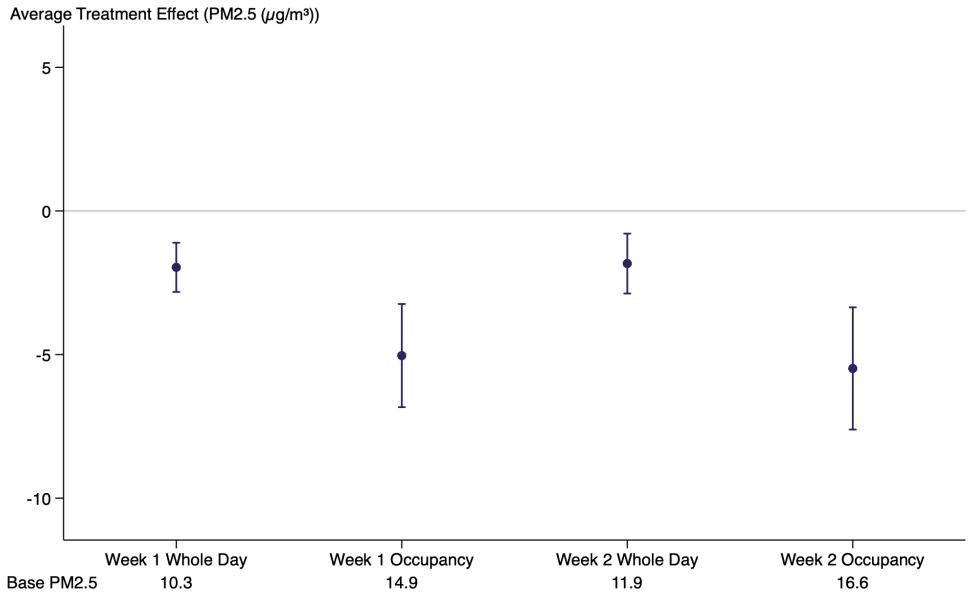
Note: This figure displays average indoor and ambient PM2.5 by income category.

Figure A16: Average Treatment Effects by Day (occupancy hours only)



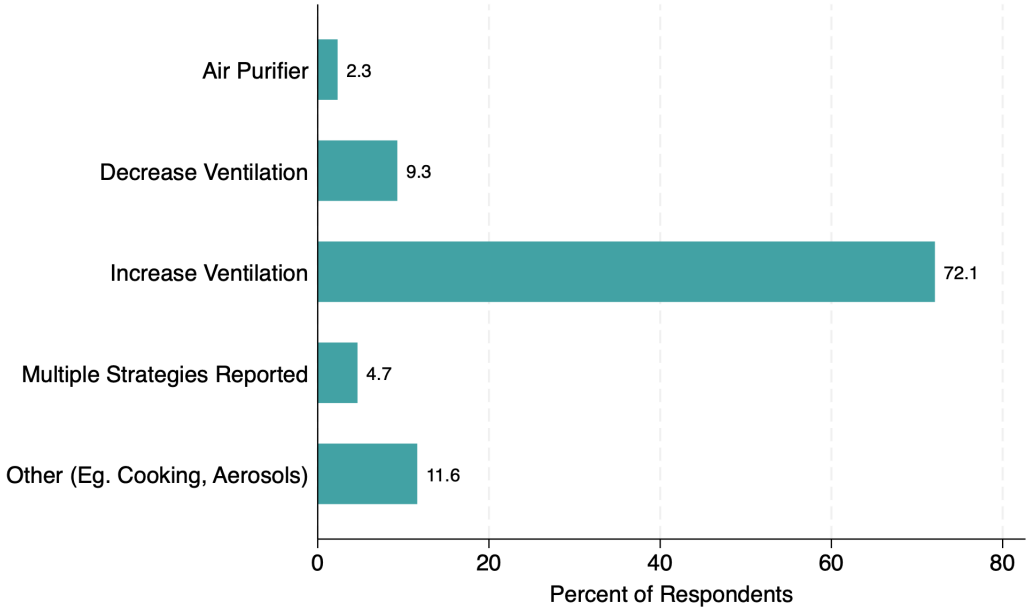
Note: This figure displays estimates of the coefficients of the average treatment effect by day respective to treatment time from 4pm-11pm. 95% confidence intervals are presented.

Figure A17: Average Treatment Effect by Week of the Experiment



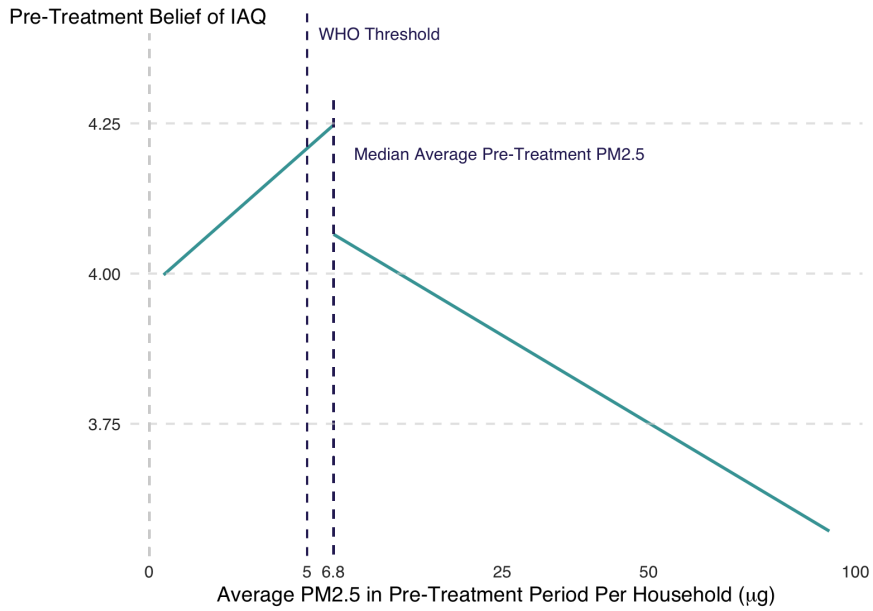
Note: This figure displays the average treatment effect by week after treatment and time of day bootstrapped 1000 times, controlling for ambient PM2.5 and day and month fixed effects. 95% confidence intervals are presented, clustered at the household and date level.

Figure A18: Changes Implemented by Treatment Group



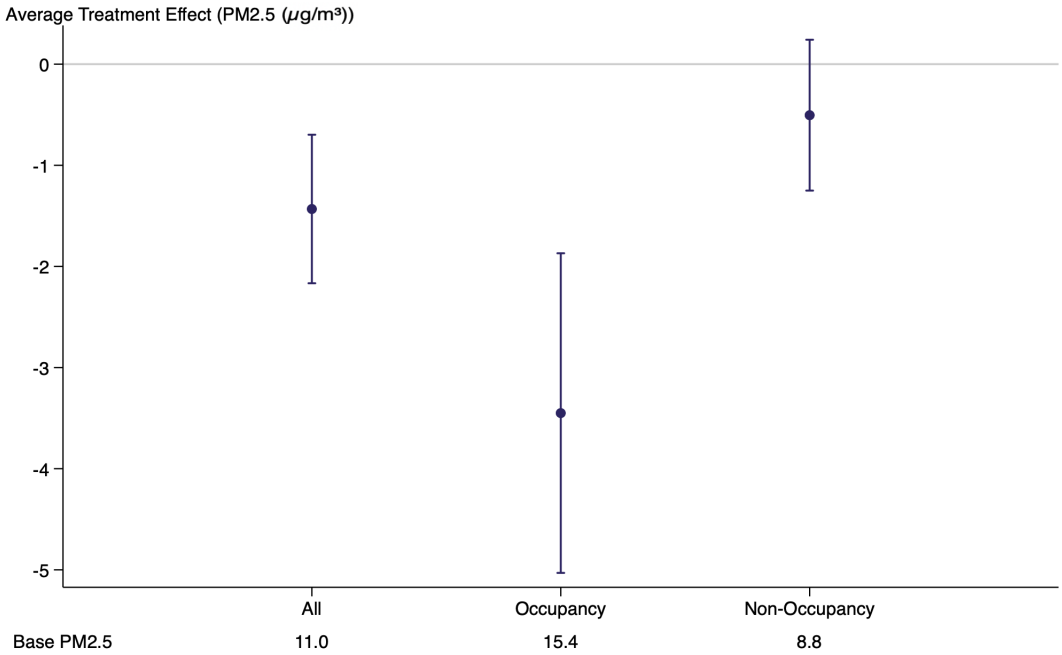
Note: This figure displays any changes implemented by the treatment group. This is analysed by the endline survey.

Figure A19: Mean PM2.5 vs Belief



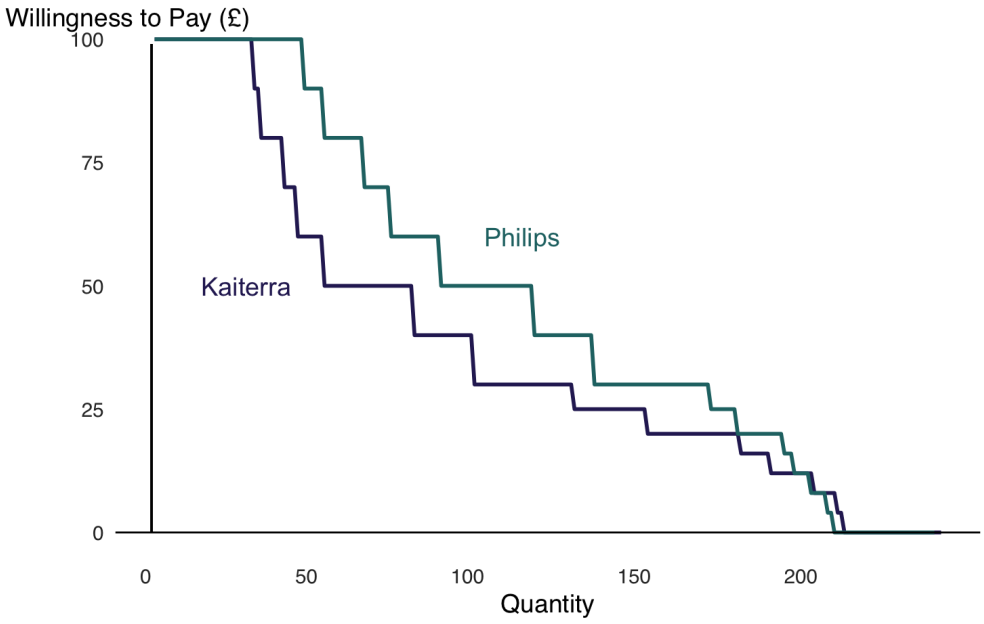
Note: This figure displays lines of best fit between the mean of PM2.5 for each household in the pre-treatment period and how the household answered in the survey about their PM2.5. The higher this belief is, the worse they believe their atmosphere is for their health. The options ranged from 1-5, with 5 being the worst. Here, we have split the graph into two sections: the first half shows households who had lower average pollution levels pre-treatment and the second half shows households who had higher average pollution levels pre-treatment. We can see from this graph that people who had better pollution in their environment actually believed it was worse.

Figure A20: Main Results with Household Fixed Effects



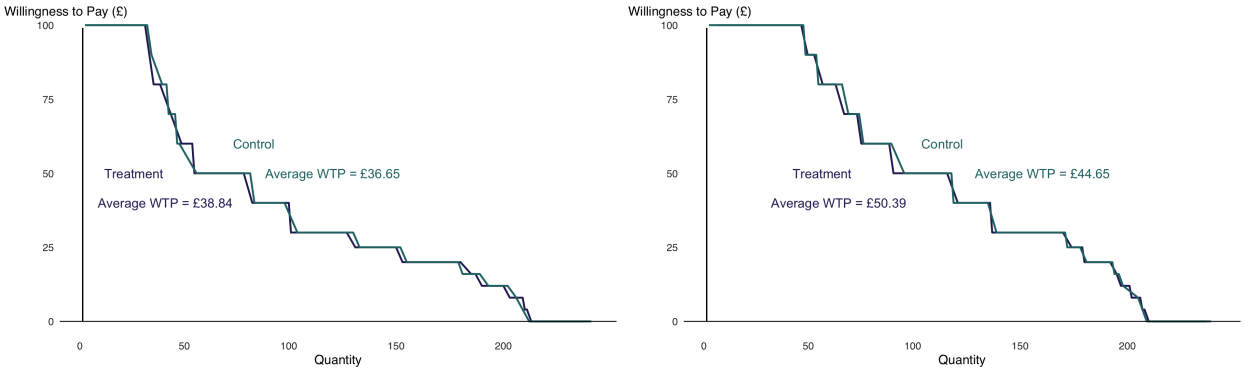
Note: This figure displays estimates of the coefficients of the average treatment effect by day respective to treatment time bootstrapped 1000 times with household fixed effects. Ambient pollution levels as well as day and month fixed effects have been controlled for. 95% confidence intervals are presented clustered at a household and date level. Occupancy time here refers to 16:00-23:00. Base PM2.5 refers to the post-treatment time control group average PM2.5 levels.

Figure A21: WTP



Note: This figure displays the WTP demand curves. The quantity demanded signifies the number of households who answered how much they were willing to pay. There were 237 households who answered this question, with answers ranging between £0-100. We used the BDM incentive compatible eliciation process here to avoid bias and mimic a second-price sealed-bid auction (Becker et al., 1964).

Figure A22: Histograms of Households by Monitor Type

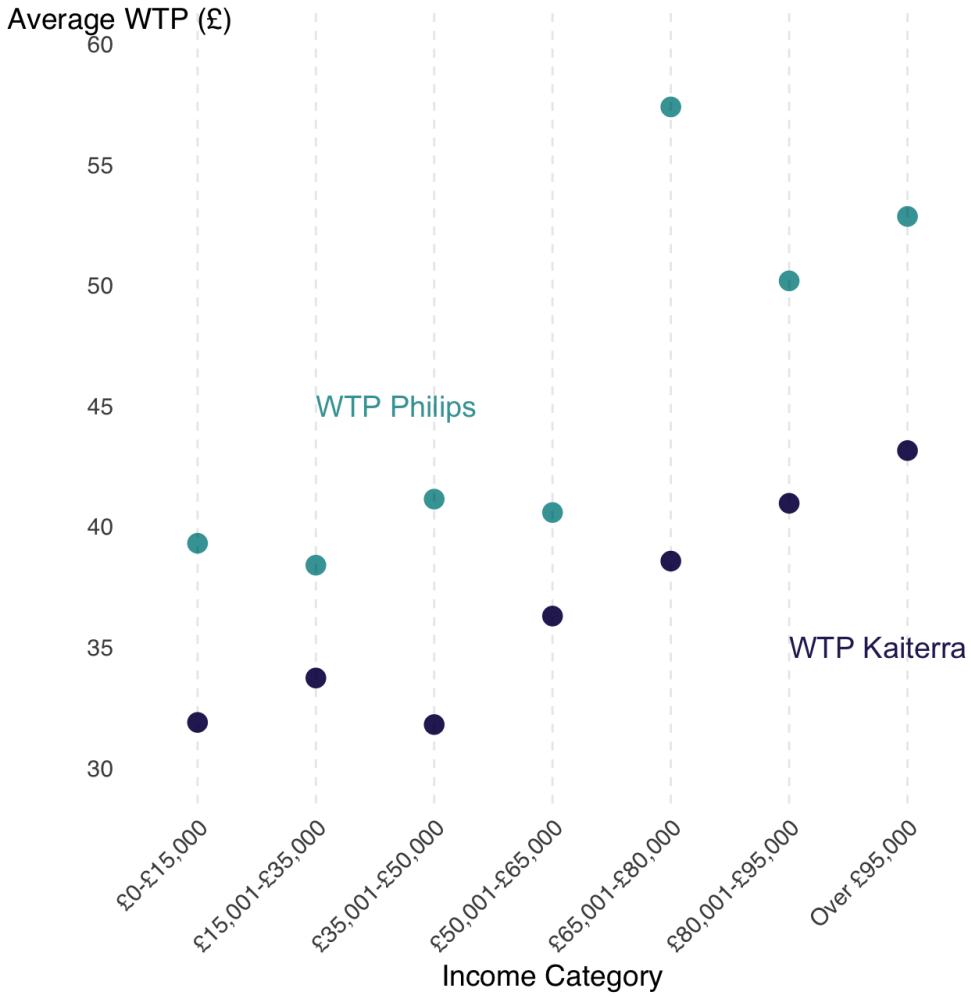


(a) WTP Kaiterra

(b) WTP Philips

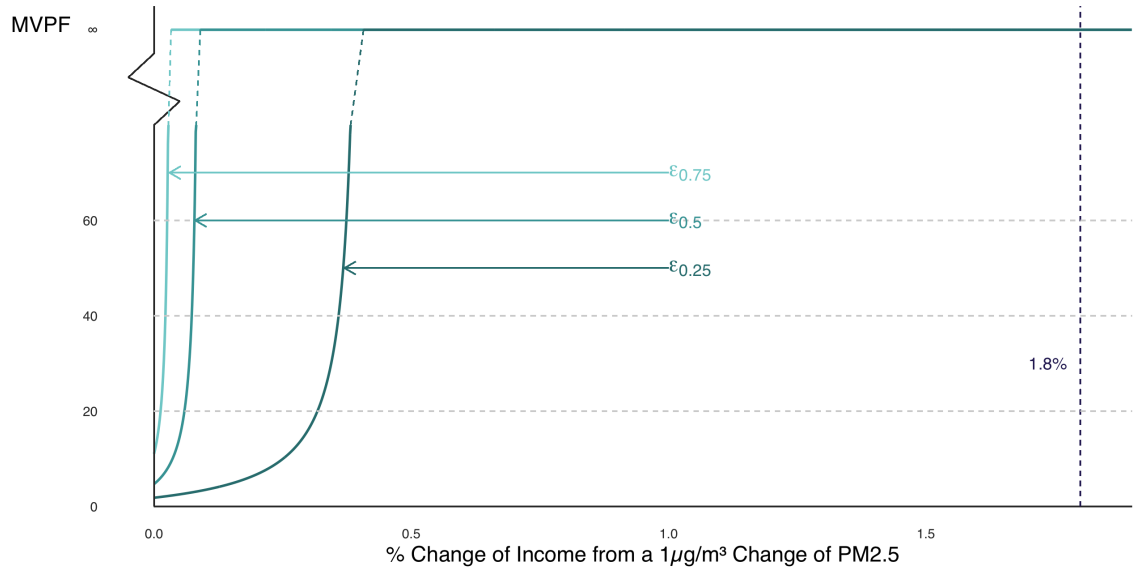
Note: This figure displays the WTP demand curves for the Kaiterra Air Pollution Monitor and the Philips Air Purifier.

Figure A23: WTP by Income



Note: This figure displays the average WTP by income level for the Kaiterra Monitor and Philips Air Purifier.

Figure A24: MVPF by the Income Change due to PM2.5 reduction and price elasticity of IAP monitor demand



Note: This figure displays how the MVPF varies with the % change of income due to PM2.5 reductions. This is for the IAP monitor over a 20 year horizon across the interquartile ranges of elasiticies with the darkest shade referring to a PED of -0.22 and the lightest shade corresponding to a PED of -1.31. This is specifically relating to the third row of Table 6, as described in the text and shown in Figure 9. With our median elasticity of 0.75, we show that MVPF = ∞ for the Borgschulte et al. (2024) estimate of 1.8%. These results are robust to a 10 year horizon too. This shows that an estimate as small as 0.083% can still have an infinite MVPF for the IAP monitor using the median elasticity.

B Tables

Table A1: Waves by Month

Wave	Dates	Number of Households
1	August-September 2022	66
2	November-December 2022	52
3	February-March 2023	56
4	May-June 2023	38
5	October-November 2023	46

Note: This table displays the time frame in which each wave of the trial was conducted. Each household has one month of data (2 weeks before treatment and 2 weeks after.

Table A2: External Validity

Characteristic	Camden	England	Our Sample
Single Household	34%	30%	24%
Socially Rented	34%	17%	13%
Homeowners	30%	63%	51%
Gas Heated	66%	73%	81%
Electric Heated	13%	8%	10%
Percent Below Median Income	45%	50%	24%
NVQ Level 4 or Above	72%	43%	86%

Note: This table displays the Camden and England averages of various household characteristics relative to the sample studied.

Table A3: Summary Statistics

	All			Pre-Treatment		
	All	Occupancy	Non-Occupancy	All	Occupancy	Non-Occupancy
Indoor PM2.5	10.539	14.281	8.664	10.815	14.649	8.899
	(36.60)	(45.92)	(30.72)	(34.92)	(46.26)	(27.36)
Indoor AQI	28.328	35.409	24.781	29.787	36.602	26.382
	(47.01)	(55.72)	(41.53)	(46.74)	(56.28)	(40.72)
Indoor Temperature	21.280	21.672	21.084	21.702	22.113	21.497
	(3.67)	(3.70)	(3.64)	(3.77)	(3.82)	(3.74)
Outdoor PM2.5	10.774	11.578	10.371	11.661	12.188	11.398
	(7.17)	(6.68)	(7.36)	(7.95)	(6.95)	(8.39)
Outdoor Temperature	12.992	15.115	11.928	13.825	16.054	12.709
	(6.11)	(6.56)	(5.58)	(6.20)	(6.66)	(5.63)

Note: This table displays the descriptive statistics from both the overall and pre-treatment periods. Standard deviations are reported in parentheses.

Table A4: Household Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Below Median Income	10.410**									9.894*
	(4.59)									(5.64)
College Educated		-0.983								5.382
		(3.47)								(4.95)
Owner Occupied			-5.884**							-3.692
			(2.55)							(2.91)
Single Household				0.000						-3.078
				(2.81)						(4.33)
Children Dummy					-2.301					-0.918
					(2.59)					(3.14)
Asthma Dummy						5.541				6.938
						(5.81)				(5.06)
Health Condition							0.857			-0.316
							(1.55)			(2.07)
Number of Residents								-0.885		-0.483
								(0.96)		(1.57)
Number of Smokers									19.894***	17.885***
									(6.92)	(6.78)
Observations	22,701	22,701	24,093	24,093	24,093	22,701	24,093	22,591	22,488	22,378

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects and ambient PM2.5 levels are controlled for. The observations are restricted to occupancy time (16:00-23:00) to reflect times when the average person is at home. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A5: Dwelling Characteristics

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
Fireplace	-3.682**												-2.238
	(1.83)												(2.72)
Purifiers		0.502											1.311
		(3.38)											(3.35)
Gas Hob			-3.560										13.291***
			(2.67)										(4.49)
Electric Hob				8.920**									19.238***
				(4.16)									(6.01)
Induction Hob					-4.167								9.811*
					(2.78)								(5.60)
Indoor Temperature						0.358							0.550*
						(0.34)							(0.31)
Number of Windows							-0.781**						-0.780***
							(0.31)						(0.29)
Open Plan Kitchen								0.639					3.158
								(2.77)					(2.72)
Stovetop									-3.768				0.801
									(4.80)				(4.43)
Oven										1.320			0.576
										(2.85)			(3.07)
Microwave											2.216		1.511
											(2.85)		(3.09)
Toaster												0.769	1.557
												(2.97)	(3.29)
Observations	24,093	22,376	20,913	20,913	20,913	24,093	21,506	22,488	22,701	22,701	22,701	22,701	20,041

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects and ambient PM2.5 levels are controlled for. The observations are restricted to occupancy time (16:00-23:00) to reflect times when the average person is at home. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A6: Main Results

	Total Indoor PM2.5	Occupancy Indoor PM2.5	Non-Occupancy Indoor PM2.5
Panel A: Bootstrapped Estimates			
Treatment x Post	-1.903***	-4.964***	-0.426
	(0.39)	(0.87)	(0.39)
	[0.001]	[0.001]	[0.257]
Treatment	-0.233	1.261**	-0.980***
	(0.27)	(0.62)	(0.24)
Post	0.827***	2.525***	-0.129
	(0.31)	(0.64)	(0.30)
Ambient PM2.5	0.557***	0.549***	0.519***
	(0.02)	(0.04)	(0.01)
Constant	-0.885*	-1.460	-0.060
	(0.53)	(1.23)	(0.52)
Panel B: Non-Bootstrapped Estimates			
Treatment x Post	-1.903	-4.964**	-0.426
	(1.34)	(2.33)	(1.06)
Treatment	-0.233	1.261	-0.980
	(1.67)	(2.65)	(1.35)
Post	0.827	2.525*	-0.129
	(0.95)	(1.51)	(0.83)
Ambient PM2.5	0.557***	0.549***	0.519***
	(0.04)	(0.08)	(0.03)
Constant	4.772***	7.385***	3.969***
	(1.03)	(1.55)	(0.97)
Base PM2.5	10.95	15.36	8.80
Observations	150,079	50,085	99,994

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. The observations that are restricted to occupancy time (16:00-23:00) are to reflect times when the average person is at home. Base presents the mean level of PM2.5 in the control group in the post-treatment period. Panel A shows estimates bootstrapped 1000 times and Panel B has non-bootstrapped elements. Square brackets display the MHT adjusted Romano-Wolf p-values for our main specification. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A7: AQI Brackets

	0-50	51-100	101-150	151-200	201-300	301-500
Panel A: Bootstrapped Estimates						
Treatment x Post	-0.000	0.029***	-0.007**	-0.013***	-0.004***	-0.005***
	(0.01)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)
Treatment	0.019***	-0.014***	-0.009***	0.006**	-0.001	-0.001
	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Post	0.006	-0.023***	0.002	0.010***	0.003**	0.002*
	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Ambient PM2.5	-0.015***	0.012***	0.001***	0.001***	0.000*	0.000***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Constant	1.129***	-0.125***	0.002	-0.006	0.000	-0.000
	(0.02)	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)
Panel B: Non-Bootstrapped Estimates						
Treatment x Post	-0.000	0.029**	-0.007	-0.013**	-0.004	-0.005
	(0.02)	(0.01)	(0.00)	(0.01)	(0.00)	(0.00)
Treatment	0.019	-0.014	-0.009	0.006	-0.001	-0.001
	(0.03)	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)
Post	0.006	-0.023**	0.002	0.010**	0.003*	0.002
	(0.02)	(0.01)	(0.00)	(0.00)	(0.00)	(0.00)
Ambient PM2.5	-0.015***	0.012***	0.001***	0.001***	0.000**	0.000**
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Constant	0.951***	-0.006	0.023**	0.018***	0.007***	0.007***
	(0.02)	(0.01)	(0.01)	(0.01)	(0.00)	(0.00)
	(1.03)	(1.55)	(0.97)			
Observations	50,085	50,085	50,085	50,085	50,085	50,085

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of AQI. This is a linear probability model, where each coefficient shows the probability that a household's AQI level will be within a certain bracket. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. The observations that restricted to occupancy time (16:00-23:00) to reflect times when the average person is at home. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A8: Below vs Above Median Base PM2.5

	Above		Below	
	Whole Day PM.5	Occupancy PM2.5	Whole Day PM.5	Occupancy PM2.5
Treatment x Post	-4.342***	-10.567***	0.819***	0.816***
	(0.75)	(1.57)	(0.16)	(0.31)
Treatment	-0.302	3.185***	-0.050	-0.277
	(0.50)	(1.16)	(0.09)	(0.20)
Post	1.687***	4.813***	-0.029	0.334
	(0.57)	(1.18)	(0.14)	(0.26)
Ambient PM2.5	0.629***	0.641***	0.425***	0.384***
	(0.02)	(0.06)	(0.01)	(0.01)
Constant	2.287***	0.099	-1.753***	-0.526
	(0.74)	(1.52)	(0.53)	(1.16)
Observations	77,644	25,933	72,435	24,152

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. The observations that are restricted to occupancy time (16:00-23:00) are to reflect times when the average person is at home. Standard errors bootstrapped 1000 times and clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A9: Belief Outcomes Above vs Below

	Above Median Base PM2.5				Below Median Base PM2.5			
	Indoor		Outdoor		Indoor		Outdoor	
	Belief	Confidence	Belief	Confidence	Belief	Confidence	Belief	Confidence
Treatment	0.040	0.207	0.081	-0.154	0.494***	0.409**	0.242	0.411**
	(0.14)	(0.18)	(0.18)	(0.20)	(0.14)	(0.19)	(0.15)	(0.19)
Constant	0.169	0.034	-0.017	0.186	-0.068	0.017	0.017	-0.034
	(0.10)	(0.13)	(0.13)	(0.14)	(0.10)	(0.13)	(0.10)	(0.13)
Observations	121	120	121	121	113	113	113	112

Note: This table displays estimates of the coefficients of treatment on the difference in air quality beliefs in the baseline and endline survey. Air quality belief represents a participants perception of indoor and outdoor air quality. Confidence represents how certain a participant is about this level of air quality. Standard errors are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A10: Household Fixed Effects

	Total Indoor PM2.5	Occupancy Indoor PM2.5	Non-Occupancy Indoor PM2.5
Treatment x Post	-1.432***	-3.450***	-0.505
	(0.37)	(0.81)	(0.38)
Post	1.016***	2.844***	-0.023
	(0.30)	(0.65)	(0.29)
Ambient PM2.5	0.587***	0.612***	0.537***
	(0.02)	(0.04)	(0.01)
Constant	-2.031***	-2.128	-1.844**
	(0.72)	(1.49)	(0.78)
Observations	150,079	50,085	99,994

Note: This table displays bootstrapped estimates of the coefficients of binary variable post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. The treatment variable is not included due to the inclusion of household fixed effects. Day, month and household fixed effects are controlled for, as well as ambient levels of PM2.5. The observations that are restricted to occupancy time (16:00-23:00) are to reflect times when the average person is at home. Standard errors (bootstrapped 1000 times) clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table A11: Whole Day Specifications

	All waves			First waves		
	Whole Day	No Ambient	No FE	Whole Day	No Ambient	No FE
Treatment x Post	-1.903***	-1.247***	-0.587	-2.019***	-1.688***	-0.318
	(0.39)	(0.37)	(0.38)	(0.55)	(0.55)	(0.52)
Treatment	-0.233	-0.582**	-0.642**	1.249***	1.073***	0.874**
	(0.27)	(0.26)	(0.27)	(0.37)	(0.38)	(0.37)
Post	0.827***	-0.814***	-0.170	0.769*	-0.215	-0.279
	(0.31)	(0.28)	(0.29)	(0.41)	(0.41)	(0.40)
Ambient PM2.5	0.557***			0.542***		
	(0.02)			(0.02)		
Constant	-0.885*	6.747***	11.124***	-2.746***	5.975***	9.335***
	(0.53)	(0.49)	(0.18)	(0.80)	(0.67)	(0.23)
Observations	150,079	150,079	150,079	65,363	65,363	65,363
Time Fixed Effects	Yes	Yes	No	Yes	Yes	No

Note: This table displays estimates of the coefficients of binary variables treatment and post as well as an interaction between the treatment and post on hourly levels of indoor PM2.5. Day and month fixed effects are controlled for, as well as ambient levels of PM2.5. Standard errors clustered two way at the household level and by date are reported in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

C Online Appendix

C.1 Impact on Health

We first focus on the intervention's impact on health. To analyze this dimension, we follow Carozzi and Roth (2023) and begin by applying the point estimates derived in our study to Concentration-Response (C-R) functions extracted from the existing literature. This synthesis of empirical data from our intervention with established C-R functions allows for a robust estimation of the health impacts due to changes in indoor air pollution. Next, we use the UK Government Value of a Prevented Fatality (VPF) and combining this with the C-R functions to quantify the potential mortality benefit of our intervention.[40]

We draw from the literature to estimate the upper and lower bound mortality C-R functions as in Fowlie et al. (2019). Lepeule et al. (2012) provided a pivotal extension of the Harvard Six Cities study and estimate that an annual increment of $10\mu g/m^3$ in PM2.5 concentrations correlates with a 14% increase in the risk of mortality from all causes compared to baseline (this is our upper bound C-R function).[41] Complementing this, Krewski et al. (2009) conducted an extensive cohort analysis,[42] and estimated that an annual increment of $10\mu g/m^3$ PM2.5 concentrations correlates with a 6% increase in the risk of mortality from all causes (this is our lower bound C-R function). While these studies focus on ambient PM2.5, there is currently no evidence to suggest that particulate matter originating indoors is less harmful than outdoor sources. As mentioned in the background section, the WHO advisory group on indoor air quality guidelines similarly concluded that there is no compelling evidence to indicate that indoor particulate matter poses a lower health risk than its outdoor counterpart (WHO et al., 2010). As such, we reasonably assume that these C-R functions are also applicable to indoor PM2.5.

More formally, we can outline the calculation as follows. We begin by defining the relationship between air pollution and mortality risk:

$$\ln(y) = \alpha + \beta \cdot PM2.5$$

where $\ln(\cdot)$ is the natural logarithm and β is the coefficient of interest which measures the esti-

[40] For more details on VPF and its relationship with the Value of a Life Year (VOLY) for the UK Government, please see:https://www.gov.uk/government/publications/valuation-of-risks-to-life-and-health-monetary-value-of-a-life-year-voly/annexe-5. Also, please note that our overall approach for estimating the health mortality impact aligns with the US EPA in their Regulatory Impact Analysis.

[41] Utilizing a Cox proportional hazards model, Lepeule et al. (2012) reported a Relative Risk (RR) of 1.14, with a 95% confidence interval of [1.07,1.22].

[42] They employed a random-effects Cox model to articulate the concentration-response relationship. They identified a lower mortality RR of 1.06, with a 95% confidence interval of [1.04,1.08], which is our lower bound C-R function.

mated average effect of PM2.5 on mortality. Defining y_0 as the baseline mortality incidence rate, we can now express the relationship between changes in $PM_{2.5}$ (ΔPM) and the mortality incidence rate (Δy) as:

$$\Delta y = y_0 \cdot \left(1 - \frac{1}{\exp(\beta \cdot \Delta PM2.5)}\right)$$

By multiplying the change in the mortality incidence rate by the relevant population, we calculate the total change in mortality. Multiplying this change by the VPF then provides a monetary estimate of the mortality benefit.

Using this approach, we estimate the monetary health benefits of our intervention and generalize them to the UK population. Utilizing the two C-R functions discussed earlier, our main estimates from Figure 3, and assuming that this effect remains constant (as suggested by our analysis in Figure A17), we evaluate the impact of providing all households in the UK with the same treatment implemented in our study.[43] Our analysis considers two key reductions in PM2.5 concentrations: a 5 $\mu g/m^3$ reduction observed during occupancy hours and a 1.9 $\mu g/m^3$ reduction calculated as the all-day average. These reflect the treatment effects measured during periods of maximum exposure (occupancy time) and across the full day, respectively.

Using these reductions, our results suggest that the annual mortality benefit of such an intervention, based on the high and low C-R functions from Lepeule et al. (2012) and Krewski et al. (2009), in conjunction with the UK VPF recommended estimate of £2.4 million (2023 GBP), would be £102.4 billion and £46.4 billion (or £1,501 and £680 per capita), respectively, for the 5 $\mu g/m^3$ reduction. For the smaller 1.9 $\mu g/m^3$ reduction, the annual mortality benefits are estimated at £39.73 billion and £17.79 billion (or £582.05 and £260.63 per capita), respectively.

Given the substantial uncertainty surrounding the C-R functions, we adopt a midpoint C-R function of 1.10. Furthermore, there is also uncertainty surrounding the correct estimated pollution reduction that we should use for our calculation, as we do not observe individuals' exposure throughout the entire day. For instance, individuals might be at school or work during certain hours, where indoor air quality may differ significantly from that observed in their homes. To address this, we adjust our ATE by weighting the observed reduction during occupancy hours (16:00–23:00), which represents one-third of the day. In other words, we divide the 5 $\mu g/m^3$ reduction we observed during occupancy time by 3, resulting in an adjusted ATE of 1.67 $\mu g/m^3$. Using this adjusted ATE and the midpoint C-R function, the estimated annual mortality benefit is £25.52 billion.

[43] We also assume that the treatment effect observed in our London-based study applies to the United Kingdom as a whole.

C.2 Impact on Economic Welfare

In this section, we estimate the willingness to pay (WTP) for an indoor air pollution (IAP) monitor and an air purifier, as well as the welfare implications of subsidizing these technologies. This setting is particularly well-suited for valuation, as treatment group households possess accurate and complete information about their indoor air quality, eliminating issues of misinformation or biased beliefs.

To mitigate common valuation biases (e.g., hypothetical bias), we used an incentive-compatible Becker-DeGroot-Marschak (BDM) mechanism (Becker et al., 1964). Participants were informed that one would be randomly selected to receive a £100 Amazon voucher and asked how much of the voucher they would forgo to acquire either: (1) a Kaiterra Laser Egg monitor, which provides real-time PM2.5 readings, or (2) a Philips 800 Series air purifier, which filters 99% of airborne particulates within 16 minutes. A link to product details was provided. If this question was selected for implementation, a computer-generated price (from a known distribution) was compared with the participant's bid: if the bid exceeded the drawn price, the participant received the item and the difference in voucher value—thus replicating a second-price auction and incentivizing truthful valuation.

Figure A21 shows demand curves for both products. Participants placed higher value on the air purifier—likely due to its direct mitigation effect—than on the monitor, which only provides information.[44,45] Figures A22a and A22b show demand by treatment status. Average WTP for the monitor was £36.65 (control) and £38.84 (treatment), while for the air purifier it was £44.65 (control) and £50.39 (treatment). Dividing WTP by mean PM2.5 concentrations (11.0 and 9.1 $\mu g/m^3$, respectively), we estimate the WTP per $\mu g/m^3$ reduction in PM2.5 as £4.06 (control) and £5.54 (treatment), with a sample average of £4.80.[46]

Reweighting these values to match the population distribution from Table A2, we find that average WTP for the air purifier ranges from £38.94–44.14 (control) and £48.91–52.01 (treatment). Corresponding WTP for the monitor ranges from £34.25–39.28 and £36.06–39.30. These reweighted values closely match sample estimates, consistent with minimal heterogeneity by income, education, or other characteristics.

To our knowledge, this is the first study to experimentally estimate WTP for reductions in indoor PM2.5. The most closely related study is Ito and Zhang (2020), which uses a revealed preference approach based on air purifier sales in China to estimate WTP for PM10. They find households are willing to pay approximately $6.30 (£5.10) per $\mu g/m^3$ of PM10. Adjusting for

[44] Note: the Philips purifier also has a real-time air quality light.

[45] We find no evidence of bunching at zero WTP, suggesting participants were not influenced by shame or stigma associated with high IAP exposure (Butera et al., 2022).

[46] Assuming linearity in WTP.

purchasing power parity (PPP), our UK WTP estimate of £4.80 corresponds to £1.91 in China, suggesting Chinese households value air pollution reduction more—potentially due to greater pollution severity and awareness. This discrepancy is unlikely to stem from pollutant differences, as PM2.5 is a subset of PM10 and filtered by most purifiers.[47]

We next estimate the marginal value of public funds (MVPF) for a £1 subsidy on each technology using the WTP estimates above, health gains from Section C.1, and the framework in Hahn et al. (2024):

$$MVPF = \frac{x, ds + V, dx + I, dx + C, dx}{x, ds + H, dx + T, dx + P, dx} = \frac{1 + \frac{V+I+C}{p}(-\epsilon)}{1 + \frac{H+T+P}{p}(-\epsilon)} \tag{2}$$

Where x is quantity, s the subsidy, p the price, and ϵ the price elasticity of demand. V, I, and C represent health, income, and crime benefits to individuals; H, T, and P represent corresponding government savings. We assume a 2% discount rate over a 10-year period and full pass-through of the subsidy. A key uncertainty is whether demand elasticity already reflects private benefits; if it does, then the numerator simplifies to $1 + \frac{C\epsilon}{p}$, but we show below that our results are robust to either assumption.

We use -0.75 as the mean demand elasticity for the monitor. Based on the ATE of –1.9 $\mu g/m^3$ (daily), and cost estimates from mortality dose-response functions (£260.63–£582.05), we calculate per capita health WTP, scaled to households using the UK average of 2.41 persons per household. Assuming a 2-year monitor lifespan and a price of £135, this yields a 10-year discounted cost of £624.57.

For the air purifier, we assume it reduces PM2.5 to zero and use baseline levels of 11.0 $\mu g/m^3$ (daily) and 15.4 (occupancy time) with a mean demand elasticity of -0.81. This yields per capita health WTP of £1469.61–£3177.42 (daily) and £2031.69–£4325.57 (occupancy). Assuming a 5-year lifespan and £25 annual filter replacement, the 10-year discounted cost is £438.69.

We also estimate income effects using Borgschulte et al. (2024), who find that a 1 $\mu g/m^3$ increase in PM2.5 reduces earnings by 1.81%. Applying this to the UK average income (£35,393) and a 23.4% marginal tax rate,[48] we estimate that each unit reduction yields £149.90 in additional annual tax revenue per person (£361.27 per household), cumulating to £3,310–£6,025 over 10–20 years.

For crime, we use Bondy et al. (2020), who show that a 10-unit AQI increase raises crime by 2.6%. We convert this to a 1.04% crime reduction per 1 $\mu g/m^3$ decrease in PM2.5. Excluding

[47] PM10 includes particles up to 10 micrometers in diameter, encompassing PM2.5.

[48] Weighted across all UK tax brackets.

labor-related costs already captured in income effects, Heeks et al. (2018) estimate remaining social costs at £36 billion (2016), or £1665 per household in 2023. Applying a 1.04% reduction and discounting over 10 years, this translates to annual per household savings of £13.18, which we scale by the ATE and incorporate via $\frac{C \cdot \epsilon}{p}$.

On the cost side, NHS savings per 1 $\mu g/m^3$ are £0.952 million per 100,000 people over 10 years, or £43.41 per household for our ATE of –1.9, rising to £352 for an ATE of –15.4. Tax revenue gains are added as a fiscal benefit, and police-related crime savings (£8.01 billion or £370 per household) are included via $\frac{P \cdot \epsilon}{p}$.

Table 6 and Figure 9 summarize MVPF estimates. Our baseline scenario—ATE of –1.9 over 20 years, low C-R function—yields total private benefits of £47.08 per £1 subsidy: £1 transfer, £0.75 health benefit, £44.83 income benefit, and £0.50 crime benefit. Government gains are £13.02 per £1 subsidy: £13.69 in additional tax revenue, £0.18 NHS savings, and £0.15 in reduced crime expenditures.

The resulting MVPF is effectively infinite. Figure 9 shows negative net government costs due to large fiscal benefits. Even omitting income effects (Borgschulte et al. (2024)) yields an MVPF of 3.4—well above typical values reported by Hahn et al. (2024). Our result remains robust across a wide range of assumptions: MVPF stays infinite even if fiscal externalities fall by 86%, and even modest ATEs (e.g., –0.14) lead to infinite MVPFs for a £1 subsidy when elasticity is median (–0.75). Figure 10 further illustrates this robustness across treatment effects and demand elasticities. Even the 25th percentile elasticity (–0.22) produces an infinite MVPF at an ATE of –0.46.

It is rare to observe infinite MVPFs. Hendren and Sprung-Keyser (2020) document such outcomes for select child health and education programs, but no environmental policy evaluated by Hahn et al. (2024) exhibits an infinite MVPF. To our knowledge, this is the first demonstration that an environmental intervention—specifically a subsidy for IAP mitigation—can pay for itself.